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China Liver Cancer Guidelines for the Diagnosis and Treatment of Hepatocellular Carcinoma (2024 Edition)

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Keywords
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Abstract
Hepatocellular carcinoma (HCC), which accounts for approximately 75–85% of primary liver cancers, ranks 4th in newly diagnosed cases among various types of cancer in China, and is the 2nd leading cause of cancer-related mortality, thereby posing a significant threat to the life and health of the Chinese population. Since the publication of the "*Guidelines for Diagnosis and Treatment of Primary Liver Cancer in China*" in June 2017, which were updated by the China's National Health Commission in December 2019 and December 2021, additional high-quality evidence from researchers worldwide regarding the diagnosis, staging, and treatment of HCC has emerged, necessitating another

update to the guidelines. The new edition (2024 Edition) was written by more than 120 multidisciplinary experts in the field of HCC in China, which not only reflects the real-world situation in China but also may reshape the nationwide diagnosis and treatment of HCC. The new guideline aims to encourage the implementation of evidence-based practice and improve the national average 5-year survival rate for patients with HCC, as proposed in the "Healthy China 2030: A Vision for Health Care."

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Overview

According to the data released by the National Cancer Center of China, in 2022, there were 367,700 new cases of primary liver cancer nationwide, ranking 4th among all types of new cancer cases (lung, colorectal, thyroid, and liver), and liver cancer has the 5th highest incidence rate (lung, female breast, thyroid, colorectal, and liver). There were 316,500 deaths due to primary liver cancer in 2022, with both the number of deaths and the mortality rate ranking 2nd (lung and liver) [1, 2]. Primary liver cancer is classified into three main pathological types: hepatocellular carcinoma (HCC), intrahepatic cholangiocarcinoma (ICC), and combined hepatocellular-cholangiocarcinoma (cHCC-CCA). These pathological subtypes of primary liver cancer vary greatly in their pathogenesis, biological behavior, pathologic histology, treatment, and prognosis, as HCC accounts for 75–85% of all cases of primary liver cancer, and ICC accounting for 10–15% of cases [3, 4]. In this guideline, the term "liver cancer" refers exclusively to HCC.

To standardize the diagnosis and treatment of liver cancer in China, the former National Health and Family Planning Commission of the People's Republic of China released the *Guidelines for the Diagnosis and Treatment of Primary Liver Cancer* (2017 Edition) in June 2017, which were updated by the National Health Commission in December 2019 and December 2021, respectively, and finally revised and released the 2019 Edition and the 2022 Edition. The *Guidelines for the Diagnosis and Treatment of Primary Liver Cancer* (2022 Edition) reflected the advancements in research, diagnosis, and comprehensive multidisciplinary treatment of liver cancer in China at that time. These guidelines helped standardize the diagnosis and treatment of liver cancer, improve patient prognosis, ensure medical service quality and safety, and optimize the use of medical resources. Since the publication of the 2022 edition, high-quality evidence has emerged from researchers worldwide regarding the diagnosis, staging, and treatment of liver cancer; many of these findings are directly applicable to clinical practice in China. In response to these developments, the National Health Commission decided to revise and update the 2022 edition to produce the *China Liver Cancer (CNLC) Guidelines for the Diagnosis and Treatment of Hepatocellular Carcinoma* (2024 Edition) (hereinafter referred to as the Guidelines) that includes the latest practices in the clinical diagnosis and treatment of liver cancer based on the latest research. The Oncology Branch of the Chinese Medical Association (CMA), in conjunction with organizations such as the Liver Cancer Committee of the China Anti-Cancer Association, the Ultrasonography Branch of the CMA, the Surgeon Branch of the Chinese Medical Doctor Association, and the Chinese College of Interventionalists, were entrusted to update the guideline by forming a nationwide committee of multidisciplinary experts in the field of liver cancer. The guideline aims to standardize the diagnosis and treatment of liver cancer, reflect the latest developments, improve the survival rate of liver cancer patients, and contribute to the achievement of the "Healthy China 2030" to increase the overall 5-year survival rate of cancer patients by 15%.

Methodology

Grading of Recommendations, Development and Evaluation (GRADE) is the most widely used system for evaluating evidence and grading recommendations [5].The GRADE system consists of two parts. The first part is the evaluation of evidence, during which the quality of evidence is classified into one of four levels: high, medium, low, and very low, based on risk of bias, inconsistency, indirectness, imprecision, and publication [6]. The second part is the grading of recommendations; based on the GRADE system, the pros and cons of medical interventions, quality of evidence, values and preferences, and resource consumption are considered to classify recommendations as either strong or weak (conditional) [7]. For any given medical intervention, a strong recommendation is typically assigned when there is a significant net benefit, higher quality of evidence, values and preferences are both clear and aligned, and a lower cost and resource consumption. Otherwise, a weak recommendation (conditional recommendation) is assigned. The assessment of the evidence underlying the evidence-based recommendations in this guideline was based on the GRADE methodology, and the *Oxford*

Table 1. Recommendation strength

Recommendation strength	Description of definition
Strong recommendation	High confidence that the true value is close to the effect estimate. Based on: high-quality research evidence supporting a net benefit (e.g., benefits outweighing harms); good consistency between findings with no or few exceptions; minor or no doubts about the quality of the study; and/or agreement of expert panel members. In other cases, high-quality evidence that convince the experts that the benefits clearly outweigh harm (including what is discussed in the literature review and analysis of the guidelines) may also support a strong recommendation
Moderate recommendation	Moderate confidence in effect estimates. Based on: good research evidence supporting net benefits (e.g., benefits outweighing harms); consistency between research findings, with minor and/or a few exceptions; minor or few doubts about the study quality; and/or agreement of the expert panel members. In other cases, moderate-quality evidence with the benefits outweighing the harms (including those discussed in the literature review and analysis of the guidelines) may also formulate a moderate recommendation
Weak recommendation	There is limited confidence in the effect estimates, and this recommendation provides the best current guidance for clinical practice. Based on: limited research evidence supporting a net benefit (e.g., benefits outweighing harms); consistent study findings with major exceptions; major doubts about study quality; and/or agreement from expert panel members. In other cases, limited evidence (including what is discussed in the literature review and analysis of the guidelines) may also lead to weak recommendations

Recommendation strength "Strong recommendation, Moderate recommendation, Weak recommendation" is expressed in the article with "A, B, C," respectively.

Centre for Evidence-Based Medicine Levels of Evidence (2011 Edition) (*OCEBM Levels of Evidence*) was used as a supporting tool for the specific grading of evidence. The transition from evidence to recommendations was mainly based on GRADE; the grading scheme employed in the American Society of Clinical Oncology (ASCO) guidelines methodology [8] was also used to modify the grading of recommendations accordingly (shown in Table 1).

The strength of recommendations was categorized into three levels: strong, moderate, and weak. A strong recommendation (recommendation A) reflects a high level of confidence from the expert group regarding a specific clinical practice, indicating that the majority, if not all, target users should adopt the recommendation. A moderate recommendation (recommendation B) reflects a moderate level of confidence from the expert group in a specific clinical practice, and while most target users should adopt the recommendation, consideration should be given to the joint decision made by the physician and the patient during clinical practice. A weak recommendation (recommendation C) reflects only some confidence from the expert group in a specific clinical practice; the recommendation should be conditionally applied to the target group with an emphasis on physician-patient joint decision-making.

Screening and Diagnosis

Screening and Monitoring Individuals at High Risk of HCC

Screening and monitoring (ultrasound [US] combined with serum alpha-fetoprotein [AFP] testing) individuals at high risk of HCC facilitates its early detection, diagnosis, and treatment [9], and can significantly reduce the risk of death [10] (evidence level 1, recommendation A). The rapid and convenient identification of patient groups at high risk of HCC are a prerequisite for large-scale screening for HCC, while stratified assessment of HCC risks in a population forms the basis for the development of HCC screening strategies. In China, populations at high risk of HCC include those with hepatitis B virus (HBV) and/or hepatitis C virus (HCV) infection, consume excessive amounts of alcohol, hepatic steatosis or metabolic dysfunction-associated liver disease, dietary exposure to aflatoxin B1, cirrhosis from other causes, and those with a family history of liver cancer, especially males >40 years of age. Although current anti-HBV and anti-HCV therapies may significantly reduce the risk of HCC, the development of HCC is not fully prevented [11]. The age-Male-AlBi-Platelets score (aMAP score), a risk assessment model developed by Chinese scholars, indicated for a

variety of chronic liver diseases and various types of HCC, can be used to categorize a population with liver diseases into risk groups for HCC: low-risk (aMAP score of 0–50), intermediate-risk (aMAP score of 50–60) and high-risk (aMAP score of 60–100) with annual HCC incidence rates of 0–0.2%, 0.4%–1.0%, and 1.6%–4.0%, respectively [12] (evidence level 2, recommendation B). Additionally, two novel liver cancer prediction models, aMAP-2 and aMAP-2 Plus, which are based on multivariate longitudinal data (aMAP score, AFP) and characteristics of circulating cell-free DNA, could identify ultra-high-risk populations with an annual HCC incidence of 12.5% [13]. Liver cancer screening should emphasize using liver cancer risk prediction score as an effective tool, and develop a novel model of integrated precision screening by the community and hospital [14]. This will effectively increase early diagnosis rate and reduce mortality rate of liver cancer. Screening for HCC is recommended at least every 6 months in high-risk populations [9] (evidence level 2, recommendation A).

Imaging Examinations for HCC

Because different imaging methods have unique advantages and disadvantages, emphasis should be placed on their integrated application to allow a comprehensive assessment.

Ultrasound

US is the most commonly used liver imaging modality in clinical practice, owing to its cost-effectiveness, convenience, real-time scanning, noninvasiveness and radiation-free characteristics. Routine gray-scale US can be used for early and sensitive detection of focal liver lesions, to identify the lesion as cystic or solid, and to provide a preliminary determination of whether the lesion is benign or malignant. Typical gray-scale US features of HCC appear as a round or oval solid intrahepatic mass with a common surrounding hypoechoic halo. Most of the internal echoes of the lesion are hypoechoic, but may also manifest as isoechoic, hyperechoic, or heterogeneous echoes. Concurrent liver cirrhosis manifestations such as coarsened liver echogenicity, reduced liver volume, uneven liver surface, and portal hypertension can also be observed on gray-scale US. A few diffusive type of liver cancers are indistinguishable from cirrhosis.

Meanwhile, gray-scale US is able to detect the HCC metastases in the liver or other abdominal cavity, and identify infiltration to intrahepatic vessels and bile ducts. Intrahepatic metastases often appear as solid nodules of varying sizes around the main tumor in the liver or in other parts of the liver, with variable numbers, diameters mostly of <3 cm, and peripheral halo sign. Tumor thrombus in the portal vein, hepatic vein, and bile duct manifests as intraluminal hypoechoicity. When the tumor thrombus completely fills the portal vein lumen, fine collateral circulation may form at the periphery of portal vein, presenting "honeycomb-like" changes on US imaging. Hepatic vein tumor thrombus can extend into the inferior vena cava or even the right atrium. When HCC directly invades adjacent organs, such as gallbladder, right kidney, etc., gray-scale US may also observe the unclear boundaries between the tumor and the aforementioned structures.

Color Doppler flow imaging is used to visualize the blood supply within a lesion, determine whether the lesion is benign or malignant, and display its adjacent relationship to important intrahepatic vessels, as well as demonstrate the intrahepatic vascular invasion. Moreover, it can be used to assess the treatment response after locoregional treatment for HCC. On color Doppler flow imaging, HCC manifests as increased blood flow signals within the lesion, presenting in various patterns, such as dot, short linear, branch, basket, and peripheral ring-like configuration. The blood vessels around the lesion can be seen to be compressed. Pulsed Doppler spectrum detects arterial blood flow signals within the lesion, with a resistance index often >0.6. Arterial flow signals may also be detected when tumor thrombus occurred in the portal vein, hepatic vein, or bile ducts.

Contrast-enhanced US allow dynamic observation of intralesional blood perfusion changes in real-time. Intraoperative contrast-enhanced US can sensitively detect small occult lesions, guide real-time locoregional treatment, and assess the postoperative efficacy/outcomes of locoregional treatment for HCC [15–23] (evidence level 2, recommendation A).

US contrast agents are administered mainly through a peripheral vein and dynamic changes in contrast agents' distribution within and around the tumor can be tracked using specific contrast-enhanced US mode. US contrast agent consists primarily of microbubbles filled with inert gas, which is safe and allergic reactions are extremely rare. Currently, the commonly used US contrast agents are sulfur hexafluoride microbubbles and perflubutane microspheres, both of which are administered intravenously. The former is a purely intravascular agent and can be used for vascular phase imaging, whereas the latter can be phagocytosed by Kupffer cells to form post-vascular phase imaging. The vascular phases include the

arterial phase (within 30 s of contrast agent injection), the portal venous phase (31–120 s after injection), and the late phase (>120 s after injection). The post-vascular phase is generally defined as 8 min after contrast agent injection. Typical contrast-enhanced US features of HCC show rapid hyperenhancement in the arterial phase of the lesion, which enhances earlier than the surrounding liver parenchyma, followed by a rapid regression to hypoenhancement in the portal venous phase and late phase, i.e., "wash-in and wash-out" enhancement pattern [22, 23] (evidence level 1, recommendation A). The diagnostic performance of contrast-enhanced US is related to the size of the lesion. Most HCCs with a diameter of >3.0 cm show the typical enhancement pattern described above, but a few HCCs with a diameter of <2.0 cm tend to have atypical imaging manifestations. The wash-out rate of contrast agent in the portal venous phase and late phase is related to the degree of tumor differentiation, typically exhibiting slower wash-out in well-differentiated tumors and more rapid wash-out in poorly differentiated tumors.

Due to the high sensitivity of contrast-enhanced US to microvascular flow, it can be used to observe the blood flow changes at different stages of hepatocarcinogenesis, such as regenerative nodule, low-grade dysplastic nodule, high-grade dysplastic nodule, high-grade dysplastic nodule with focal carcinogenic changes, early HCC, and advanced HCC. Therefore, contrast-enhanced US can also be used in the screening of individuals at high risk of HCC and for monitoring the evolution of intrahepatic nodules [18]. Contrast-enhanced US liver imaging reporting and data system (LI-RADS) may be considered in patients at high risk for HCC to improve the specificity of HCC diagnosis (evidence level 3, recommendation B).

Tumors in the late phase or post-vascular phase on contrast-enhanced US often show hypoenhancement and are clearly demarcated from the surrounding liver parenchyma. Consequently, contrast-enhanced US is particularly well-suited for the detection of multiple small intrahepatic lesions and for monitoring post-ablation or postsurgical patients for early detection of recurrent lesions. When the intrahepatic lesion shows hypoenhancement in the late phase or the post-vascular phase, US contrast agents can be re-injected after 10 min to observe the enhancement of the lesion in the arterial phase, and then determine the presence and nature of the lesion. Contrast agents with post-vascular phase (e.g., perfluorobutane) are suitable for the detection of lesions due to their extended imaging duration (30–120 min) [23].

Contrast-enhanced US can be used for tumor pre-ablation planning, puncture guidance, immediate post-ablation assessment, and subsequent follow-up. The immediate post-ablation evaluation is particularly important, as it can promptly identify any residual lesions that were not completely ablated, thereby allowing for timely additional treatment. Quantitative contrast-enhanced US enables the measurement of various parameters such as contrast agent arrival time, time to peak, transit time, peak enhancement, perfusion volume, etc., which can be used to assess the efficacy of systemic antitumor therapy (chemotherapy, targeted therapy, immunotherapy, etc.) and to predict patient responsiveness to systemic antitumor therapy at an early stage, thus assisting in clinical decision-making [24, 25].

US combined with dynamic contrast-enhanced computed tomography (CT) and magnetic resonance imaging (MRI) scanning provides an effective image navigation technique for precise localization of HCC, especially occult HCC that cannot be visualized by conventional US imaging [26–28] (evidence level 3, recommendation B). US fusion image navigation is valuable in preoperative planning, intraoperative monitoring and determination of safe margins, and immediate postoperative efficacy assessment in HCC ablation. The use of contrast-enhanced US in fusion navigation can further improve accuracy, especially in assessing the extent of ablation for occult lesions, isoechoic lesions, and larger lesions [28, 29].

US shear wave elastography allows the quantitative measurements of the tissue stiffness of liver tumors and the extent of fibrosis/cirrhosis of the surrounding liver parenchyma to provide valuable information for optimizing HCC treatment regimens [30, 31] (evidence level 3, recommendation B). Quantitative US techniques can measure intrahepatic fat content in metabolic dysfunction-associated steatotic liver disease, offering adjunctive information for early warning of metabolic dysfunction-associated steatotic liver disease-associated HCC [32–36]. The application of multiparameter US techniques plays an important role in the precise preoperative diagnosis, intraoperative localization, and postoperative evaluation of HCC. Intraoperative US (contrast-enhanced US) and laparoscopic US (contrast-enhanced US) are becoming popular in liver surgery, which can help detect occult microscopic lesions and determine the extent and margins of surgical resection [37]. High-frequency US is useful in detecting occult lesions in subcapsular or superficially located lesions [16]. US-based radiomics plays an important role in the differential diagnosis of HCC, prediction of biological

behaviors such as microvascular infiltration of HCC, and selection of treatment regimens [38]. With the development of artificial intelligence technology, an intelligent prediction model of HCC can be established by integrating patients' clinical and tumor imaging information to accurately predict the recurrence and metastasis of HCC, which is expected to provide scientific and rational decision-making basis for the clinical choices of ablation or surgical treatment [39, 40].

CT and MRI

Dynamic contrast-enhanced CT and MRI scan are the imaging methods of choice for the diagnosis of HCC in those with abnormal USA and/or serum AFP levels during screening. With the advantages of radiation-free, high tissue resolution, multiplanar and multisequence dynamic contrast-enhanced imaging capabilities, and its comprehensive imaging ability with morphological combination function (including diffusion-weighted imaging, etc.), dynamic contrast-enhanced MRI has become the preferred imaging technique for clinical detection, diagnosis, staging of HCC and to evaluate responses. Dynamic contrast-enhanced MRI is more accurate for detecting and diagnosing HCC ≤2.0 cm in size than dynamic contrast-enhanced CT [41, 42] (evidence level 1, recommendation A). Dynamic contrast-enhanced MRI is superior to dynamic contrast-enhanced CT in evaluating the invasion of HCC into the portal vein, the main trunk, and branches of the hepatic vein, and in identifying abdominal or retroperitoneal lymph node metastasis. The modified response evaluation criteria in solid tumor (mRECIST) in combination with T2-weighted imaging and diffusion-weighted imaging are recommended for evaluating the response to treatment in patients with HCC using dynamic contrast-enhanced MRI scans.

Dynamic contrast-enhanced CT or MRI utilizing nonspecific gadolinium-based contrast agents involves several key phases: the late arterial phase, typically imaged about 35 s post-contrast injection; the portal venous phase, imaged 60–90 s post-injection, characterized by full enhancement of the portal vein, visible contrast filling in the hepatic veins, and peak parenchymal enhancement; and the delayed phase, captured approximately 3 min after contrast injection, during which both the portal vein and hepatic veins show enhancement, albeit less than in the portal venous phase, with the liver parenchyma also demonstrating reduced enhancement compared to the portal venous phase.

Quadruple-phase contrast-enhanced MRI with the hepatocyte-specific agent gadolinium ethoxybenzyl-diethylenetriamine pentaacetic acid (Gd-EOB-DTPA) includes the following phases: the late arterial phase, as previously described; the portal venous phase, also as previously described; the transitional phase, typically imaged 2–5 min after Gd-EOB-DTPA injection, during which the signal intensity of the hepatic vessels and hepatic parenchyma is similar, reflecting a combination of intracellular and extracellular contrast contributions; and the hepatobiliary phase, imaged 12–20 min post-Gd-EOB-DTPA injection, where the parenchymal signal is hyperintense relative to the hepatic vessels due to contrast excretion via the biliary system. In patients with normal liver function, the hepatobiliary phase is typically captured at 12–15 min, whereas in those with significantly impaired liver function, imaging may be delayed to 20 min to account for the reduced biliary excretion rate.

Diagnosing HCC with imaging is primarily based on the "wash-in and wash-out" enhancement pattern of dynamic enhancement scans [43–45] (evidence level 1, recommendation A). On dynamic contrast-enhanced CT and MRI scans, liver tumors exhibit significant homogeneous or heterogeneous enhancement in the arterial phase, particularly in the late arterial phase. In contrast, tumors appear hypointense compared to the surrounding parenchyma during the portal venous and/or delayed phases. "Wash-in" refers to noncircular enhancement, while "wash out" refers to the non-peripheral aspect. "Wash-in" is observed in the late arterial phase, and "wash-out" is observed in the portal venous and delayed phases. When using Gd-EOB-DTPA, the "wash-out" sign may usually be observed in the portal venous phase, and the "wash-out" sign in the transitional and hepatobiliary specific phases may be used as auxiliary features of malignancy.

Gd-EOB-DTPA-enhanced MRI showed significant enhancement of liver tumors in the arterial phase, with hypointensity compared with the parenchyma during the portal venous phase, and often showed significant hypointensity in the hepatobiliary phase. However, approximately 5–12% of well-differentiated HCC, especially small HCC (≤2.0 cm in diameter), exhibit slight hyperintensity for the visible tumor in the hepatobiliary specific phase associated with contrast agent uptake [46].

Diagnosing HCC using dynamic contrast-enhanced MRI, especially HCC ≤2.0 cm in diameter, requires the integration of additional characteristic imaging features (e.g., capsule-like enhancement, moderate hyperintensity on T2-weighted imaging, and diffusion restriction) and suprathreshold growth (50% increase in the maximum diameter of the lesion within 6 months)

[47] (evidence level 3, recommendation B). Capsule-like enhancement is defined as smooth, homogeneous, well-defined borders that mostly or completely encircle the lesion, especially during the portal venous, delayed, or transitional phases in which circumferential enhancement is observed.

HCC with a diameter of ≤1.0 cm is defined as subcentimeter hcc (scHCC). According to the literature, the 5-year survival rate after local resection of scHCC was 97.3%, which was significantly higher than that of small HCC with a diameter of 1.0–2.0 cm (5-year survival rate of 89.5%) [48]. In high-risk populations, Gd-EOB-DTPA-enhanced MRI is recommended for the diagnosis of scHCC after exclusion of defined benign lesions [49–54] (evidence level 2, recommendation B). This approach is particularly useful for patients with combined liver cirrhosis, and aids in discrimination scHCC from precancerous lesions, such as high-grade dysplastic nodules [55].

Currently, CT scans and dynamic contrast-enhanced imaging are not only commonly used for the clinical diagnosis and staging of liver cancer but also for evaluating the efficacy of local treatments for liver cancer. Particularly, they have certain advantages in observing the deposition of iodized oil and tumor viability following transcatheter arterial chemoembolization (TACE), which is especially helpful in determining the need for further TACE treatment [56–58]. Preoperative CT-based radiomic can also be used to predict the efficacy of the first TACE treatment [59]. Meanwhile, post-processing techniques for CT scans may be used to perform 3D vascular reconstruction and to measure liver volume and tumor volumes. 3D reconstruction technology can be used for liver lobe segmentation, preoperative surgery simulation, and assisting doctors to develop the optimal surgical plan.

Fusion models utilizing dynamic contrast-enhanced CT and/or MRI data for HCC may improve clinical decision-making. This includes the selection of treatment regimens and evaluating, and predicting treatment responses [60]. Imaging features to predict microvascular invasion (MVI) preoperatively in HCC are highly specific but relatively insensitive. Artificial intelligence (including radiomics and deep machine learning models) is a possible breakthrough point for preoperative prediction of MVI [61–63] (evidence level 3, recommendation B).

Digital Subtraction Angiography

Digital subtraction angiography (DSA) is a mandatory examination before intra-arterially interventional therapy for HCC patients, which is often performed through selective or super-selective catheterization of the hepatic artery. DSA not only provides a clear delineation of the anatomy and variations of the hepatic artery but also enhances the visualization of tumor catheterization and contrast staining. Furthermore, DSA can also evaluate the number, size, and the vascularity of HCC [64]. DSA combined with cone beam CT (CBCT) can show tumor lesions more clearly, increase the detection rate of small HCC, determine the 3D spatial relationships of the feeding artery branches, and guide the super-selective catheterization of these arterial branches [65] (evidence level 2, recommendation A). Indirect portal vein angiography via the superior mesenteric artery or splenic artery, allows for the evaluation of portal venous flow and tumor thrombi within the main portal vein or its first-order branches.

Nuclear Medicine Imaging

Positron emission tomography/CT (PET/CT): whole-body 18F-fluorodeoxyglucose (18F-FDG) PET/CT have the following advantages: (1) tumor staging, a single procedure allows for a comprehensive evaluation of lymph node metastasis and distant organ metastasis [66, 67] (evidence level 1, recommendation A); (2) restaging, PET/CT functional imaging can accurately depict tumor recurrence or metastases that occur in areas with altered or complex anatomical structures, as it is not obscured by anatomical variations [68] (evidence level 2, recommendation B); (3) more sensitive and accurate in evaluating the efficacy of targeted drugs that inhibit tumor activity [69, 70] (evidence level 2, recommendation A); (4) guiding biologic target volume delineation for radiation therapy and aids in determining sites for puncture biopsies [68]; (5) evaluation of the extent of malignancy and prognosis [71–73] (evidence level 2, recommendation B). PET/CT has limited sensitivity and specificity to diagnose HCC, it can be used as an adjunct and supplement to other imaging examinations, with advantages in staging, restaging, and efficacy evaluation of HCC. PET imaging with Carbon-11 acetate (11C-acetate) or choline (11C-choline) enhances sensitivity for the diagnosis of well-differentiated HCC and complements 18F-FDG PET/CT [74, 75]. Gallium-68 or fluorine-18 labeled fibronectin inhibitor-04 (68 Ga-DOTA-FAPI-04/18 F-NOTA-FAPI-04) PET/CT can effectively improve the diagnostic sensitivity of primary and metastatic lesions, especially middle- and high-grade differentiated HCC and ICC, thus compensating for the limitations of 18F-FDG PET/CT imaging [76, 77] (evidence level 3, recommendation C).

Single-photon emission computed tomography/CT (SPECT/CT): SPECT/CT has gradually become the standards for nuclear medicine single-photon imaging, supplanting conventional SPECT. With SPECT/CT, lesions identified through whole-body planar imaging can be subjected to regional SPECT/CT fusion imaging. This approach significantly enhances diagnostic accuracy by providing both SPECT and diagnostic CT images of the lesion site simultaneously [78] (evidence level 3, recommendation B).

PET/MRI: a single PET/MRI examination can simultaneously obtain anatomical structure, information of dynamic contrast-enhanced MRI and functional metabolism information of PET, which improves the sensitivity of HCC diagnosis [79] (evidence level 4, recommendation C).

Hematological Molecular Markers for HCC

Serum AFP is a commonly used biomarker for the diagnosis of HCC and for monitoring treatment response. Serum AFP ≥400 μg/L is highly suggestive of HCC after excluding pregnancy, chronic or active liver disease, embryonal tumor of the gonads, and other digestive system tumors. For patients with mildly increased serum AFP, imaging examinations should be combined and dynamic monitoring should be performed. Cross comparison with changes in liver function should also be performed to facilitate diagnosis. Abnormal prothrombin, protein induced by vitamin K absence/antagonist-II (PIVKA II) or des-gamma carboxyprothrombin (DCP), plasma cell-free microRNA [80] and serum lens culinaris agglutinin-reactive fraction of AFP (AFP-L3) can also be used as early diagnostic biomarkers for HCC, especially individuals with negative serum AFP. The sensitivity and specificity of the GALAD model, based on sex, age, AFP, PIVKA II, and AFP-L3, were 85.6% and 93.3%, respectively, for diagnosing early stage HCC, which facilitates the early diagnosis of AFP-negative HCC [81] (evidence level 1, recommendation A). Optimized GALAD-like models, such as C-GALAD, GALAD-C, and C-GALAD II, developed using data from large samples of the Chinese population, have been utilized for the early diagnosis of HCC. In addition, simplified GAAD models constructed based on gender, age, AFP, PIVKA II [82] and ASAP model [83] had similar diagnostic efficacy to the GALAD model (evidence level 1, recommendation A). The test kit, which utilizes a panel of seven microRNAs for the diagnosis of HCC, demonstrated sensitivity and specificity of 86.1% and 76.8%, respectively. In the subgroup of AFP-negative HCC patients, the sensitivity and specificity were 77.7% and 84.5%, respectively [80] (evidence level 1, recommendation A). In recent years, "liquid biopsies" has shown significant value in early diagnosis and efficacy evaluation of tumors, including circulating cell-free microRNA, circulating tumor cells (CTC) [84], circulating cell-free DNA [][85], circulating tumor DNA [][86, 87], cell-free mitochondrial DNA, free viral DNA, and extracellular vesicles, etc.

Liver Biopsy for HCC

Diagnostic liver biopsy is typically unnecessary in patients presenting with space-occupying lesions that exhibit typical imaging features and meet the clinical criteria for the diagnosis of HCC [44, 88–90] (evidence level 1, recommendation A), especially in patients with HCC who have surgical indications. For patients with resectable HCC or who are scheduled for liver transplantation (LT), preoperative liver biopsy is not recommended, in order to reduce the risk of tumor rupture, hemorrhage, and dissemination. For space-occupying lesions without typical imaging characteristics, liver biopsy can provide a definitive pathologic diagnosis. Liver biopsy can also be used to determine the nature of the lesion and the molecular classification of HCC [91] and can provide valuable information on the cause of liver disease to guide treatment, determine prognosis, and conduct research. Therefore, the need for liver lesion biopsy should be assessed based on the patient benefit, potential risks, and the operating experience of the physician.

Liver biopsy should be performed under the guidance of US or CT with a 16- or 18-gauge needle. The primary risks associated with liver biopsy include bleeding and the potential for needle tract implantation. It is essential to assess platelet count and coagulation profiles preoperatively. Liver biopsy should generally be avoided in patients exhibiting hemorrhagic tendencies. Normal liver tissues should be passed by when selecting the puncture tract to avoid direct puncture of nodules located on the surface of the liver. The biopsy site should be chosen within the tumor or at its periphery, where imaging indicates active tumor involvement. The integrity of the tissue sample retrieved should be visually inspected to enhance diagnostic accuracy. Pathologic diagnosis using liver lesion biopsy is associated with a certain false-negative rate due to multiple factors including the size of lesion, especially for lesions with a diameter of ≤2 cm. Therefore, a negative result from liver biopsy cannot exclude the possibility of HCC. Observation and regular follow-ups are required. For patients with limited biopsy specimens and a negative

pathological diagnosis, but who are clinically highly suspected of HCC, it is recommended to perform a repeat liver biopsy and/or conduct close follow-up.

Summary:

1. US combined with serum AFP testing is used for early screening for HCC. Monitoring at least every 6 months is recommended for individuals in high-risk populations.
2. Dynamic contrast-enhanced CT, MRI scans, Gd-EOB-DTPA dynamic contrast-enhanced MRI examination, and contrast-enhanced US are the first-choice imaging methods for the diagnosis of HCC in patients with abnormal US and/or serum AFP levels during screening.
3. The characteristic "wash-in and wash-out" enhancement pattern is the primary basis for the imaging diagnosis of HCC.
4. The preferred imaging technique for the clinical diagnosis, staging, and evaluation of treatment response for HCC is dynamic contrast-enhanced MRI.
5. PET/CT scans possess significant value in the staging of HCC and in the comparative evaluation of treatment efficacy both before and after therapy.
6. Serum AFP is a commonly used and important biomarker for the diagnosis of HCC and monitoring treatment response. In the serum AFP-negative population, DCP, test kit based on a combination of seven microRNAs, and AFP-L3 may be useful for the early diagnosis of HCC.
7. Liver biopsy for diagnostic purposes is usually not necessary in patients with space-occupying lesions that have typical imaging characteristics and are evaluable using the clinical criteria for the diagnosis of HCC.

Pathologic Diagnosis of HCC

Pathologic Diagnostic Terminology in HCC

Primary liver cancer: malignant tumors originating from the hepatocytes and the epithelial cells of the intrahepatic bile duct, mainly including HCC, ICC, and cHCC-CCA.

1. HCC: a malignant neoplasm occurring in hepatocytes. The use of the pathologic diagnosis terms "hepatocellular liver cancer" or "hepatocellular-type liver cancer" is not recommended.
2. ICC: a malignancy of the epithelial cells covering the intrahepatic bile duct branches; adenocarcinoma is the most common form. The 2019 WHO Classification of Tumors of the Digestive System does not recommend the use of the pathologic diagnostic terminology of cholangiocellular carcinoma for ICC, and the use of the pathologic diagnostic terminology of cholangiolocellular carcinoma for fine bile duct carcinoma is no longer recommended [92]. ICC may be histologically divided into four subtypes: (1) large duct type ICC: originating in the large bile ducts above the bile canaliculus of the liver lobules and the adjacent portal area, with large and irregular openings of the glandular ducts, and mucus glands can be seen in the surrounding area; (2) small duct type ICC: originating from the bile ducts of the liver lobules and septal bile ducts, the openings of the glandular ducts is smaller and the arrangement is regular; (3) cholangiolocarcinoma: originating from the hepatic ductal system or fine bile duct, the carcinoma cells are in the shape of a small cube, which are loosely arranged in angular small ducts or branches within the collagen fiber interstitium; (4) ductal plate malformation type ICC: the tumor glandular ducts are irregularly cystic dilated, and the lumen of the ducts contains papillary protrusions. Some studies have shown that the biological behavior and genotypic characteristics of large duct type of ICC are different from those of other types of ICC, with a worse clinical prognosis [93].

The clinical and pathological implications of molecular typing in HCC and ICC continue to be investigated. Recent research indicates that Epstein-Barr Virus-associated ICC exhibits distinctive clinical pathological features, immune microenvironment, and molecular characteristics, potentially responding exceptionally well to immune checkpoint therapy. Therefore, the use of Epstein-Barr virus encoded ribonucleic acid in situ hybridization assay is recommended for screening patients who may benefit from immune checkpoint inhibitor therapy [94]. A high expression of triosephosphate isomerase in ICC tissues is a useful indicator for assessing the risk of postoperative recurrence and metastasis [95].

3. cHCC-CCA: it refers to the presence of both HCC and ICC in the same tumor node, excluding collision carcinoma. Some scholars have suggested that ≥30% of each of the two tumor components should be used as the pathological diagnostic criteria for cHCC-CCA [96], but there is no internationally unified pathological diagnostic standard for the percentage of two histological components of cHCC-CCA, which is yet to be studied in a large-sample multicenter study. For this reason, it is recommended that the ratio of the

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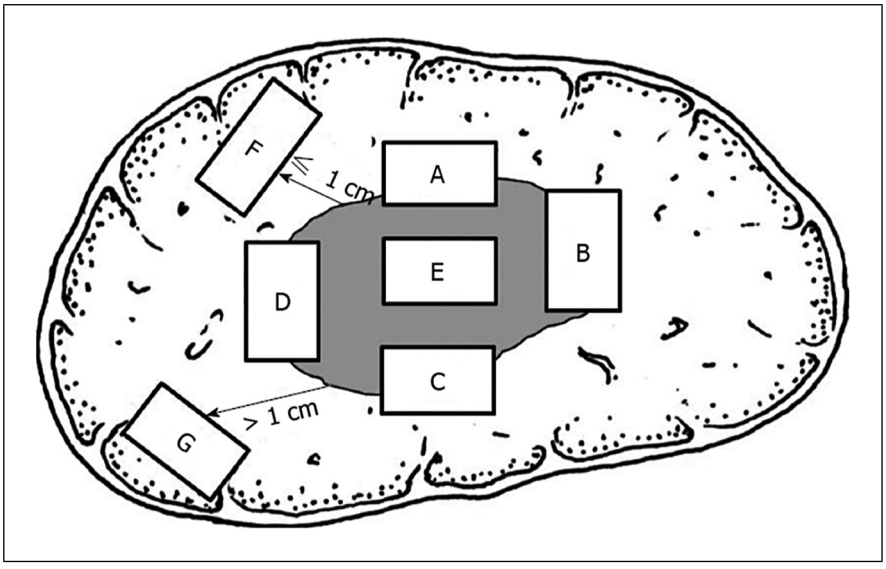


Fig. 1. Schematic diagram of the recommended baseline specimen sampling protocol for liver tumors. A, B, C, and D indicators denote the 12, 3, 6, and 9 o'clock positions, respectively, along the boundary between cancer and adjacent nonneoplastic liver tissue; E, tumor area; F, proximal nonneoplastic adjacent liver tissue; G, distal nonneoplastic adjacent liver tissue.

two tumor components be labeled in the pathological diagnosis of cHCC-CCA, and attention should be paid to the histological grading and subtyping, MVI grading, and lymphovascular invasion of the HCC and ICC components, respectively, for clinical evaluation of the biologic characteristics of the tumors and the formulation of treatment plans. The diagnosis of cHCC-CCA should be exercised with caution when a particular tumor component constitutes a very small proportion of the lesion.

Guidelines for the Pathologic Diagnosis of HCC

The guidelines for the pathological diagnosis of HCC encompass several key procedures: specimen handling, sampling of the specimen, histological examination, and the compilation of the pathology report [91, 96]. Key points of specimen handling: (1) the surgeon should indicate the site, type and quantity of the specimen sent for examination, whether the patient has received conversion/neoadjuvant therapy, as well as the treatment regimen and cycles on the pathology examination request form. To facilitate the identification of surgical margins and significant lesions, they can be marked with dyes or sutures. (2) Where possible, the intact tumor specimen should be delivered to the pathologist for dissection and fixation within 30 min after removal. Tissue bank technicians should collect specimens under the guidance of the pathology department to ensure that sampling accuracy prioritizes the requirements of pathological diagnosis. (3) Tissue samples should be fixed in 10% neutral buffered formalin solution for 12–24 h.

Key points of specimen sampling: the area adjacent to HCC is the representative area for the biologic features of tumor. To this end, the "7-point" baseline sampling method (shown in Fig. 1) should be employed, that is, specimens are collected in a ratio of 1:1 in the 12, 3, 6, and 9 o'clock positions along the boundary between neoplastic and adjacent nonneoplastic liver tissues. At least one tissue sample should be obtained from within the tumor. Additionally, samples should be collected from nonneoplastic liver tissues adjacent to the tumor, with one sample taken from the region ≤1 cm (proximal) and another from the region >1 cm (distal) away from the tumor boundary. The nearest surgical margin of the liver and the portal vein embolus should be sampled separately. The surgical margin of the bile duct should also be sampled for ICC specimens. For solitary tumors with a diameter of 3 cm or less, the entire tumor should be sampled for pathological examination. The actual site and number of samples collected may be increased based on the tumor's diameter and the number of tumors present [97, 98] (evidence level 3, recommendation B).

Key Points of Pathological Examination of HCC

Macroscopic description of specimens [99]: all surgical samples submitted should be thoroughly inspected, and the following details should be specifically described: size, number, color, and texture of tumors; their relationship with blood vessels and bile ducts; encapsulation status; lesions in the nonneoplastic liver tissue; type of liver cirrhosis; distance between tumor and incisal margin; and status of the incisal margin.

Microscopic observation and description [99]: all specimens collected should be thoroughly observed, and the pathologic diagnosis may be based on the 2019 WHO Classification of Tumors of the Digestive System for HCC [96]. The following information should be specifically described: the degree of differentiation of tumor cells may be described according to the internationally used Edmondson-Steiner grading system (I~IV) or the highly, moderately, and poorly differentiated classification recommended by the WHO. The histological morphology of HCC is usually divided into microtrabecular, macrotrabecular, pseudoglandular, and compact types. Special histological subtypes of HCC: include fibrolamellar, cirrhotic, clear cell, fatty change, chromophobe cell, neutrophil-rich, lymphocyte-rich, and undifferentiated types. Biphenotypic HCC exhibits characteristic features of HCC in clinical presentations, imaging studies, serological analyses, cancer cell morphology, and histopathology. However, immunohistochemical staining reveals the expression of both hepatocellular and biliary epithelial markers, indicating that this subtype of HCC is more aggressive [100], and may be sensitive to regorafenib [101]. The degree and extent of tumor necrosis and mesenchymal reaction should also be observed in surgical specimens after conversion/neoadjuvant therapy. The growth pattern of HCC: including perineoplastic infiltration, capsule invasion, or breakthrough, MVI, and the presence of satellite nodules, etc.

Evaluation of chronic liver disease: HCC is often accompanied by different varying degrees of chronic viral hepatitis or liver cirrhosis. The use of the Scheuer scoring system, which is more convenient, or the Chinese Criteria for Histologic Grading and Staging of Chronic Viral Hepatitis is recommended [102–104], and attention should be paid to the description of the proportion of hepatocellular steatosis, as well as the diagnosis and differential diagnosis of fatty liver disease.

Diagnosis of MVI: MVI refers to the presence of clusters of cancer cells in the vascular lumen with endothelial cell linings under the microscope [105], which is most commonly seen in the invasion of the branches of the portal vein in HCC (including the intracapsular blood vessels). Invasion to the lymphatic vessels is observed in ICC. The pathologic grading of MVI [106]: M0: no MVIs detected; M1 (low-risk group): ≤5 MVIs, which occur in proximal nonneoplastic adjacent liver tissue (≤1 cm); M2 (high-risk group): M2a was defined as >5 proximal nonneoplastic MVI and no distal nonneoplastic MVI; M2b was defined as MVIs occurring in distal nonneoplastic adjacent liver tissue (>1 cm) (shown in Fig. 2). The risk of postoperative recurrent and metastasis increased and clinical prognosis decreased in each MVI group of patients in turn [97, 107]. MVI and satellite lesions may be considered different developmental stages during the intrahepatic metastasis of HCC. Satellite lesions in nonneoplastic adjacent liver tissues should be included in the MVI grading in cases where it is difficult to distinguish satellite lesions from MVI. MVI has a great impact on the evaluation of recurrence and metastasis risk and on the selection of appropriate treatment strategy [91, 96, 108–110] and should, therefore, be used as an indicator for routine histopathologic examination. MVI pathological grading is based on the "7-point" baseline sampling. An explanation should be provided during the grading of MVI when the submitted specimens for examination cannot meet the "7-point" baseline sampling (evidence level 2, recommendation A).

Immunohistochemical Examination

The main purposes of immunohistochemical examination for HCC are to differentiate between benign and malignant tumors in liver; between HCC, ICC, and other special types of liver tumors; and between primary and metastatic liver tumor. Due to the high heterogeneity of histological types of liver cancer, there are deficiencies in the diagnostic specificity and sensitivity of the existing liver cancer markers. Appropriate combinations of diagnostic examinations and assessments are often necessary, and the concurrent use of biomarkers for other systemic tumors may also be warranted.

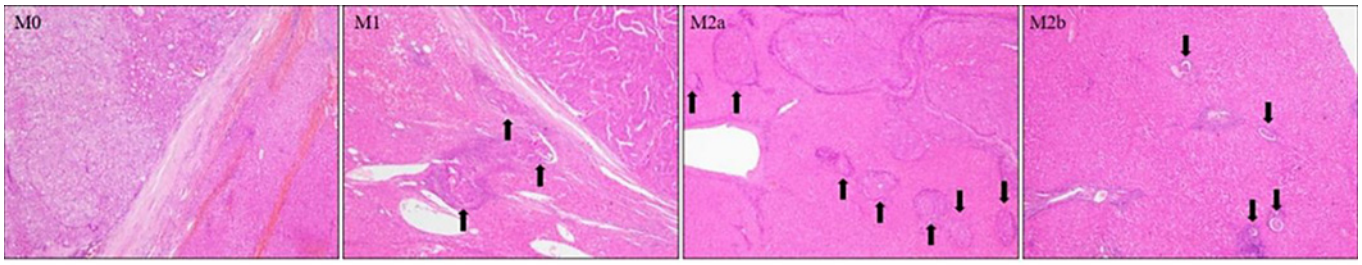
Immunohistochemical Biomarkers for HCC

A positive result for the following hepatocyte-specific biomarkers may indicate hepatocyte-origin tumors; however, these results are insufficient to differentiate between benign and malignant hepatocellular neoplasms.

1. Arginase-1: hepatocyte plasma/nucleus staining.
2. Hepatocyte antigen: hepatocyte plasma staining.
3. Specific staining antibodies for the bile canaliculus of the hepatocyte membrane: CD10, polyclonal carcinoembryonic antigen, and bile salt export pump, can localize specifically to the hepatocyte membrane's bile canaliculus, aiding in the confirmation of HCC.

The following biomarkers may assist in the differentiation of benign and malignant hepatocellular tumors.

1. Glutamine synthetase: a strong cytoplasmic positivity in a diffuse pattern is mostly observed for HCC. Some hepatocellular adenomas, especially β-catenin



MVI grading	Diagnostic criteria
M0	no MVIs detected
M1 (low-risk group)	≤5 MVIs, occur in proximal non-neoplastic adjacent liver tissue (≤1 cm)
M2a (high-risk group)	>5 MVIs, occur in proximal non-neoplastic adjacent liver tissue (≤1 cm)
M2b (high-risk group)	MVIs occurring in distal non-neoplastic adjacent liver tissue (>1 cm)

Fig. 2. Pathological grading criteria for microvascular invasion (MVI).

mutated-activated hepatocellular adenomas, may also exhibit diffuse positivity. Glutamine synthetase staining in high-grade dysplastic nodules often exhibits moderate focal staining with a positive cell count <50%. A characteristic irregular pattern of staining is observed in focal nodular hyperplasia of the liver. In normal liver tissue, staining is typically limited to hepatocytes surrounding the central vein. These distinctive features can assist in differential diagnosis.
2. Phosphatidylinositol protein-3: HCC plasma and cell membrane staining.
3. Heat shock protein 70: HCC plasma or nucleus staining.
4. CD34: although CD34 immunohistochemical staining does not directly label tumor parenchymal cells, it can reveal microvascular density and distribution patterns across various liver tumor types. For instance, CD34 staining exhibits a diffuse pattern in HCC, a sparse pattern in ICC, a patchy pattern in hepatocellular adenoma, and a strip pattern in focal nodular hyperplasia. These histologic staining patterns may facilitate the differentiation and diagnosis of benign and malignant liver tumors.

Immunohistochemical Biomarkers for ICC

1. ICC universal immunohistochemical biomarkers: cytokeratin 7, cytokeratin 19, mucin 1 (MUC1), and epithelial cell surface glycoprotein (MOC31).
2. Large duct type ICC: S100 calcium-binding protein, mucin 5AC (MUC5AC), and others.
3. Small duct type of icc: c-reactive protein, neural cadherin, neural cell adhesion molecule (CD56).

Immunohistochemical Biomarkers for cHCC-CCA

Both HCC and ICC components express the above-mentioned biomarkers of the respective tumors. In addition, positive expression of biomarkers such as CD56, CD117 and epithelial cell adhesion molecule may suggest that the tumor is characterized by stem cell differentiation and is more aggressive.

Molecular Pathology Examination

The main purposes of molecular pathology examination for liver cancer are: assisting in the diagnosis of special subtypes of HCC; screening of commonly used targeted/immunotherapy drugs for ICC. Molecular pathology testing in the diagnosis of liver cancer serves multiple purposes. It can identify specific subtypes of HCC and screen for targeted and immunotherapy drugs for ICC. This approach aids in guiding clinical prognosis monitoring and personalized treatment. Furthermore, it enhances the pathological understanding of special subtypes of HCC and ICC. Clinically, patients can be stratified and managed based on molecular pathological changes.

Commonly used molecular pathologic diagnostic biomarkers for HCC: (1) Fibrolamellar HCC: with DNAJB1-PRKACA gene fusion [92]; (2) Scirrhous HCC: with mutations in the tuberous sclerosis complex 1/2 gene [92].

Targeted/immunotherapy biomarkers commonly used in ICC: (1) large duct type ICC: common human

epidermal growth factor receptor 2 gene amplification, BRAF V600E gene mutation, neurotrophic tyrosine receptor kinase gene fusion, RET gene fusion, high microsatellite instability, high tumor mutation burden, etc. [110]. (2) Small duct type ICC: common fibroblast growth factor receptor 2 gene mutation (rearrangement or fusion) [111] with isocitrate dehydrogenase 1/2 gene mutations [112].

Pathologic Evaluation of Resected HCC Specimens after Conversion/Neoadjuvant Therapy

Specimen sampling: when handling resected HCC specimens, regardless of whether the resection follows conversion or neoadjuvant therapy, the following procedures should be conducted: Incise the tumor bed along the plane of the largest diameter and measure the three-dimensional dimensions. Provide a detailed description of the extent of necrosis and any residual tumors, including the percentage of necrosis. For small HCC with a diameter of ≤3 cm, the whole tumor should be collected. For tumors with a diameter of >3 cm, sections should be cut at 0.5–1 cm interval along the side with the longest diameter, and the most representative section with residual tumor should be selected for complete sampling, while the other sections should be selectively sampled. The tumor bed and surrounding liver tissue should be obtained for cross-referencing. Macroscopic photography of the specimen may also be obtained as reference for histologic observations.

Microscopic assessment: to determine the proportions of the three components of the tumor bed in the resected specimen of HCC, i.e., surviving tumor cells, necrotic tumor cells, and tumor stroma (fibrous tissue and inflammation). The sum of these three areas of the tumor bed is equal to 100%. The number of wax blocks obtained should be indicated in the pathology report. The total percentage of residual tumor should be determined by taking the mean value of the percentages of the three components above in each section and indicated in the pathology report.

The assessment of pathologic complete response and major pathologic response (MPR) is an important pathologic indicator for evaluating the efficacy of preoperative treatments and exploring rational postoperative interventions.

1. athologic complete response: it is defined as the absence of any viable tumor cells upon thorough histologic examination of the tumor bed specimen following preoperative treatment.
2. MPR: it is defined as a reduction in viable tumor cells to levels below the threshold that significantly impacts clinical prognosis following preoperative treatment. MPR is often defined in lung cancer studies as a reduction of residual tumor cells in the tumor bed to ≤10% [113], which is also consistent with studies showing correlation between the degree of tumor necrosis and prognosis after preoperative treatment with TACE for HCC [114]. However, for surgical samples of HCC received with conversion/neoadjuvant therapy, the clinical significance of the reduction of residual tumor cells in the tumor bed to how much is not clearly established, and it is generally believed that it should be at least to 50% or more. Expansion of the area of tumor specimen collection is recommended for those with initially diagnosed as MPR, for further confirmation.

For HCC specimens treated with immune checkpoint inhibitors, pay attention to describing the degree of lymphocyte infiltration and the structure of the third-order lymph nodes [115]. Meanwhile, attention should be paid to the presence of immune-related liver injuries around the cancer, including hepatocyte damage, intrahepatic inflammation, and cholangitis.

Pathologic Diagnostic Report of HCC

A typical pathologic report should include a gross description of specimens, microscopic descriptions, the results of immunohistochemical examination, the final pathologic diagnosis and MVI grading etc., with notes and recommendations to physicians if necessary. In addition, the results of molecular examination related to the clonal origin of multinodular/recurrent HCC, drug target testing, biologic behavior evaluation, and prognosis assessments may be attached for clinical reference.

Summary:

1. Standardized handling and timely delivery of biopsy/resected tissue samples are of great significance for tumor tissue preservation and correct pathologic diagnosis.
2. The "7-point" baseline sampling method should be followed when collecting HCC specimens to facilitate obtaining a representative pathobiological indication of HCC.
3. The content of pathological diagnosis report for HCC should be both standardized and comprehensive. It is essential to emphasize the description of critical prognostic factors in HCC, including histological type, degree of differentiation, invasive growth patterns, MVI grading, and molecular pathological assessments of targets relevant for guiding targeted therapies.

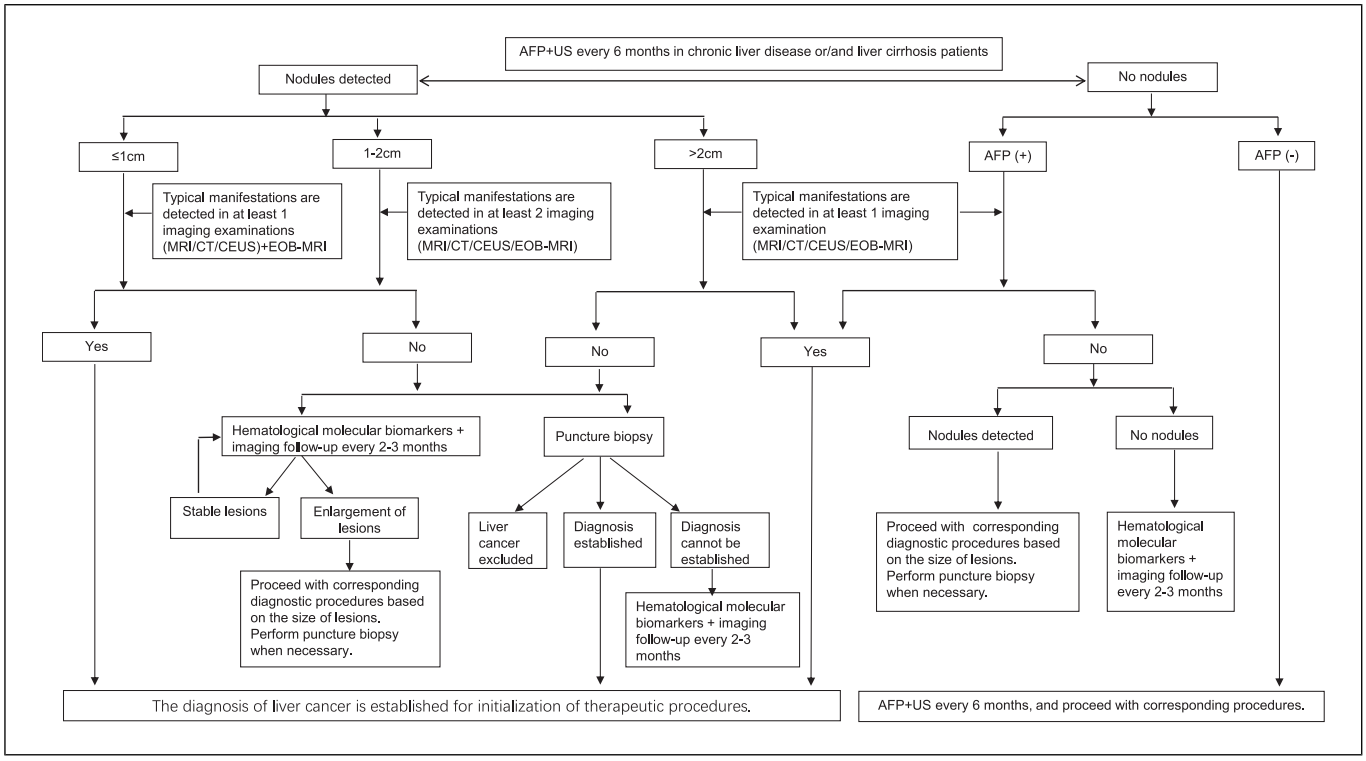


Fig. 3. Diagnostic pathway for the diagnosis of HCC. Typical presentation: significant enhancement of the lesion in the arterial phase (late major arterial phase), accompanied by decreased enhancement during the portal venous, delayed or migratory phases in a "wash-in, wash-out" pattern. Atypical presentation: absence of lesion enhancement in the arterial phase, lack of clearance during the portal venous, delayed or migratory phases, or even persistent enhancement, etc. Hematological molecular markers include serum AFP, DCP, and a combination of seven microRNAs. AFP (+): serum AFP test exceeding normal value. MRI, magnetic resonance imaging; CT, computed tomography; US, ultrasound; CEUS, contrast-enhanced ultrasound; EOB-MRI, MRI scans enhanced with the hepatocyte-specific agent gadolinium ethoxybenzyl-diethylenetriamine pentaacetic acid (Gd-EOB-DTPA).

4. Focus on the pathological evaluation of HCC resection specimens following conversion or neoadjuvant therapy.

Clinical Diagnostic Criteria and Diagnostic Roadmap for HCC

A clinical diagnosis of HCC should be established in accordance with the steps shown in the following pathway, taking into account high-risk factors for HCC, imaging characteristics, and hematological molecular markers (shown in Fig. 3). For patients at high risk for HCC, screening using US and serum AFP testing should be performed at least every 6 months. For patients with intrahepatic nodules ≤1 cm in diameter observed during follow-up, a clinical diagnosis of HCC can be established by observing typical characteristics of HCC on the "wash-in and wash-out" enhancement pattern, in at least one of the following three imaging modalities: dynamic contrast-enhanced MRI, dynamic contrast-enhanced CT, or contrast-enhanced US, including Gd-EOB-DTPA-enhanced MRI. If the above requirements are not met, follow-up imaging examinations can be performed every 2–3 months and combined with serum AFP, DCP, and a combination of seven microRNAs to definitive diagnosis, and puncture biopsy of the liver lesion can be performed if necessary.

For patients at high risk for HCC and intrahepatic nodules 1–2 cm in diameter observed during follow-up, a clinical diagnosis of HCC can be established by observing typical characteristics of HCC on at least two of the four following imaging examinations: dynamic contrast-enhanced MRI, dynamic contrast-enhanced CT, contrast-enhanced US or Gd-EOB-DTPA-enhanced MRI. If there is no or only one typical

characteristic of HCC in the above four imaging examinations, follow-up imaging examinations can be performed every 2–3 months and combined with serum AFP, DCP, a combination of seven microRNA to definitive diagnosis, and puncture biopsy of the liver lesion can be performed if necessary.

For patients at high risk for HCC and intrahepatic nodules >2 cm in diameter observed during follow-up, a clinical diagnosis of HCC can be established by observing typical characteristics of HCC on at least one of the four following imaging examinations: dynamic contrast-enhanced MRI, dynamic contrast-enhanced CT, contrast-enhanced US or Gd-EOB-DTPA-enhanced MRI. If there is no typical characteristics of HCC in the above four imaging examinations, follow-up imaging examinations can be performed every 2–3 months and combined with serum AFP, DCP, a combination of seven microRNA to definitive diagnosis, and puncture biopsy of the liver lesion can be performed if necessary.

People at high risk of HCC with elevated serum AFP, especially persistently elevated, should be performed imaging examination to definitive diagnosis of HCC; a clinical diagnosis of HCC can be established if at least one typical characteristic of HCC in the four imaging following examinations: dynamic contrast-enhanced MRI, dynamic contrast-enhanced CT, contrast-enhanced US or Gd-EOB-DTPA-enhanced MRI. If no intrahepatic nodule is detected by the above four imaging examinations, imaging review should be performed every 2–3 months under the premise of excluding pregnancy, chronic or active liver disease, germline embryogenic tumors, and other digestive system tumors, and the changes of serum AFP, DCP, and a combination of seven microRNA should be closely followed up at the same time.

Staging of HCC

The staging of HCC is crucial to assess prognosis and select the appropriate treatment. International options for staging include the Barcelona Clinic Liver Cancer (BCLC), TNM, Japan Society of Hepatology (JSH) [57], and the Asia Pacific Association for the Study of the Liver (APASL) staging systems. Based on the Chinese domestic context and clinical practice, the CNLC staging system considers the patient's performance status (PS), as well as the status of liver tumors and liver function. The CNLC staging system is divided into stages Ia, Ib, IIa, IIb, IIIa, IIIb, and IV and shown in Figure 4.

- CNLC stage Ia: PS score 0–2, Child-Pugh class A/B, a solitary tumor ≤5 cm in diameter, and absence of vascular invasion or extrahepatic metastasis on imaging examinations.
- CNLC stage Ib: PS score 0–2, Child-Pugh class A/B, a solitary tumor >5 cm in diameter, or 2–3 tumors with a maximum diameter ≤3 cm, and absence of vascular invasion or extrahepatic metastasis on imaging examinations.
- CNLC stage IIa: PS score 0–2, Child-Pugh class A/B, 2–3 tumors with a maximum diameter >3 cm, and absence of vascular invasion or extrahepatic metastasis on imaging examinations.
- CNLC stage IIb: PS score 0–2, Child-Pugh class A/B, ≥4 tumors irrespective of diameter, and absence of vascular invasion or extrahepatic metastasis on imaging examinations.
- CNLC stage IIIa: PS score 0–2, Child-Pugh class A/B, presence of vascular invasion irrespective of tumor diameter and number, but absence of extrahepatic metastasis on imaging examinations.
- CNLC stage IIIb: PS score 0–2, Child-Pugh class A/B, and the presence of extrahepatic metastasis on imaging examinations irrespective of tumor diameter, tumor number, and vascular invasion.
- CNLC stage IV: PS score 3–4, or Child-Pugh class C, regardless of tumor diameter, tumor number, vascular invasion, and extrahepatic metastasis on imaging examinations.

Treatment

The treatment of HCC is characterized by multidisciplinary participation and the coexistence of multiple therapeutic approaches, which include hepatectomy, LT, ablation therapy, endovascular therapy, radiation therapy, systemic antitumor therapy, traditional Chinese medicine therapy. Each treatment method has its unique advantages and limitations, and the indications overlap with each other. Standardized and accurate treatment decisions should be based on guidelines and high-level evidence-based medical evidence. At the same time, consideration should also be given to the latest developments and research results in various fields. However, the knowledge updates of treatment methods in a single discipline may have limitations and lagging effect in other fields. Therefore, the diagnosis and treatment of HCC should emphasize communication and cooperation among multidisciplinary team (MDT) to ensure the most appropriate treatment decision for patients and to

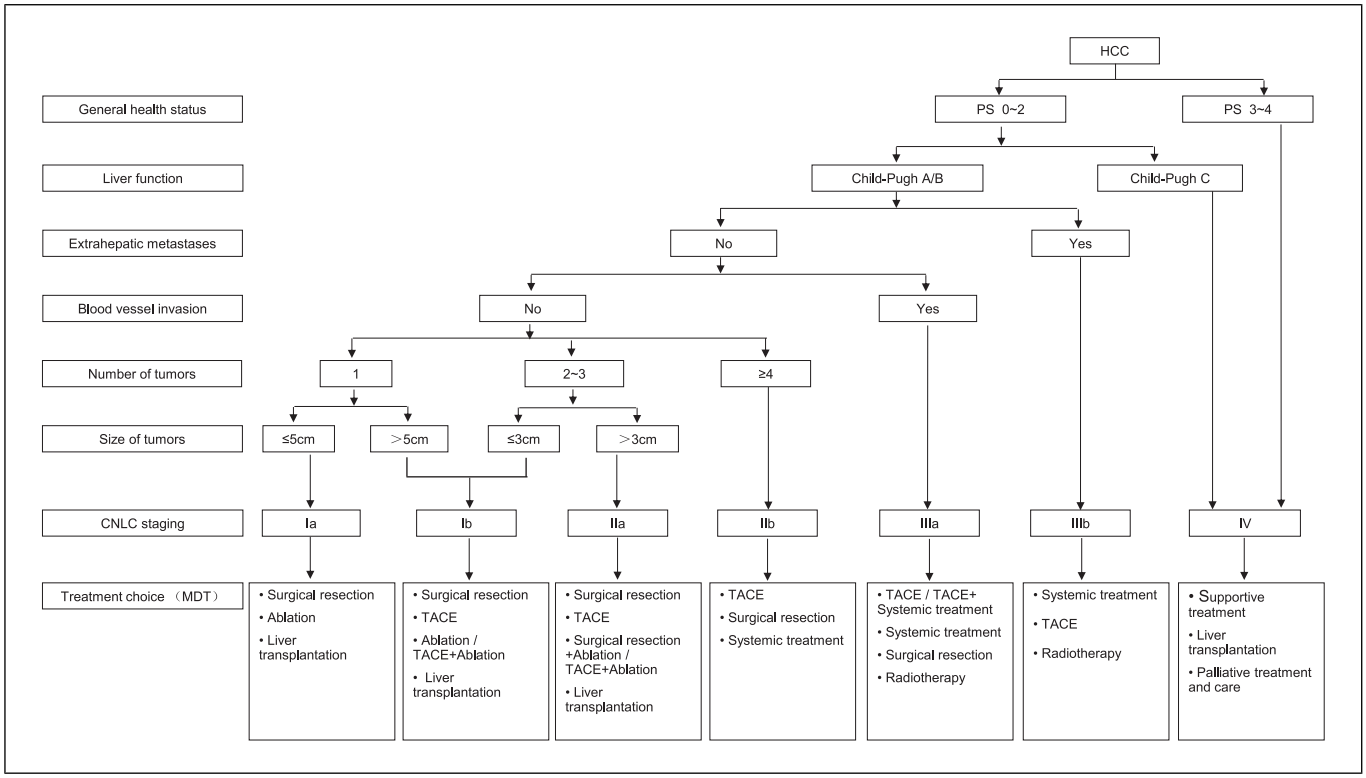


Fig. 4. CNLC staging and treatment pathway for HCC. Systemic antitumor therapy includes first-line and second-line therapy. First-line therapy: atezolizumab+bevacizumab, sintilimab+a bevacizumab biosimilar, apatinib+camrelizumab, donafenib, lenvatinib, tislelizumab, sorafenib, and FOLFOX4. Second-line therapy: regorafenib, apatinib, ramucirumab (serum AFP level ≥400 μg/L), pembrolizumab, camrelizumab, tislelizumab. CNLC, China liver cancer; MDT, multidisciplinary team; TACE, transcatheter arterial chemoembolization.

promote the progress of HCC treatment. Currently, the importance and necessity of MDT for HCC have become a widely recognized consensus in the medical field. However, due to practical constraints, there are still significant differences in the implementation methods and levels of MDT for HCC among different regions and institutions. It is suggested that hospitals at all levels carrying out diagnosis and treatment of HCC should incorporate MDT management into the medical quality management system, which should be jointly managed by the medical administrative department and the designated MDT manager. This should be conducted in a multidisciplinary consultation mode with fixed time, location, and personnel, and be carried out in the mode of multidisciplinary consultation model with a fixed time, location, and personnel. If primary hospitals have difficulties in organizing MDT on their own due to limitations in, they can implement MDT through methods such as "medical consortiums" or "telemedicine." With the publication and implementation of "Quality Control Indices for Standardized Diagnosis and Treatment of Liver Cancer in China (Version 2022)" by the National Cancer Center of China, it will further promote the standardization and homogenization of HCC diagnosis and treatment nationwide.

Surgical Treatment

Surgical treatment provides the best opportunity for long-term survival in patients with HCC (evidence level 2, recommendation A) and mainly comprises hepatectomy and LT.

Basic Principles of Hepatectomy

Thoroughness: complete resection of tumor tissue, ensuring that the surgical margin is free of residual tumor. Safety: preservation of a sufficient volume of functional liver tissue (with good blood supply as well as

blood and bile outflow) to compensate for reduced liver function and minimize surgical complications and postoperative mortality.

Evaluation of Patient's General Condition, Liver Function Reserve, and Follow-Up

Preoperative evaluation of patients' general condition, liver function reserve (LFR) and liver tumor status (stage and location) are mandatory. The Eastern Cooperative Oncology Group (ECOG) PS score is commonly used to evaluate a patient's general condition. The Child-Pugh Score, indocyanine green (ICG) clearance test, transient elastography for measuring liver stiffness or model for end-stage liver disease (MELD) is used to evaluate LFR [116–121]. Accurate evaluation of the degree of portal hypertension is helpful to screen patients suitable for surgical resection, as studies have suggested that selected patients with HCC and portal hypertension can still undergo hepatectomy, with long-term postoperative survival superior to that of those receiving other treatments [122–125] (evidence level 3, recommendation B). If the preservation of a low volume of liver tissue is expected, it can be determined by CT, MRI, or 3D reconstruction of the liver to measure future liver remnant (FLR) and calculate the percentage of FLR to standardized liver volume (SLV) [117]. A Child-Pugh class A and ICG retention test (ICG-R15) score <30% are generally considered prerequisites for successful surgical resection. The FLR that accounts for >40% of the SLV (for patients with chronic diseases, parenchymal liver damage, or liver cirrhosis) or >30% (for patients without liver fibrosis and liver cirrhosis) is another prerequisite for surgical resection. A greater FLR should be preserved in patients with hepatic impairment. Patients with HCC caused by nonalcoholic steatohepatitis have better surgical prognosis than those caused by alcoholic fatty liver disease [126] (evidence level 2, recommendation B).

Postoperative patients with HCC should have a follow-up once within 1–2 months, and then closely monitored every 3 months for changes in imaging (ultrasonography, if necessary, dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI scans, and Gd-EOB-DTPA-enhanced MRI scans) and oncologic markers such as serum AFP, DCP, and a combination of seven microRNAs. After 2 years, follow-up can be appropriately prolonged to 3–6 months, and the duration of which is recommended to be lifelong (evidence level 3, recommendation B). Current evidence does not support the notion that more frequent follow-up visits confer survival benefits [127].

Indications for Hepatectomy

1. The recommended treatment for patients with CNLC stages Ia, Ib, and IIa HCC and enough LFR is surgical resection. Previous studies have shown the overall survival (OS) of surgical resection is similar or slightly better than radiofrequency ablation (RFA) for HCC ≤3 cm in diameter [128–130] (evidence level 1, recommendation A). Meanwhile, some studies have shown the local recurrence rate after surgical resection is significantly lower than that after RFA [131–135]. For recurrent HCC, the prognosis following surgical resection remains better than that following RFA [136] (evidence level 2, recommendation B).
2. For most patients with CNLC stage IIb HCC, surgical resection is not recommended; instead, nonsurgical approaches such as TACE are recommended. However, when tumors are localized to the same liver segment or the ipsilateral hemi-liver, intraoperative RFA may be used to treat lesions outside the resected area. Hepatectomy may be superior to other treatment approaches after MDT discussion, even for multiple tumors (>3) [137–140]; in these cases, surgical resection is also recommended (evidence level 2, recommendation B).
3. For CNLC stage IIIa HCCs, surgical resection is not recommended in most cases, especially for those with tumor thrombi in the main of the portal vein. Instead, nonsurgical treatment approaches such as TACE or TACE combined with systemic antitumor therapy is preferred. However, some studies have suggested that compared with sorafenib, OS, and progression-free survival (PFS) of hepatectomy for advanced non-metastatic HCC were significantly better [141]. Most of the data on surgical resection for CNLC stage IIIa HCC come from Asian countries [142, 143], with a small portion from Western countries [144, 145]. Surgical resection may be considered after MDT discussion if the following conditions are met: (1) in HCC patients with portal vein tumor thrombus (Cheng type I/II), if the tumor is localized to the hemi-liver or the ipsilateral side of liver, surgical resection of the tumor and the tumor thrombus may be considered. Postoperatively, TACE, portal vein chemotherapy, or other systemic treatments can be performed [146, 147] (evidence level 3, recommendation C). Preoperative 3D conformal radiotherapy is associated with improved postoperative survival in these patients [148] (evidence level 2, recommendation B). Patients undergoing surgical resection of tumor thrombi located in the main trunk of the portal

vein (Cheng's classification type III) exhibit elevated short-term postoperative recurrence rates and suboptimal survival outcomes. Consequently, the presence of tumor thrombi in the main trunk of portal vein should not be regarded as an absolute indication for surgical intervention [149] (evidence level 3, recommendation B). (2) Surgical resection may be considered in patients presenting with tumor thrombi in the bile duct and resectable intrahepatic lesions. (3) Patients exhibiting partial invasion of the hepatic vein with resectable intrahepatic lesions are also deemed appropriate candidates.

4. For patients diagnosed with CNLC stage IIIb exhibiting metastasis to the hilar lymph nodes, surgical resection of the tumor combined with hilar lymph node dissection or postoperative external radiation therapy may be considered after MDT discussion. Additionally, surgical resection may also be considered for patients whose adjacent organs are involved and can be excised concurrently.

Furthermore, for patients with HCC deemed unsuitable for resection during surgical exploration, alternative treatments may be considered. These include hepatic artery and portal vein catheterization chemotherapy, other intraoperative locoregional treatments such as ablation therapy, as well as postoperative nonsurgical therapies like TACE and systemic antitumor treatments, once the patient has recovered from the surgical trauma.

Criteria for Curative Resection of HCC

Intraoperative criteria: (1) no macroscopic tumor thrombi noted in the hepatic vein, portal vein, bile duct, and inferior vena cava; (2) no adjacent organ involvement, portal lymph node, or distal metastases; (3) a distance between the surgical margin and tumor boundary ≥1 cm; or if the surgical margin is <1 cm, histologic examination of the cross section of the resected liver is free of residual tumor cells, i.e., negative surgical margin.

Postoperative criteria: (1) US, CT, and MRI (at least two of the three are mandatory) performed 1–2 months after surgery to confirm the absence of tumor lesions; (2) postoperative quantitative testing of serum tumor markers should be performed 2–3 months after surgery to confirm that their levels have normalized, especially if markers such as serum AFP, DCP, and a panel of seven microRNAs were elevated preoperatively. The rate of decline in serum AFP levels can serve as an early predictor of the completeness of the surgical resection [150].

Hepatectomy Techniques

Commonly used hepatectomy techniques include hepatic inflow and outflow control techniques, liver transection techniques, and hemostatic techniques. Preoperative 3D visualization technology allows individualized liver volume calculation and virtual liver resection, which aids in planning a more precise resection margin and trajectory, ensuring the protection of vessels within the liver remnant and preserving an adequate FLR [151–153] (evidence level 2, recommended A).

In recent years, there have been significant advancements in minimally invasive surgery (including laparoscopic hepatectomy and robot-assisted hepatectomy). Laparoscopic hepatectomy offers the benefits of reduced invasiveness and is associated with expedited postoperative recovery [154, 155] (evidence level 2, recommendation B). The 5-year OS for early stage HCC patients undergoing laparoscopic hepatectomy was comparable to that of patients undergoing laparotomy [156] (evidence level 2, recommendation B). Compared with laparotomy, laparoscopic hepatectomy for elderly HCC patients (≥65 years) has superior surgical outcomes with comparable oncologic outcomes [157] (evidence level 2, recommendation B). Although the indications and contraindications for laparoscopic hepatectomy are generally consistent with those for laparotomy, it is still recommended to conduct a comprehensive evaluation and exercise caution based on factors such as tumor size, tumor location, number of tumors, underlying liver diseases, and the skill level of the surgical team. In cases of giant HCC, multiple HCCs, HCC located in difficult sites or in the central region adjacent to important ducts, and HCC combined with severe cirrhosis, it is recommended that the procedure be undertaken by an experienced surgeon after rigorous selection. Laparoscopic hepatectomy is not recommended for HCC patients with portal vein macroscopic tumor thrombus and tumor rupture hemorrhage. The use of laparoscopic US combined with ICG tumor imaging may help detect microscopic lesions and mark the extent of resection required to obtain negative tumor margins [158]. Preliminary studies indicate that the efficacy and safety of robot-assisted hepatectomy for HCC are comparable to laparotomy [159] (evidence level 3, recommendation C).

Both anatomic and nonanatomic resection are widely employed techniques, and both require adequate surgical margins to achieve optimal oncological outcomes. In patients with HCC exhibiting MVI, anatomic resection is associated with lower local recurrence rates than nonanatomic resection, despite no difference in OS [160,

161] (evidence level 3, recommendation B). Studies have shown that hepatectomy with wide surgical margins (≥1 cm) is associated with better outcomes than hepatectomy with narrow surgical margins [162, 163] (evidence level 2, recommendation A), particularly in patients preoperatively assessed for MVI [164]. Compared to nonanatomic resection, laparoscopic anatomic hepatectomy has a significantly higher 5-year disease-free survival rate for HCC [165] (evidence level 2, recommendation B). For large liver tumors, an anterior approach hepatectomy without dissecting perihepatic ligaments may be considered [166]. For multiple liver tumors, surgical removal in combination with intraoperative locoregional ablation may be performed [167] (evidence level 3, recommendation C). For patients with portal vein tumor thrombi, the portal venous flow of the unaffected side should be temporarily interrupted during portal vein embolectomy to avoid disseminating the tumor thrombi [168]. For patients with tumor thrombi in the hepatic vein or vena cava, complete hepatic blood flow occlusion can be performed to remove the tumor thrombus as completely as possible [169]. For patients with HCC and tumor thrombi in the bile duct, resecting both the involved bile ducts and HCC optimize chances for curative intervention [170, 171] (evidence level 3, recommendation C).

A Multimodality Treatment Based on Surgery

Despite the suboptimal OS after surgery in patients with intermediate-to-advanced HCC (CNLC stages IIb, IIIa, and IIIb) based on data from previous study, advances in systemic antitumor therapy and multimodality treatment have made it possible to provide more possibilities for radical resection, reduce postoperative recurrence and metastasis, and improve prognosis in patients with intermediate-to-advanced HCC [172] (evidence level 4, recommendation B). There is a growing consensus on moderately expanding surgical indications. Consequently, exploring innovative strategies for surgery-based integrated management of intermediate and advanced HCC has emerged as a pivotal focus in recent years.

Conversion Therapy for HCC

Conversion therapy refers to patients with HCC who are initially unsuitable for surgical resection but may become eligible for surgery following intervention. Intervention strategies include functional FLR conversion and oncological conversion, etc.Surgical resection remains a critical approach for achieving long-term survival in patients who have undergone successful conversion therapy. However, further evidence from randomized controlled trials is necessary to substantiate this claim (evidence level 4, recommendation C).

Conversion Therapy for Functional FLR

Insufficient FLR is a significant factor preventing the surgical resection of HCC. For these patients, the goal of conversion therapy is to transform from insufficient FLR to sufficient functional FLR [173].

Portal Vein Embolization. Portal vein embolization (PVE) should be performed to the hemi-liver in which the tumor is located for compensatory hypertrophy of the remaining liver before resection, if HCC patients are reasonably selected, the success rate of conversion therapy is 60–80%, with a complication rate of approximately 10–20%. The period required for remnant liver hyperplasia following PVE is relatively lengthy, typically ranging from 4 to 6 weeks. During this interval, more than 20% of patients may lose the opportunity for surgery due to tumor progression or inadequate FLR hypertrophy while awaiting hyperplasia [174, 175] (evidence level 3, recommendation B). For these individuals, current treatment strategies include combined with TACE [176], hepatic vein embolization [177], and arterial ligation [178], with the aim of further promoting FLR hyperplasia and control tumor progression. Alternatively, salvage two-stage hepatectomy with associating liver partition and portal vein ligation for staged hepatectomy (ALPPS) can be employed for tumor removal [179]. Contraindications for PVE include the presence of tumor thrombus in the main or first-order branches of the portal vein, extensive tumor metastasis, as well as severe portal hypertension and coagulation dysfunction. Caution should be exercised for patients with an anticipated longer period of functional FLR hyperplasia (e.g., severe cirrhosis, older patients) and the potential for rapid tumor progression.

Associating Liver Partition and Portal Vein Ligation for Staged Hepatectomy. As a recent major innovation in liver surgery, ALPPS typically induces rates of residual hepatic hyperplasia of up to 47–192% within approximately 1–2 weeks, which is significantly higher than that achieved with PVE. Due to the short interval between the two stages of surgery, the risk of tumor progression can be minimized, with a tumor resection rate of 95–100% [180, 181]. With the advancement of surgical techniques and accumulation of experience, the surgical complications and mortality associated with ALPPS have been greatly reduced compared to the early stage of ALPPS. In recent years, several ALPPS modifications have emerged, which mainly focused on the partitioning operations of

the hepatic section in stage I surgery (RFA, microwaves, and tourniquets can be used for partitions) and on ALPPS using a minimally invasive laparoscopic approach, further improved the safety of ALPPS. If the functional FLR is still insufficient for surgical resection at 2 weeks after stage I ALPPS, transarterial embolization (TAE) can be performed, a procedure known as TAE-salvaged ALPPS, which achieves a stage II surgical resection rate of nearly 100% after 1 week [182]. Some RCTs have demonstrated that ALPPS has a significant advantage over PVE in its ability to rapidly induce FLR hyperplasia [183] (evidence level 2, recommendation A). Generally, candidates for ALPPS should be limited to the following patients: <65 years, normal liver function (Child-Pugh class A, ICG-R15 <20%), FLR insufficiency (FLR/SLV <30% in those with normal livers and <40% in those with chronic liver disease and liver damage), good general status, good surgical tolerance, no severe cirrhosis, no severe fatty liver, and no severe portal hypertension.

Oncological Conversion Therapy for HCC

Locoregional Therapy. Locoregional therapies present significant opportunities for surgical resection in patients with initially unresectable HCC, potentially translating into survival benefits, including TACE [184], hepatic arterial infusion chemotherapy (HAIC) [185] and radiotherapy (evidence level 3, recommendation B). The combination of TACE or HAIC with systemic antitumor therapy can improve conversion rate [186–188]. For Some patients with large tumor burdens, and classified as CNLC II-III who are unlikely to respond to or benefit from TACE, consideration should be given to combining it with systemic therapies to promote conversion therapy. The positive results of the Phase III EMERALD-1 trial [189] investigating TACE combined with durvalumab and bevacizumab were recently presented. For HCC patients with large tumor burden or combined portal vein tumor thrombus (especially in the main trunk of the portal vein), who are temporarily unfit for surgery, several clinical studies have shown that HAIC has a high objective response rate (ORR), and some patients experience tumor shrinkage or regression of portal vein tumor thrombus after HAIC, which improves the success rate of conversion therapy (evidence level 3, recommendation B). HAIC combined with TACE [190], radiotherapy [191], molecular targeted drugs, and/or immunotherapy may further improve the success rate of conversion therapy.

Systemic Antitumor Therapy. Antiangiogenic therapy combined with immunotherapy, as well as molecular targeted agents used in combination with or without immunotherapy, have become important treatment strategies for unresectable or intermediate-to-advanced HCC, and are also utilized for conversion therapy in HCC (evidence level 4, recommendation B). In selecting systemic antitumor therapy, it is crucial to identify the reasons for non-radical resection. Prioritizing etiological management and closely monitoring the duration and depth of the tumor response, based on MDT assessments, is essential. It is also important to closely monitor the toxicity of systemic antitumor therapy and its potential impact on conversion therapy. Actively exploring changes in the tumor immune microenvironment before and after treatment is vital. We will also actively conduct large-scale clinical studies with high-level evidence-based medicine, aiming to maximize patient benefits.

Neoadjuvant Therapy for HCC

Neoadjuvant therapy refers to preoperative local treatment or systemic antitumor therapy for HCC patients who are candidates for surgical resection but have a high risk of postoperative recurrence and metastasis (CNLC Ib-IIa and part of CNLC IIb and IIIa). The primary objective was to eliminate microscopic lesions, reduce the rate of postoperative recurrence and metastasis, and prolong survival (evidence level 4, recommendation C). The factors of high-risk recurrent and metastasis for preoperatively evaluation include: vascular invasion, single tumor diameter >5 cm, multiple tumors, involvement of adjacent organs, high preoperative AFP levels, and high preoperative serum HBV-DNA load. While neoadjuvant therapy offers potential benefits, it also carries inherent risks; thus, careful selection of appropriate patient populations is essential. Moreover, the treatment regimen must align with the objectives of neoadjuvant therapy. When selecting an appropriate treatment regimen, it is essential to consider not only the ORR but also to prioritize regimens that demonstrate a higher disease control rate. This approach aims to minimize the risk of losing surgical opportunities due to disease progression. Additionally, treatments that are characterized by relative safety and have a lower incidence of adverse effects should be favored to reduce the likelihood of complications.

Adjuvant Therapy for HCC

The 5-year recurrence and metastasis rate after surgical resection of HCC is as high as 50–70% [192, 193]. Adjuvant therapy serves as a crucial strategy to mitigate the risk of tumor recurrence and metastasis while enhancing patient survival outcomes (evidence level 2,

recommendation B). Compared to neoadjuvant therapy, adjuvant therapy facilitates more precise identification of the treatment population and allows for tailored therapeutic approaches based on postoperative pathology and molecular subtypes, without postponing surgical intervention. The primary candidates for adjuvant therapy are HCC patients who are suitable for surgical resection and have a high risk of recurrence and metastasis. Although the definitions of high-risk factors for recurrence and metastasis differ across studies, the factors for postoperative evaluation generally include: tumor rupture, tumor diameter >5 cm, multiple tumors, MVI, macrovascular invasion, lymph node metastasis, positive margins or narrow margins, and tissue differentiation of Edmondson grade III-IV [194–196].

In patients at a high risk of postoperative recurrent and metastases, there is no established international standard for adjuvant therapy. Two randomized controlled studies confirmed the effectiveness of postoperative TACE therapy to reduce recurrence and metastases and prolong survival [197, 198] (evidence level 1, recommendation A). HAIC with fluorouracil/oxaliplatin/leucovorin (mFOLFOX) regimen can reduce recurrent and metastasis and improve survival in HCC patients with MVI [199] (evidence level 1, recommendation B). In HCC patients who have undergone surgical resection, RFA, or percutaneous ethanol injection (PEI), activated CIK cell therapy can also significantly extend median recurrence-free survival [200] (evidence level 2, recommendation B). Immunomodulators (e.g., thymalfasin [201]) also exhibit comparable effects. Furthermore, a prospective multicenter randomized controlled phase III clinical study demonstrated that Chinese herbal medicine, such as Huaier granules following hepatectomy can reduce the risk of postoperative recurrence and metastasis, and can also prolong OS [202] (evidence level 1, recommendation A). In patients with HCC and HBV infection, antiviral therapy with nucleoside analog can help reduce postoperative recurrent and metastasis and should be taken long-term [203] (evidence level 1, recommendation A). Additionally, in patients with viral hepatitis-associated, adjuvant therapy with pegylated interferon can improve both OS and recurrence-free survival without serious adverse effects [204] (evidence level 1, recommendation B). In patients with HCC and HBV infection, the use of direct-acting antiviral agents (DAAs) aims to achieve a sustained virologic response. However, there is currently insufficient conclusive evidence to establish an association between DAA treatment and either an increased or decreased risk of tumor recurrence and metastasis following HCC surgery. Additionally, there is no conclusive evidence regarding differences in the timing of recurrence or postoperative survival [205, 206] (evidence level 3, recommendation C). In addition, the concurrent use of portal vein catheterization chemotherapy and TACE after surgery may also prolong survival in patients with portal vein tumor thrombosis [147] (evidence level 3, recommendation B). In recent years, systemic antitumor therapy has been studied for its role in the adjuvant treatment of HCC, with multiple treatment regimens being actively explored for their effectiveness in advanced HCC. Among these, the IMbrave050 study and the STORM study both significant contributions through their valuable attempts. However, the final results of neither study demonstrated a statistically significant improvement in recurrence-free survival (RFS) [207–209].

Summary:

1. Hepatectomy is an important means of achieving long-term survival in patients with HCC.
2. Comprehensive preoperative assessments of liver reserve function and oncological evaluation are paramount. Generally, Child-Pugh class A and ICG-R15 <30% are accepted as prerequisites for surgical resection. Furthermore, the FLR accounting for >40% of the SLV (for patients with chronic liver diseases, parenchymal liver damage, or liver cirrhosis) or >30% (for patients without liver fibrosis and cirrhosis) is another prerequisite for surgical resection. Patients with hepatic impairment require preservation of greater FLR. Additional preoperative evaluation methods include measurement of liver stiffness and the degree of portal hypertension.
3. The preferred treatment for CNLC stage Ia, Ib, and IIa HCC with adequate liver reserve function is surgical resection. In patients with CNLC stage IIb and IIIa HCC, there remains a possibility for certain individuals to derive benefits from surgical resection following the MDT assessment.
4. Techniques for controlling hepatic inflow (via the hepatic artery and portal vein) and outflow (through the hepatic vein) are routinely employed during hepatectomy. The application of preoperative 3D visualization technology enhances the precision of surgical procedures. Laparoscopic techniques offer significant advantages, including minimized surgical trauma and expedited postoperative recovery. Nevertheless, in cases of large HCCs, multiple HCCs, HCC located in challenging anatomical sites or centrally adjacent to critical ducts, as well as HCCs

associated with severe cirrhosis, it is advisable that the procedure be performed by an experienced surgeon after rigorous selection.

5. Postoperative patients with HCC should undergo meticulous monitoring every 3 months for any alterations in imaging studies (ultrasonography and, if necessary, dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI scans, and Gd-EOB-DTPA-enhanced MRI scans) and oncological markers such as serum AFP, DCP, and a combination of seven microRNAs. Following 2 years of surveillance, the follow-up interval may be appropriately extended to every three to 6 months, with lifelong monitoring recommended.
6. Conversion therapy refers to patients with HCC who are initially unsuitable for surgical resection but may become eligible for surgery following intervention. Intervention strategies include functional FLR conversion and oncological conversion, etc. For HCC patients with insufficient FLR, ALPPS or PVE is recommended in the appropriate population to increase functional FLR in the short term. Notably, ALPPS demonstrates a superior conversion rate compared to PVE. Systemic antitumor therapies and/or local treatments emerged as critical approaches for unresectable or intermediate-to-advanced HCC, as well as for conversion therapy of HCC.
7. Neoadjuvant therapy for HCC involves preoperative locoregional treatment or systemic antitumor therapy for patients who are candidates for surgical resection but are at a high risk of postoperative recurrence and metastasis. The primary goal is to eliminate microscopic lesions, reduce the rate of postoperative recurrence, and prolong survival. However, neoadjuvant therapy has inherent risks, making it crucial to carefully select appropriate patient populations. The treatment regimen should be tailored to meet the specific objectives of neoadjuvant therapy.
8. For patients with HCC who are eligible for surgical resection and exhibit a high risk of postoperative recurrence and metastasis, adjuvant treatments – including antiviral therapy, TACE, HAIC, and radiotherapy – may be associated with a reduction in the rates of postoperative recurrence and metastasis, thereby potentially extending OS.

Liver Transplantation

Indications for LT for HCC

LT is one of the radical treatments for HCC, particularly suited for patients with small HCC and HCC who are not candidates for surgical resection or ablative therapy (evidence level 2, recommendation A). Adherence to the appropriate indications for LT in HCC is crucial. This approach enhances the efficacy of LT, ensures equitable and judicious use of the scarce donor liver resources, and helps address the disparities in prognosis between patients with tumors and those without [210] (evidence level 3, recommendation B).

The Milan criteria, the University of California San Francisco (UCSF) criteria and the United Network for Organ Sharing (UNOS) criteria are widely recognized within the international community for evaluating the eligibility of patients with HCC for LT. In China, however, no standardized criteria have been established; a number of criteria have been proposed by multiple entities and scholars, including the Shanghai Fudan criteria [211], Hangzhou criteria [212], West China criteria [213], and the Sanya consensus [214]. While these frameworks share common factors such as the absence of macrovascular invasion, lymph node metastasis, and extrahepatic spread; the criteria diverge in the classification based on tumor size and number. These domestic criteria broaden the eligibility for LT for HCC, allowing a larger cohort of HCC patients to benefit from LT without significantly compromising the postoperative OS. Nevertheless, multicenter collaborative studies are still necessary to validate their application and to generate higher quality evidence. The expert panel endorses the UCSF criteria, namely: the diameter of a solitary tumor ≤6.5 cm, ≤3 tumors with a maximum tumor diameter ≤4.5 cm and the sum of tumor diameters ≤8.0 cm, with no macrovascular involvement. The fundamental principles and core policies governing human organ allocation and sharing in China include guidelines for LT in cases of HCC. These guidelines specify that recipients with liver cancer may apply for a special case score for early stage HCC, which allows them to achieve a model for end-stage liver disease score of 22 for candidates aged 12 years or older on the waiting list. The possibility exists to renew this special case score every 3 months.

Patients with HCC who fulfill the criteria for LT may undergo bridging therapies to manage tumor progression while awaiting a donor liver, thereby safeguarding their eligibility for LT. However, there is limited evidence regarding whether bridging treatment effectively reduces the likelihood of recurrence [215, 216] (evidence level 2, recommendation B). Currently, locoregional treatments are advocated as bridging therapies, including TACE, yttrium-90 radioembolization, ablation therapy, and stereotactic body radiation therapy (SBRT). Further investigation is warranted to ascertain whether

pretransplantation immune checkpoint inhibition elevates the risk of postoperative rejection and graft loss [217].

HCC patients whose tumor burden exceeds the criteria for LT may qualify by achieving a reduction in tumor load through downstaging therapy. The prognosis of patients with HCC post-LT after successful downstaging therapy is superior to that of those who do not undergo LT [218, 219] (evidence level 2, recommendation B). Recent prospective multicenter studies have further substantiated the feasibility of tumor downstaging prior to LT and its associated survival benefits [220]. Additionally, the process of downstaging can serve as a selection tool to identify LT candidates with favorable tumor biological characteristics. It is crucial to underscore the risk of hepatic decompensation associated with downstaging.

The advancement of surgical techniques has resulted in an increased availability of donor livers. The criteria for living donor LT for HCC can be further broadened [221, 222] (evidence level 4, recommendation C). The prognosis following living LT remains intricately linked to the careful selection of recipient [223–225].

Prevention and Treatment of Post-Transplantation Recurrence and Metastasis

Tumor recurrence and metastasis represent significant concerns following after LT for HCC [226]. Contributing risk factors encompass tumor stage, tumor vascular invasion, preoperative serum AFP level, and the dosing regimen of immunosuppressive therapy. The early withdrawal or absence of postoperative hormone-containing regimens [227], along with dose reduction of calcineurin inhibitors during the early post-transplantation period, is associated with lower rates of tumor recurrence and metastasis [228] (evidence level 3, recommendation B). Furthermore, employing the use of immunosuppressive therapies with mammalian target of rapamycin inhibitors, such as rapamycin and everolimus, following LT has been associated with reduced tumor recurrence and metastasis [229–233] (evidence level 2, recommendation B).

Approximately 75% of cases of tumor recurrence or metastasis following LT occur within the first 2 years posttransplant. When this occurs, the disease typically progresses rapidly, with a median survival time of about 1 year for these patients [234, 235]. Patient survival may be extended through a combination of modifications to immunosuppressive regimens, reoperation, TACE, ablation therapy, radiotherapy, and systemic treatment based on multidisciplinary diagnosis and treatment [236, 237] (evidence level 3, recommendation B). Caution is warranted regarding the use of immune checkpoint inhibitors posttransplantation for HCC [238] (evidence level 4, recommendation C). Compared with standard tacrolimus therapy, everolimus combined with reduced tacrolimus in living donor LT recipients demonstrated similar rates of recurrence and metastasis, but resulted in higher glomerular filtration rates [239]. The most common sites of recurrent and metastasis after LT for HCC are the lungs (approximately 40%) and the liver (33%). Close surveillance post-LT, as well as receiving potentially curative treatment, is associated with improved survival after recurrent and metastases [240].

Summary:

1. LT represents a radical therapeutic strategy for HCC, particularly advantageous for patients with small HCC who exhibit hepatic decompensation and are not candidates for surgical resection and ablation therapy.
2. It is recommended to follow the UCSF criteria as the Chinese criteria for LT indication in cases of HCC.
3. Premature discontinuation or absence of hormone-based treatments, reduced doses of calcineurin inhibitors during the early posttransplant period, and the use of immunosuppressive regimens featuring mammalian target of rapamycin inhibitors like rapamycin and everolimus post-LT are associated with a lower risk of tumor recurrence and metastasis.
4. Locoregional treatment serves as a crucial strategy for downstaging or bridging therapy in LT for HCC.
5. Following tumor recurrence and metastasis post-LT, the disease generally exhibits a rapid progression. Combination therapy on the basis of multidisciplinary diagnosis and treatment is associated with extended survival.

Ablation Therapy

Currently, ablation therapy has been recognized as a radical treatment for small HCC other than surgical resection. Ablation therapy with minimal impact on liver function is associated with similar efficacy to surgical resection in some patients with early stage HCC.

Ablation therapy for HCC is guided by medical imaging technology and targets tumor lesions, directly, and locally killing tumor tissues by physical or chemical methods. Local ablation techniques mainly include RFA, microwave ablation (MWA), PEI, cryoablation (CRA), high-intensity focused US ablation (HIFU), laser ablation (LA), irreversible electroporation (IRE). Local

ablation is often performed under the guidance of US, which is easy to use, provides real-time results, and is highly efficient. CT and MRI may be used for observing and guiding the ablation therapy for lesions that are invisible on conventional US. CT or MRI guidance may also be used in the ablation of metastases in the lung, adrenal glands, and bones.

Ablation can be performed using percutaneous, laparoscopic, laparotomic, or endoscopic approaches. Percutaneous ablation is cost-effective, easy to perform, and minimally invasive method for most small HCC lesions. However, there are high risks associated with ablation for subcapsular HCC, particularly in lesions protruding beyond the liver capsule. For HCC located at sites that are difficult to visualize using imaging technology or considered high risk for percutaneous ablation (such as close to the heart, diaphragm, gastrointestinal tract, or gallbladder), ablation by laparoscopic, laparotomic, or water isolation approaches may be considered.

Ablation therapy is recommended for patients with CNLC stage Ia HCC and some patients with stage Ib HCC (i.e., solitary tumor ≤5 cm in diameter; or 2–3 tumors ≤3 cm in diameter). Curative outcomes can be obtained in patients with no invasion of blood vessels, bile ducts, or adjacent organs, or distal metastasis, and with Child-Pugh class A/B [134, 241–245] (evidence level 1, recommendation A). For patients with inoperable solitary tumors or multiple tumors with a diameter 3–7 cm, ablation therapy combined with resection or TACE yields better outcomes compared to ablation alone [246–250] (evidence level 2, recommendation B).

Commonly Used Ablation Therapy

1. RFA: RFA is a commonly used, minimally invasive ablation method for HCC that is easy to use with good control over the ablation range, requires only a short hospital stays, and has proven efficacy. RFA is particularly suitable for older patients and patients with comorbidities, severe cirrhosis, tumors located in deep positions in the liver or central HCC. For patients with resectable early stage HCC, RFA is associated with similar or slightly lower OS than surgical resection but has a lower incidence of complications and shorter hospital stay [129, 134, 241, 243–245, 251] (evidence level 1, recommendation A). For solitary HCC (particularly central solitary HCC) ≤2 cm in diameter, RFA has similar or superior efficacy to surgical resection [252, 253] (evidence level 2, recommendation A). RFA permits the ablation of an entire tumor while maintaining a sufficient safety margin and minimizing damage to normal liver tissues. Prerequisites for RFA include accurate assessment of the range of tumor infiltration and the identification of satellite lesions before the procedure. Therefore, the importance of multimodal imaging before precise treatment and imaging evaluation pre- and posttreatment are emphasized. Contrast-enhanced US enables accurate assessment of tumor size and shape, determination of the extent of tumor infiltration, and detection of micro and satellite lesions. This provides reliable data to inform ablation protocols aimed at completely inactivating tumors during US-guided ablation.
2. MWA: MWA has been commonly used ablation method in recent years, is not significant different from RFA in terms of local efficacy, complication rates, and long-term survival [254–258] (evidence level 1, recommendation A). MWA is characterized by high efficiency, short ablation duration, and a reduced heat-sink effect compared to RFA. It is especially effective for achieving comprehensive tumor ablation in challenging scenarios, such as tumors adjacent to blood vessels, those with a rich blood supply, and larger tumors. Establishing a temperature monitoring system is crucial. It helps regulate parameters like power and duration, defines the effective thermal field range, protects surrounding tissues from thermal damage, and enhances the overall safety of the MWA procedure. The selection of MWA or RFA should be based on the size and position of tumors.
3. PEI: PEI has proven efficacy against tumors ≤2 cm in diameter, with similar long-term efficacy to RFA despite having a higher local recurrence rate than RFA for tumors >2 cm in diameter [259] (evidence level 2, recommendation B). PEI offers the advantage of safety, making it well-suited for tumors in high-risk locations, such as those near the hepatic hilum, gallbladder, and gastrointestinal tract. However, due to the need for intra-tumoral drug diffusion, repeated PEI sessions with multipoint punctures are often necessary. Consequently, PEI is currently used as an adjuvant to thermal ablation therapies.

CRA: CRA treatment can be performed using percutaneous, laparoscopic, or laparotomic approaches, the same as RFA and MWA. The effectiveness of CRA treatment for HCC ≤2 cm is similar to that of MWA and RFA [260] (evidence level 2, recommendation B).

Basic Technical Requirements

1. The physician must be adequately trained and have sufficient clinical experience to consider the advantages and disadvantages of the various ablation

techniques and determine their suitability for each patient. A thorough evaluation of the patient's general PS, liver function, and coagulation functions, as well as the evaluation of the size, position, and number of tumors, and the relationship to adjacent organs, should be performed prior to ablation. An appropriate puncture tract and ablation plan should be determined, and a postoperative care plan should be formulated to achieve an effective ablation range while ensuring safety.
2. An appropriate imaging guidance (e.g., US or CT) and ablation technique (RFA, MWA, CRA, or PEI) should be selected based on the size and position of the tumor. Multimodal image fusion guidance may be applied when available.
3. Caution should be exercised in the ablation of HCC adjacent to the hepatic hilar region or near the first- and second-order bile ducts to avoid complications such as the bile ducts damage. In such cases, PEI alone or RFA/MWA combined with PEI is safe. If thermal ablation is applied, at least 5 mm should be allowed between the tumor and the first- and second-order hepatic ducts, and low power, short duration, intermittent radiation should be used. The use of temperature monitoring methods is recommended for ablation equipment where available.
4. The range of ablation should cover at least 5 mm beyond the peritumoral liver tissue to ensure a safety margin for complete ablation. For tumors that are ill-defined or exhibit irregular infiltration, it is recommended to extend the ablation zone appropriately, provided that adjacent liver tissues and structures allow for such an approach.

Treatment Recommendations for Tumors 3–5 cm in Size

Several randomized controlled trials and retrospective analyses support surgical resection as a recommended treatment for HCC with a diameter of 3–5 cm [135, 244, 261] (evidence level 1, recommendation A). In clinical practice, the initial treatment should be selected after a thorough consideration of the patient's general PS, liver function, the size, number, and position of tumors, as well as the skill and experience of the physician. Surgical resection is the preferred treatment option for patients who can tolerate hepatectomy, particularly when tumors are located in superficial areas, the peripheral liver, or high-risk sites that are not suitable for ablation. However, the latest multicenter retrospective study in China showed that MWA can achieve OS similar to laparoscopic hepatectomy, but disease-free survival is slightly inferior to resection for solitary HCC with a diameter of 3–5 cm [242]. Ablation therapy, or surgical resection in combination with ablation therapy, is the recommended choices for patients with 2–3 lesions located in different areas or for deeply located or central tumors.

Assessment and Follow-Up after Ablative Therapy for HCC

Dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI, or contrast-enhanced US are recommended for assessing the local response to ablation approximately 1 month postoperatively. Dynamic changes in serum tumor biomarkers should also be monitored. The response to ablation can be categorized as follows [262]: (1) Complete ablation: follow-up imaging with dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI, or contrast-enhanced US shows no enhancement in the ablated area of the tumor in the arterial phase, which indicates complete necrosis of the tumor. (2) Incomplete ablation: follow-up imaging with dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI, or contrast-enhanced US shows local enhancement in the ablated region of the tumor in the arterial phase, which is suggestive of residual tumor tissue. Repeat ablation is suggested for patients with residual tumors after treatment. Ablation therapy should be abandoned and substituted with other treatments if the presence of residual tumors is confirmed after two consecutive ablation sessions. Periodic follow-ups are required after complete ablation. Generally, serum tumor marker testing and imaging examination with dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI, or contrast-enhanced US should be performed every 2–3 months to screen for possible local recurrence and new intrahepatic lesions. Ablation therapy may be employed to manage tumor progression, offering the benefits of minimal invasiveness, safety, and the feasibility of repeated applications.

Adjuvant Therapy for Prevention of Recurrence and Metastasis after Ablation of HCC

Some studies have demonstrated that ablation therapy not only enhances the release of tumor-associated antigens and neoantigens but also stimulates HCC-associated antigen-specific T-cell responses, thereby activating or enhancing the overall antitumor immune response [263–265]. Therefore, the combination of ablation therapy and immunotherapy can produce synergistic antitumor effects [263, 266, 267]. Several relevant clinical studies are currently underway to investigate these effects.

Summary:

1. Ablation therapy is suitable for HCC patients with CNLC stage Ia and some patients with stage Ib (i.e., solitary tumors with a diameter of ≤5 cm or 2–3 tumors with maximum diameter ≤3 cm) to obtain a curative outcome. Ablation combined with TACE or resection may be used for inoperable solitary or multiple tumors with a diameter of 3–7 cm.
2. For HCC with a diameter ≤3 cm, the OS rates of ablation therapy are similar to, or slightly lower than, those of surgical resection, but the complication rate and length of hospital stay are lower compared to surgical resection. For solitary HCC lesion ≤2 cm in diameter, the efficacy of ablation therapy is similar to that of surgical resection, especially for central HCC.
3. No significant differences in local efficacy, incidence of complications, or long-term survival have been reported between MWA and RFA; selection should be based on the size and position of tumors.
4. PEI has similar long-term efficacy to RFA for tumors with a diameter ≤2 cm. The advantage of PEI is its safety and, in particular, PEI is suitable for lesions in high-risk locations such as lesions near the hepatic hilar region, gallbladder, and gastrointestinal tracts. However, multiple and multipoint punctures are required for the intra-tumoral diffusion of the drug.
5. CRA treatment can be performed under laparoscopy or direct vision, guided by US, CT, etc. The effectiveness of CRA for HCC ≤2 cm is similar to that of MWA and RFA.
6. Regular follow-up with dynamic contrast-enhanced CT, dynamic contrast-enhanced MRI scan, contrast-enhanced US, and serum tumor markers after ablation therapy should be performed to evaluate ablation outcomes.

Transarterially Interventional Therapy

There are four techniques categorized by the type of transarterial chemotherapy and embolization: (1) transarterial infusion chemotherapy: chemotherapy drugs are infused through the feeding arteries, including HAIC. Commonly used chemotherapy drugs for this technique include anthracycline, platinum, and fluorouracil. The compatibility and dose should be selected based on the patient's tumor burden, body surface area, liver and kidney function status, blood cell level, PS, previous medication, and comorbidities. The concentration and duration of the perfused drugs should be decided according to the pharmacokinetic characteristics of the chemotherapeutic drugs [268]. (2) TACE: a lipiodol emulsion-containing chemotherapy drugs supplement granular embolic agent (gelatin sponge particles, blank microspheres, and polyvinyl alcohol particles) or drug-eluting microspheres is infused through the feeding arteries. (3) TAE: the feeding arteries of a liver tumor are embolized with granular embolic agents alone. (4) Transarterial radioembolization (TARE): intra-arterially injected radioactive microspheres through the feeding arteries of tumors. TACE is the most commonly used transarterially interventional therapy for HCC [269–274].

Basic Principles for TACE

(1) The procedure should be performed under the guidance of a DSA system. (2) The indications and contraindications must be well understood and strictly followed. (3) Super-selective catheterization of the branches of the feeding arteries before treatment must be ensured. (4) The patient's liver function must be properly reserved. (5) The procedure must be performed in a standardized and personalized manner. (6) Switching to TACE treatment regimen or combining with other treatments such as local ablation, systemic antitumor therapy and radiation therapy and surgery should be considered if the tumor continues to progress after 3–4 sessions of TACE.

Indications for TACE

(1) Patients with CNLC stage IIb and IIIa HCC, which is recommended as the preferred treatment. (2) Patients with CNLC stage Ia, Ib, and IIa HCC who are indicated for surgical resection or ablation therapy but are unable or unwilling to receive these procedures for nonsurgical reasons such as advanced age, inadequate liver function reserve, or high-risk tumor sites. (3) Patients with CNLC stage IIIb HCC who are expected to benefit from controlling intrahepatic tumor growth through TACE treatment. (4) Patients with incomplete obstruction of the main portal vein, or those with abundant compensatory collateral branches resulting from portal vein stenting, even in cases of complete obstruction. (5) Patients with portal hypertension-related bleeding as a result of hepatic artery-portal venous shunt. (6) Patients at high risk of recurrence – characterized by factors such as larger or multiple tumors, combined macroscopic or microscopic tumor thrombosis, palliative surgery, and a failure of postoperative AFP and other tumor markers to return to the normal range – may benefit from adjuvant TACE following surgical resection. (7) Patients with HCC who are initially unresectable but can undergo

preoperative conversion therapy to create opportunities for surgical resection, LT, or ablation therapy. (8) Bridging and downstaging treatments for HCC patients during the waiting period for LT. (9) Patients with spontaneous rupture of HCC.

Contraindications for TACE

(1) Severe liver dysfunction (Child-Pugh class C), including severe jaundice, hepatic encephalopathy, refractory ascites, or hepatorenal syndrome. (2) Serious coagulation dysfunction cannot be corrected. (3) Complete obstruction of the main portal vein by tumor thrombi, and the collateral circulation of the portal vein are insufficiently compensated, and the hepatopetal flow of the portal vein cannot be effectively restored through portal vein angioplasty. (4) The presence of active hepatitis or serious infection that cannot be effectively controlled. (5) Tumor diffuse or distant extensive metastases with an expected survival <3 months. (6) Patients with ECOG PS score >2, cachexia or multiple organ failures. (7) Tumor burden ≥70% of total liver volume (Note: not absolutely contraindicated, e.g., fractionated embolization may be considered in the case of basically normal liver function). (8) Significant reduction in peripheral white blood cell (WBC) and platelet, with a WBC level $<3.0 \times 10^9$/L and a platelet level $<50 \times 10^9$/L (Note: not absolutely contraindicated, e.g., chemotherapy-induced myelosuppression should be excluded in patients with hypersplenism). (9) Renal insufficiency (blood creatinine >176.8 μmol/L or blood creatinine clearance rate <30 mL/min). (10) Patients with severe allergies to iodine contrast media.

Procedure and Classification for TACE

1. Arteriography: The success of TACE is based on comprehensive and standardized arteriography. Hepatic arteriography is commonly performed via femoral artery (or radial artery) using the Seldinger technique and the catheter is usually placed in the celiac or common hepatic artery for DSA imaging [275, 276]. The acquisition of images should include arterial, hepatic parenchymal and venous phases to evaluate the location, size, number and feeding arteries of tumors. The arteriography manifestations should be carefully analyzed in combination with preoperative imaging. If blood vessels in the liver area are sparse or absent, or if there is incomplete tumor staining, arteriography of various arteries, such as the superior mesenteric artery, the left gastric artery, the inferior phrenic artery, the renal artery, the internal thoracic artery, the intercostal artery, or the lumbar artery, may be necessary. This can help identify an ectopic origin of hepatic arteries or the presence of extrahepatic collateral feeding vessels supplying the tumor [64]. DSA combined with CBCT is recommended to improve the rate of tumor lesion visualization and the accuracy of determining the branching of the feeding arteries [65] (evidence level 2, recommendation A). For severe cirrhosis, tumor thrombus in the main and first-order branches of the portal vein, indirect portal venography via the superior mesenteric artery or splenic artery is recommended for evaluating portal venous blood flow.
2. Embolic agents may be categorized as conventional-TACE (cTACE) and drug-eluting beads-TACE (DEB-TACE). cTACE refers to the use of lipiodol emulsion-containing chemotherapy drugs as the main method of embolization followed by gelatin sponge particles, blank microspheres, or polyvinyl alcohol. First, a fraction of the chemotherapy drug is infused over a period of ≥20 min, followed by embolization with the emulsion mixture consisting of the chemotherapy drugs and lipiodol. The lipiodol and chemotherapeutic drugs should be fully emulsified. The single dose of lipiodol is usually 5–20 mL and should not exceed 30 mL. The endpoint for treatment is determined by the presence of dense lipiodol accumulation within the tumor region and the fluoroscopic visualization of small portal vein branches encircling the tumor. Granular embolic agents are used after embolization with lipiodol emulsion. The embolization of normal liver tissues as a result of agent reflux or the entry of the agents into nontarget organs should be avoided. DEB-TACE refers to the embolization with mainly drug-eluting microsphere loaded with chemotherapeutic drugs (usually loaded with anthracycline chemotherapeutic agents). The drug-eluting microspheres can embolize the feeding arteries to HCC lesions, resulting in ischemia and necrosis of the tumor. Meanwhile, as a carrier of chemotherapy drugs, allowing for sustained and stable release of the drugs to maintain a high locoregional concentration around the tumor. Different size microspheres should be selected according to the tumor size, blood supply, and the therapeutic purpose, with 100–300 μm and 300–500 μm being most commonly used. The injection speed of 1 mL/min is recommended for DEB-TACE. Attention must be given to the redistribution of microspheres post-embolization to ensure complete embolization of the

distal tumor-feeding arteries. This process should aim to preserve the proximal blood supply branches near the tumor while minimizing damage to the surrounding healthy liver tissue due to potential microsphere reflux [277, 278]. There is no significant difference in OS between cTACE and DEB-TACE, but DEB-TACE is more advantageous in terms of objective tumor response [279] (evidence level 1, recommendation B).

3. Precision TACE: Precision TACE is advocated to reduce the differences in the efficacy of TACE due to the heterogeneity of tumors. Precision TACE includes: (1) standardized arteriography; (2) superselective catheterization using a microcatheter to the branch of the tumor-feeding artery for embolization [280]; (3) the use of CBCT to assist the precise catheterization of the target vessel and monitoring of the efficacy after embolization during the procedure is recommended [281]; (4) appropriate combination of embolization materials, including iodized oil, gelatin sponge particles, blank microspheres, and drug-eluting microspheres [282]; (5) different embolization endpoints should be used according to the patient's tumor status, PS, liver function, and therapeutic objectives. It is crucial to determine individualized TACE goals before treatment. For HCC confined to a liver segment or tumors that are less than 5 cm in diameter, the objective was to achieve complete devascularization of the tumor and, if possible, visualize the small branches of the peripheral portal vein. This approach aims to achieve the desired double embolization effect, involving both the hepatic artery and the portal vein [283, 284]. For massive HCC, it is necessary to achieve maximum devascularization of the tumor depending on the patient's liver function, PS, and patency of the portal vein. Fractionated TACE can be recommended for patients with tumors involving the entire liver and a high tumor burden. The treatment approach should initially target tumors within liver lobes that have a high tumor burden. Subsequently, any remaining tumors should be treated at an interval of 2–4 weeks, depending on the recovery of the patient's PS and liver function. This strategy aims to minimize liver function damage and enhance the safety of TACE treatment.

Common Adverse Effects and Complications of TACE

Post-embolization syndrome is the most common adverse effect associated with TACE, which mainly manifests as fever, pain, nausea, and vomiting. The fever and pain experienced by patients are primarily due to ischemia of the tumor and surrounding hepatic tissues resulting from hepatic artery embolization, while nausea and vomiting are mainly side effects of chemotherapy. Additionally, other common adverse effects may include transient liver function abnormalities, renal impairment, and myelosuppression. Adverse effects after TACE treatment usually last 5–7 days, and most patients can fully recover after symptomatic treatment. Complications were as follows: acute hepatic and renal impairment; gastrointestinal bleeding; cholecystitis and perforation of the gallbladder; hepatic abscesses and biloma; ectopic embolization of embolic agents (including pulmonary and cerebral embolisms, perforation of the gastrointestinal tract, spinal cord injury, and diaphragm injury).

Evaluation of Response to TACE

The mRECIST and/or EASL criteria are most commonly used to evaluate efficacy of TACE [90, 285]. The evaluation is divided into short-term and long-term efficacy assessments. Short-term efficacy evaluation parameters include ORR, PFS, TTP, etc., and long-term efficacy evaluation parameter is OS. ORR based on the mRECIST criteria is correlated with OS. Patients who exhibit an objective tumor response early in the treatment course, particularly those achieving a complete response, tend to have a more favorable prognosis [286–288] (evidence level 2, recommendation B).

Major Factors Affecting the Efficacy of TACE

(1) Tumor staging; (2) tumor burden; (3) integrity of the tumor capsule; (4) tumor blood supply; (5) tumor pathologic subtype; (6) serum AFP level; (7) degree of liver cirrhosis; (8) liver function status; (9) serum HBeAg status and HBV-DNA level of patients with chronic HBV infection; (10) ECOG PS score; (11) whether combining ablation, molecular targeted therapy, immunotherapy, radiation therapy, and surgical procedures.

Follow-Up and Treatment during the Interval between TACE Sessions

Assessment by dynamic contrast-enhanced CT and/or MRI scan, tumor markers, liver and renal function tests, and routine blood tests are usually recommended 4–6 weeks after the first session of TACE. Repeated sessions of TACE can be performed as on-demand [289, 290], which may be postponed if the imaging examination shows thick lipiodol deposition in the liver tumor, necrosis of tumor tissues, with the absence of tumor enhancement and new lesions. The need and frequency

of subsequent TACE should be determined based on follow-up results, which mainly include the response to previous sessions of treatment, liver function, and changes in the patient's general condition. Follow-up may be performed every 1–3 months or longer time intervals. The response of the liver tumor should be evaluated by dynamic contrast-enhanced CT and/or MRI to determine the need for repeated TACE. For large/huge liver tumors, 3–4 sessions of TACE are often required and TACE in combination with other treatments is recommended to achieve tumor control, improve quality of life, and prolong survival.

TACE-Based Combination Therapy

The focus should be on integrating TACE with other locoregional therapies and/or systemic antitumor therapies to enhance the overall efficacy of TACE treatments [269].

1. Combining TACE with portal vein stenting, especially when augmented with iodine-125 seed strips or iodine-125 seed-embedded portal vein stents, can facilitate the restoration of hepatopetal flow within the portal vein. This approach can alleviate portal hypertension and effectively manage tumor thrombi within the main portal vein [291–293] (evidence level 2, recommendation B). The tumor thrombi in the first-order branches of the portal vein may be treated with Iodine-125 seed strips or Iodine-125 seed implantation via direct puncture [294–296] (evidence level 2, recommendation B).
2. TACE combined with ablation therapy: to improve the efficacy of TACE, an appropriate combination of TACE therapy with ablation therapy is recommended, including RFA, MWA, and CRA [297, 298]. The effect is better than that of ablation alone for single or multiple tumors with a diameter of 3–7 cm in patients with unresectable CNLC stage Ib and IIa HCC [247, 248, 299] (evidence level 2, recommendation B). Currently, there are two clinical approaches of TACE combined with thermal ablation therapy: (1) Sequential ablation: TACE followed by local ablation therapy, separated by an interval of 1–4 weeks; (2) Concurrent ablation: local ablation therapy is performed during TACE, which results in significantly improved clinical efficacy and reduced hepatic impairment [297, 300].
3. TACE combined with external radiotherapy: TACE combined with external radiotherapy can improve the local tumor response rate. The combination therapy is commonly used for limited large HCC after TACE, CNLC stage III HCC accompanied by portal or inferior vena cava tumor thrombus and extrahepatic metastasis [301, 302] (evidence level 2, recommendation B).
4. TACE combined with surgical treatment: Bridging, downstaging, or conversion therapy prior to surgical treatment: TACE can effectively reduce tumor burden, potentially transforming patients initially beyond liver transplant criteria to meet the criteria, thereby improving LT rates and reducing the rates of post-transplant recurrence and metastasis. Patients who received LT after downstaging therapy had OS similar to those within the criteria [218, 303–305] (evidence level 2, recommendation B). TACE can create potential opportunity for surgical resection in patients with initially unresectable HCC, which can be achieved a long-term survival benefit [184]. TACE in combination with locoregional treatments such as HAIC and radiotherapy, as well as targeted/immunotherapy, can improve success rates of conversion therapy [306, 307] (evidence level 2, recommendation B).

   Adjuvant TACE for postsurgical patients with high risk factors of recurrent and metastases: adjuvant TACE can promptly detect residual or recurrent HCC lesions after surgery and provide effective treatment. Postoperative adjuvant TACE is recommended for patients with high risk factors for postoperative recurrence, including preoperative tumor rupture, diameter >5 cm, multiple lesions, MVI/visual tumor thrombus, positive surgical margins, poor differentiation, and postoperative tumor marker levels that have not been declined to normal range after surgery [197, 198, 308–310] (evidence level 1, recommendation A).
5. TACE combined with HAIC: for HCC patients with portal vein tumor thrombus (especially in the main of the portal vein), obvious or extensive arteriovenous fistulae that cannot be well embolized by TACE, large tumors, and who are resistant to multiple TACE treatments, the efficacy of TACE can be further improved by TACE combined with mFOLFOX-HAIC [311, 312] (evidence level 3, recommendation B).
6. TACE combined with systemic antitumor therapy: for patients with moderate or high tumor burden and resistance to or failure of TACE therapy, the combination therapy with molecularly targeted drugs should be used as early as possible [313, 314] (evidence level 2, recommendation B). Currently, the efficacy of TACE combined with molecularly targeted drugs is superior to that of molecularly targeted agent alone [187, 315] (evidence level 2, recommendation B). The use of TACE in combination with targeted

and immunotherapy can prolong the survival of patients with intermediate-to-advanced HCC, with better efficacy than TACE treatment alone [316–318] (evidence level 2, recommendation A). Recently, two phase 3, randomized, placebo-controlled trials – LEAP-012 [319] (TACE + lenvatinib+ pembrolizumab vs. TACE + dual placebo) and EMERALD-1 [189] (TACE + durvalumab + bevacizumab vs. TACE + placebo) – have demonstrated statistically significant improvements in the primary endpoint of median PFS for intermediate-stage and unresectable HCC patients eligible for embolization (LEAP-012: mPFS = 14.6 vs. 10.0 months, HR = 0.66; EMERALD-1: mPFS = 15.0 vs. 8.2 months, HR = 0.77). These findings suggest that TACE combined with PD-L/PD-L1 inhibitors and molecular targeted agents has the potential to improve clinical outcomes compared to TACE alone.

7. TACE combined with antiviral therapy: antiviral therapy should be actively performed in combination with TACE in HCC patients with a history of HBV/HCV infection [320–322] (evidence level 1, recommendation A).

Hepatic Arterial Infusion Chemotherapy

As a form of transarterial interventional therapy, HAIC has not yet formed a unified treatment protocol. In recent years, scholars in China have used mFOLFOX-based infusion regimen to improve the efficacy of HAIC in patients with advanced HCC [323]. Studies have shown that mFOLFOX-HAIC is more effective than sorafenib in HCC patients who are resistant to multiple TACE treatments demonstrate portal vein tumor thrombus and extrahepatic oligometastases [185, 186, 324] (evidence level 1, recommendation A). In addition, HAIC combined with sorafenib is more effective than sorafenib alone in patients with HCC with portal vein tumor thrombus (CNLC stage IIIa) [186] (evidence level 1, recommendation A).

Summary:

1. TACE is the most commonly used intra-arterially interventional therapy for HCC, mainly for patients with CNLC stage IIb, IIIa and selected patients with stage IIIb HCC.
2. TACE must be administered based on standardized and individualized regimens, and precision TACE is advocated to minimize outcome variability attributed to tumor heterogeneity.
3. HCC with thrombus in the main or first-order branch of the portal vein may be treated with portal vein stenting in combination with Iodine-125 seed strips or Iodine-125 seed portal vein stenting or Iodine-125 seed implantation via direct puncture.
4. TACE in combination with ablation therapy, radiation therapy, surgery, HAIC, molecularly targeted drugs, immunotherapy and antiviral therapy should be advocated to further improve the efficacy of TACE.
5. mFOLFOX-HAIC is superior to sorafenib in HCC patients who are resistant to multiple TACE treatments demonstrate portal vein tumor thrombus and extrahepatic oligometastases.

Radiation Therapy

Radiation therapy (abbreviated as radiotherapy) is categorized into external radiotherapy and internal radiotherapy. External radiotherapy involves directing beams of photons or particle radiation from a radiotherapy device at the tumor, originating from outside the body. Internal radiotherapy, on the other hand, involves implanting radionuclides directly into the tumor via natural body tracts or percutaneous needle insertion. HCC is known to be sensitive to radiation, and a favorable tumor response can be achieved with moderate doses of radiotherapy.

External Radiotherapy

Indications for External Radiotherapy

(1) In patients with CNLC stage Ia, and a proportion of patients with stage IIb HCC, stereotactic body radiation therapy may be considered a viable alternative treatment option when surgical resection or local ablation therapy are not clinically indicated, or if patients decline invasive treatment [325–332] (evidence level 2, recommendation B). (2) In patients with CNLC stage IIa and IIb HCC, TACE in combination with external radiotherapy may be appropriate as there is evidence that TACE in combination with external radiotherapy is associated with an improved local control rate, prolonged survival, and better efficacy than monotherapy with TACE or sorafenib or TACE in combination with sorafenib [301, 327, 333–339] (evidence level 2, recommendation B). (3) In patients with CNLC stage IIIa HCC, preoperative neoadjuvant radiotherapy or postoperative adjuvant radiotherapy for resectable HCC with tumor thrombosis in the portal vein may prolong survival [148, 340, 341] (evidence level 2, recommendation B); for patients with unresectable HCC, palliative radiotherapy or a combination of radiotherapy and TACE, especially pre-TACE radiotherapy, can significantly

extend patient survival [301, 338, 339, 342, 343] (evidence level 2, recommendation B). (4) Patients with CNLC stage IIIb HCC: for a proportion of patients with oligometastatic disease, SBRT may be performed to prolong survival. External radiotherapy may also be used to reduce pain, obstruction, or bleeding caused by lymph node, lung, bone, brain, or adrenal metastasis [302, 344–346] (evidence level 3, recommendation B). (6) Radiotherapy combined with surgery can improve the survival of some HCC patients. External radiotherapy may also be used as a bridging therapy while waiting for LT [347, 348] (evidence level 2, recommendation B). Similar to HCC with resectable portal vein tumor thrombus, preoperative neoadjuvant radiotherapy can be performed for some central HCC (closely adjacent to the hepatic hilum, less than 1 cm away from the major vascular structures, usually located in Couinaud I, IV, V, VIII, or at the junction of central segments), with better efficacy and tolerability [349]. Radiation therapy combined with interventional or immune checkpoint inhibitor can enable some patients with unresectable HCC to be translated into surgical resection after tumor shrinkage or downstaging [302, 329] (evidence level 3, recommendation B). For patients with HCC exhibiting MVI and narrow surgical margins – defined as ≤1 cm from the tumor – on postoperative pathology, adjuvant radiotherapy following surgery can reduce the risk of local recurrence or distant metastasis and prolong tumor-free survival [350–353] (evidence level 2, recommendation B). (6) Radiotherapy combined with molecular targeted drugs and immune checkpoint inhibitors: for some patients with CNLC stage III or with poor prognostic factors such as MVI, the use of sorafenib or other molecularly targeted drugs for pre- or post-radiotherapy may prolong survival [354]. However, caution should be exercised when using these drugs concurrently [355–359]. The combination of immune checkpoint inhibitors and SBRT for HCC has been suggested to have potential synergistic effects [360–364] (evidence level 3, recommendation C).

Contraindications to External Radiotherapy

External Radiotherapy is not recommended for HCC patients with diffusely distributed intrahepatic lesions or CNLC stage IV HCC.

Principles and Key Points for External Radiotherapy

The principle of administering external radiotherapy for HCC involves a comprehensive consideration of the radiation dose required by the tumor, the dose threshold of the surrounding normal tissues, and the specific radiotherapy technique employed. Key considerations for performing external radiotherapy in HCC include:

1. During preparation of the radiotherapy plan, intrahepatic lesions should be defined by contrast-enhanced CT and, if necessary, a wider range of radiographic images such as MRI should be consulted. The regenerative ability of normal liver tissues should also be considered. During radiotherapy, a proportion of normal liver tissue should be preserved without being irradiated to obtain for proliferation.
2. The irradiation dose is closely related to survival time and local control rate and is predominantly dependent on the tolerable dose for peripheral normal tissues [191, 365]. Favorable outcomes [326] for HCC radiotherapy may be achieved with a recommended irradiation dose of at least 45–60 Gy, delivered in 3–10 fractions (Fx) for SBRT [366], with a bioequivalent dose of radiotherapy of ≥80 Gy (10 Gy is taken as the α/β ratio). The recommended dose is 50–75 Gy for conventional fractionation radiotherapy and 3 Gy × 6 Fx for neoadjuvant radiotherapy for tumor thrombi in the portal vein [148]. When image-guided radiation therapy (IGRT) is feasible, hypofractionated radiotherapy may be utilized for targeted intrahepatic lesions, tumor thrombi, or extrahepatic metastases in lymph nodes, lung, and bone. This approach allows for an increased single dose and a shortened radiation treatment duration, without compromising or potentially enhancing efficacy [367–369]. For non-SBRT hypofractionated external beam radiotherapy, the biologically effective dose can be calculated using established models. In patients with HBV infection, the α/β ratio for hepatocytes is considered to be 8 Gy, while for tumor cells it ranges from 10 to 15 Gy. These ratios may serve as reference values for dose conversion [191, 302, 370].
3. The tolerance radiation dose for nontumorous liver tissues is associated with several factors, including the method of radiotherapy segmentation, Child-Pugh classification, normal liver (liver tumor) volume [371], blood stasis of the gastrointestinal tract, and coagulation function.
4. Radiotherapy techniques for HCC: use of three-dimensional conformal or intensity-modulated radiotherapy, IGRT, or SBRT is recommended. IGRT is superior to non-IGRT techniques [365]. Helical tomographic radiotherapy is suitable for HCC patients with multiple lesions. Respiratory motion is the primary cause of liver tumor movement and deformation during radiotherapy. To mitigate the effects of

respiratory motion, various techniques may be employed, including respiratory gating, real-time tumor tracking, respiration control, and the use of internal target volume determination techniques, such as abdominal compression in conjunction with 4D CT [372].

5. Currently, no high-level clinical evidence is available to support the superiority of proton radiotherapy compared with photon radiotherapy in terms of survival rate in patients with HCC [327].

Evaluation of Response to Radiotherapy

The EASL or mRECIST criteria can be used to evaluate the efficacy of radiotherapy for HCC [327]. Contrast-enhanced CT and/or dynamic contrast-enhanced MRI scans are commonly used to evaluate the efficacy of HCC after radiotherapy. In the early period (3 weeks-3 months) after radiotherapy, the tumor is mostly coagulative necrosis, with slow shrinkage, and the efficacy evaluation is mostly stable [373]. In the late period of radiotherapy (about 3–6 months later), the tumor necrosis is gradually absorbed, showing a significant reduction in tumor size, with an average reduction of at least 36%. The enhancement in the arterial phase is significantly reduced or non-enhanced.

Major Complications of External Radiotherapy

Radiation-induced liver disease (RILD) is a key dose-limiting complication of external radiotherapy for HCC and can be divided into typical and atypical RILD: (1) Typical RILD: Typical RILDs present as increased alkaline phosphatase >2 times the upper limit of normal (ULN), jaundice-free ascites, and hepatomegaly. (2) Atypical RILD: Atypical RILDs present as alkaline phosphatase >2 times the ULN, ALT >5 times the ULN or the pretreatment level, and a reduction of ≥2 points in Child-Pugh Score, but with the absence of hepatomegaly and ascites. Diagnosis of RILD must exclude clinical symptoms and liver dysfunction caused by progression of liver tumors, virus activation or drug toxicities [302, 371].

Focal radiation-induced liver injury observed in clinical imaging corresponds to the area of the liver exposed to a higher dose of radiation. On this basis, RILD may only be diagnosed when RILD-related clinical symptoms or abnormal laboratory indicators, and clinical interventions are required. In the early stage, focal radiation-induced liver injury is mostly characterized by intrahepatic well-defined low-density area on CT, low signal on T1-weighted imaging/high signal on T2-weighted imaging on MRI, with decreased enhancement in the arterial phase on T1-weighted imaging [373]. In advanced stage, enhanced CT or MRI showed localized atrophy, heterogeneous enhancement in the arterial and delayed phases of the lesion, which exited slowly [374–377]. It is necessary to differentiate between local recurrence or progression during follow-up.

Proton Beam Radiotherapy and Internal Radiotherapy

Proton beam radiotherapy (PBT) has similar efficacy for postoperative recurrent or residual HCC lesions (≤2 lesions each <3 cm in size) to that of RFA [378, 379] (evidence level 2, recommendation B).

Internal radiotherapy is a method of local treatment for HCC, including Y-90 microsphere therapy, iodine-131 monoclonal antibody, radioactive lipiodol, iodine-125 seed implantation [345, 346]. Sequential treatment with iodine-131-metuximab following RFA for HCC is associated with a reduced rate of local recurrence after RFA treatment and improved patient survival [380] (evidence level 2, recommendation B). Particle implantation techniques include interstitial implantation, portal vein implantation, inferior vena cava implantation, and bile duct implantation. Strontium chloride (89-SrCl2) emits β rays and can be used for the targeted treatment of bone metastasis from HCC [381] (evidence level 3, recommendation C).

Summary:

1. For patients with CNLC stage IIIa HCC combined with resectable portal vein tumor thrombus, preoperative neoadjuvant radiotherapy or postoperative adjuvant radiotherapy may be performed to prolong survival. For patients with unresectable HCC, palliative radiotherapy, or a combination of radiotherapy and TACE, etc., can be adopted to prolong patient survival.
2. For patients with CNLC stage IIIb HCC, external radiotherapy may be used to reduce pain, obstruction, or bleeding caused by lymph node, lung, bone, brain, or adrenal metastases.
3. Liver tumor irradiation dose: the general recommended radiation dose is ≥45–60 Gy in 3–10 Fx for SBRT, and 50–75 Gy for conventional fractionation radiotherapy. The irradiation dose is closely related to the survival of patient. Hypofractionated radiotherapy may be adopted to treat some intrahepatic lesions and extrahepatic metastases with increased single dose and shortened radiation treatment duration.
4. The tolerated radiation dose of peripheral normal liver tissue must account for the radiotherapy

segmentation method used, Child-Pugh classification, normal liver (liver-tumor) volume, blood stasis of the gastrointestinal tract, and coagulation function.
5. IGRT is superior to 3D conformal or intensity-modulated radiotherapy. SBRT must be performed under the guidance of IGRT.
6. Internal radiotherapy is a method for the local treatment of HCC.

Systemic Therapy

Systemic therapy primarily encompasses antitumor therapy, which include molecular targeted therapies, immune checkpoint inhibitors, chemotherapy, and traditional Chinese herbal medicine. Additionally, it addresses the underlying conditions associated with HCC, such as antiviral treatments, liver-protective and choleretic therapies, as well as supportive symptomatic management.

Due to the insidious onset of liver cancer, fewer than 30% of liver cancer patients are candidates for radical curative treatment at the time of initial diagnosis. Systemic antitumor therapy plays a crucial role in managing intermediate-to-advanced HCC. This therapeutic approach can effectively control disease progression and extend patients survival, with some individuals achieving partial or complete responses. The indications for systemic antitumor therapy in unresectable intermediate-to-advanced HCC include: (1) HCC patients with CNLC stage IIIa and IIIb; (2) patients with CNLC stage IIb who are not suitable for surgical resection or TACE; (3) patients exhibiting resistance to TACE or experiencing failure following TACE.

First-Line Systemic Antitumor Therapy

1. Combination of Atezolizumab and Bevacizumab: This combination therapy has been approved for patients with unresectable HCC who have not received prior systemic therapy (evidence level 1, recommendation A). Results of the IMbrave150 global multicenter Phase III study [382] showed that the median OS and PFS were significantly longer in the atezolizumab and bevacizumab combination group compared to the sorafenib group, with a 34% lower risk of death and a 35% lower risk of disease progression. A significant clinical benefit in the combination therapy group was also observed in a Chinese patient subgroup, with a 47% lower risk of death and a 40% lower risk of disease progression compared with sorafenib. In addition, the combination therapy delayed patient-reported median time to deterioration of quality of life. Common adverse effects included hypertension, proteinuria, abnormal liver function, diarrhea, and decreased appetite.
2. Combination of Sintilimab and a Bevacizumab biosimilar (IBI305): This combination therapy has been approved in China as a first-line treatment for patients with unresectable or metastatic HCC who have not received prior systemic antitumor therapy (evidence level 1, recommendation A). The results of the domestic multicenter phase III study ORIENT-32 [383] showed that sintilimab in combination with a bevacizumab biosimilar (IBI305) had significantly better efficacy than sorafenib, with a 43% decreased risk of death and a 44% decreased risk of disease progression. The combination regimen also had a favorable safety profile, with the most common adverse effects being proteinuria, thrombocytopenia, elevated glutamate transaminase, hypertension, and hypothyroidism.
3. Combination of Apatinib and Camrelizumab: The combination therapy has been approved in China as a first-line treatment for patients with unresectable or metastatic HCC (evidence level 1, recommendation A). The results of the international multicenter phase III CARES-310 study [384] showed that apatinib combined with camrelizumab had a 38% lower risk of death and a 48% lower risk of disease progression compared with sorafenib monotherapy group. Grade ≥3 adverse events in the combination therapy group were mainly hypertension, hand-foot syndrome, and elevated transaminases.
4. Donafenib: Donafenib has been approved in China for patients with unresectable HCC who have not received prior systemic antitumor therapy (evidence level 1, recommendation A). Compared to sorafenib, donafenib can significantly prolong median OS in patients with advanced HCC, with a 17% decreased risk of death. Median PFS was similar in the donafenib and sorafenib groups, but donafenib had favorable safety and tolerability profiles [385]. The most frequent adverse effects were hand-foot skin reactions, elevated glutamine transaminase, elevated total bilirubin, decreased platelets, and diarrhea.
5. Lenvatinib: Lenvatinib is indicated for patients with unresectable advanced HCC with liver function Child-Pugh class A (evidence Level 1, recommendation A). The phase III, randomized, non-inferiority REFLECT trial [386] showed that the OS for lenvatinib was non-inferior to that of sorafenib, and met the non-inferiority endpoint (HR: 0.92, 95% CI:

0.79–1.06). The median PFS was significantly better in the lenvatinib group than sorafenib group, with a 34% lower risk of disease progression and a higher ORR. Common adverse effects included hypertension, proteinuria, diarrhea, decreased appetite, fatigue, hand-foot syndrome, and hypothyroidism.

6. Tislelizumab: Tislelizumab has been approved in China as a first-line therapy for patients with unresectable or metastatic HCC (evidence level 1, recommendation A). Results of the RATIONALE-301 global multicenter phase III study [387] showed that the prespecified primary study endpoint was met with a non-inferiority OS (HR: 0.85, 95% CI: 0.71–1.02) and a 15% lower risk of death compared to sorafenib. Common adverse effects included elevated glutamine aminotransferase, elevated alanine aminotransferase and elevated total bilirubin.
7. Sorafenib: Sorafenib was the first molecularly targeted drug approved for systemic antitumor therapy of HCC. Numerous clinical studies have demonstrated that sorafenib offers significant survival benefits in patients with advanced HCC across various countries and regions, irrespective of the underlying liver diseases [388, 389] (evidence level 1, recommendation A). Sorafenib can be used in patients with liver function Child-Pugh class A/B. Sorafenib provides a more significant survival benefit in patients with Child-Pugh class A compared to patients with Child-Pugh class B [390]. Efficacy assessment and monitoring for toxicity should be performed regularly during treatment. Common adverse events include diarrhea, hand-foot syndrome, rash, hypertension, poor appetite, and fatigue, all of which usually occur within 2–6 weeks after start of therapy. Blood pressure should be closely monitored during treatment, while liver and kidney functions, HBV-DNA, blood counts, coagulation function, and urine protein should be tested regularly. The risk of myocardial ischemia should be taken into account during treatment, and the required monitoring and related tests should be performed, especially in elderly patients.
8. Systemic chemotherapy: The FOLFOX4 regimen has been approved in China as the first-line therapy for locally advanced and metastatic HCC that was unsuitable to surgical resection or locoregional therapy [391, 392] (evidence level 1, recommendation A). In addition, arsenic trioxide has been shown to have a palliative effect on intermediate-to-advanced HCC [393] (evidence level 3, recommendation C). However, hepatorenal toxicity should be monitored and prevented during clinical application.
9. Progress in other first-line systemic anti-tumor therapy: Dual-immune combination therapy has also been successful in the field of first-line treatment of unresectable intermediate-to-advanced HCC. The results of the global phase III HIMALAYA study [394] showed that durvalumab (PD-L1 inhibitor) in combination with tremelimumab (CTLA-4 inhibitor) (STRIDE regimen: a single dose of tremelimumab 300 mg plus durvalumab 1500 mg administered at regular intervals every 4 weeks) provided superior clinical efficacy and a manageable safety profile compared to sorafenib, with a 22% reduction in the risk of death. The STRIDE regimen reduced the risk of death by 29% in Asia patients (excluding Japan) and 34% in HBV-positive patients. The STRIDE regimen has been approved by the US Food and Drug Administration (FDA), the European Union, and Japan for the treatment of patients with unresectable HCC. However, it has not yet been approved for the first-line treatment for unresectable HCC in China. The phase III, randomized CheckMate 9DW trial [395] demonstrated that the combination of nivolumab and ipilimumab (nivolumab 1 mg/kg + ipilimumab 3 mg/kg Q3W [up to 4 cycles] followed by nivolumab 480 mg Q4W) resulted in a 21% lower risk of death compared to the monotherapy groups treated with lenvatinib or sorafenib. The findings from the CheckMate 9DW trial support the use of nivolumab in conjunction with ipilimumab as a potential new first-line therapeutic option for unresectable HCC.

In addition, immune checkpoint inhibitor monotherapy had also made progress in the first-line treatment for HCC. In the HIMALAYA study [394], durvalumab met the pre-specified non-inferiority endpoints compared with sorafenib for the first-line treatment of advanced HCC, with a 14% reduction in the risk of death. Currently, multiple immune checkpoint inhibitor/dual-immune combination anti-angiogenic drugs [396], immune checkpoint inhibitors combined with chemotherapy, and the combination of multiple immune checkpoint inhibitors for the first-line treatment of unresectable HCC are underway, and the results are promising.

Second-Line Systemic Antitumor Therapy

1. Regorafenib: Regorafenib has been approved for use in the treatment of HCC patients who have been previously treated with sorafenib (evidence Level 1,

recommendation A). The international multicenter phase III RESORCE study [397] evaluated the efficacy and safety of regorafenib in HCC patients who had disease progression after sorafenib treatment. The results showed those patients in the regorafenib group had a significant 37% decrease in the risk of death and a 54% decrease in the risk of disease progression compared to placebo. Common adverse effects included hypertension, hand-foot skin reactions, malaise and diarrhea. The adverse effects are similar to those of sorafenib, making regorafenib unsuitable for patients who are intolerant to sorafenib.

2. Apatinib: Apatinib mesylate is a novel small-molecule targeted drug independently developed in China and has been approved for using as monotherapy in patients with advanced HCC who have failed or are intolerant to at least one line of prior systemic antitumor therapy (evidence level 1, recommendation A). Results from a phase III clinical study of apatinib as a second-line treatment for advanced HCC in China [398] showed that compared to placebo, apatinib significantly prolonged the median OS of patients with advanced HCC receiving second or later-line therapy, with a 21.5% reduction in the risk of death and a 52.9% reduction in the risk of disease progression. Common adverse effects include hypertension, proteinuria, leukopenia, and thrombocytopenia. Patients should be closely followed up for adverse effects during the course of apatinib treatment, and necessary dose adjustments need to be made according to patient tolerance.
3. Ramucirumab: Ramucirumab has been approved for the treatment of patients with HCC who have previously received sorafenib therapy and have an AFP ≥400 ng/mL (evidence level 1, recommendation A).The results of the REACH-2 study [399] showed that, in patients with advanced HCC who had failed prior sorafenib therapy and had an AFP ≥400 ng/mL, ramucirumab significantly improved OS and PFS compared to placebo, with a 29.0% reduction in the risk of death and a 54.8% reduction in the risk of disease progression. The results of the REACH-2 China study [400] demonstrated that ramucirumab reduced the risk of death by 14.6% and the risk of disease progression by 51.2% compared to placebo in Chinese HCC patients previously received sorafenib and with an AFP ≥400 ng/mL. Common adverse effects include fatigue, peripheral edema, hypertension, and decreased appetite.
4. Pembrolizumab: Pembrolizumab monotherapy has been approved in China for the treatment of patients with HCC who have previously received sorafenib or oxaliplatin-containing chemotherapy (evidence level 1, recommendation A).The KEYNOTE-394 study [401] evaluated the efficacy of pembrolizumab versus placebo, when combined with best supportive care, in Asian patients with advanced HCC who had previously received sorafenib or oxaliplatin-containing chemotherapy, with 80% of the participants being from China. The results showed a 21% lower risk of death and a 26% lower risk of disease progression in the pembrolizumab group compared to placebo. Common adverse effects include elevated aspartate aminotransferase, elevated alanine transaminase, rash, pruritus, and elevated blood bilirubin.
5. Camrelizumab: Camrelizumab has been approved for the treatment of patients with advanced HCC who have previously received sorafenib therapy and/or oxaliplatin-containing systemic chemotherapy (evidence level 3, recommendation B). The results of a phase II clinical study [402] of camrelizumab in Chinese patients with HCC who had previously received systemic antitumor therapy showed an ORR of 14.7%, a 6-month OS rate of 74.4%, and a 12-month OS rate of 55.9%. The common adverse effects include reactive capillary hyperplasia, elevated aspartate aminotransferase, elevated alanine transaminase, hypothyroidism, and malaise. Several clinical studies have shown [403, 404] that the incidence of reactive capillary hyperplasia was significantly reduced with the combination of camrelizumab and apatinib.
6. Tislelizumab: Tislelizumab has been approved for the treatment of patients with advanced HCC who have previously received systemic therapy with sorafenib or lenvatinib, or oxaliplatin-containing chemotherapy (evidence level 3, recommendation B). Results from a global, multicenter, phase II study (RATIONALE-208) [405] evaluating the efficacy and safety of tislelizumab in patients with unresectable HCC who have previously received at least one systemic therapy reported a median PFS of 2.7 months and a median OS of 13.2 months. The ORR for the total population was 13%. The safety profile was favorable, with the main adverse events comprising elevated glutamine transaminase, elevated glutamine aminotransferase, weakness, pruritus, and hypothyroidism.
7. Other Second-Line Systemic Anti-Tumor Therapy Regimens: FDA has approved nivolumab in combination with ipilimumab for patients with HCC who have disease progression after prior treatment with

sorafenib, or patients who are intolerant to sorafenib [406], and cabozantinib for HCC patients who have progressed after first-line systemic antitumor therapy [407]. Combination regimens of immune checkpoint inhibitor therapy with targeted agents, chemotherapeutic agents, locoregional therapy, and dual-targeted immune checkpoint inhibitors (PD-1/CTLA-4 bispecific antibody and PD-1/VEGF bispecific antibody) for second-line treatment of HCC are also being explored.

Selection of Systemic Antitumor Therapy

According to the results of the IMbrave150, ORIENT-32, and CARES-310, atezolizumab combined with bevacizumab, sintilimab combined with bevacizumab biosimilar, camrelizumab combined with apatinib have better short-term and long-term efficacy than sorafenib monotherapy in the first-line treatment of liver cancer, and can be used as the preferred options for the first-line therapy. The guidelines and consensus such as AASLD/NCCN/JSH updated in 2023 [408–410] recommend atezolizumab combined with bevacizumab or durvalumab combined with tremelimumab (STRIDE regimen) as preferred first-line systemic antitumor therapy options. STRIDE regimen has been approved for first-line treatment of HCC in the European Union, the United States, and Japan, with the advantage of not increasing the risk of bleeding caused by portal hypertension. However, this regimen has not yet been approved in China.

Gastroscopy is recommended prior to treatment with atezolizumab combined with bevacizumab or sintilimab combined with bevacizumab biosimilars, to assess for high-risk varices or other gastrointestinal bleeding, and to manage according to relevant guidelines, due to the potential risk of bleeding associated with bevacizumab. For patients with gastrointestinal bleeding within the last 6 months and endoscopy showing high-risk bleeding, esophagogastric fundal varices should be adequately treated prior to bevacizumab or alternative treatment plans should be considered. Limited data are available on systemic antitumor therapy in patients with moderate-to-severe hepatic dysfunction (Child-Pugh score of 7 or above), and such patients should use these therapies cautiously under the guidance of physician, with close monitoring of liver function.

The current approval indications for second-line treatment agents are based on phase III clinical studies comparing the agent with placebo after the failure of first-line treatment with sorafenib or systemic chemotherapy. For patients receiving first-line treatment with immune-based combination regiment, immune monotherapy, or other tyrosine kinase inhibitor monotherapy, second-line treatment may choose approved second-line treatment agents or previously unused first-line treatment agents based on the type of disease progression and the specific first-line treatment.

Response Evaluation for Systemic Antitumor Therapy

Currently, RECIST 1.1 is mainly used for response evaluation in patients receiving systemic therapy. mRECIST may be used in combination to evaluate treatment response in patients receiving antiangiogenic molecular targeted therapies. iRECIST may be used to evaluate response in patients receiving treatment with immune checkpoint inhibitors [411].

Traditional Chinese Herbal Medicine

Within the clinical medicine framework that integrates traditional Chinese medicine and Western medicine, characterized by syndrome differentiation for treatment [412], a combined approach of disease and syndrome [413] is employed for the clinical diagnosis and management across various stages of HCC, including early stage, intermediate-to-advanced stage and end-stage. This clinical diagnostic and therapeutic model emphasize the core pathological mechanism of liver cancer known as "the fluctuation of cancer toxicity." It comprehensively utilizes traditional Chinese medicine, including herbal prescriptions, modern Chinese pharmaceuticals, and specialized diagnostic techniques, while synergistically integrating with contemporary medical technologies. This result in a systematic and standardized protocol aimed at collaboratively combating liver cancer, enhancing treatment tolerance, minimizing postoperative complications, preventing recurrence and metastasis, reducing adverse reactions, and ultimately prolonging survival. Based on the staging of liver cancer, traditional Chinese medicine employs a phased approach to treatment.

Modern Chinese Medicine Preparation Therapy

Icaritin (Epimedium) soft capsule is an innovative single-molecule drug derived through the extraction, isolation, and purification of the natural medicinal plant Epimedium, a traditional herbal remedy known for its multiple effects such as tumor growth inhibition and immune regulation. It has received conditional approval in China for treating advanced liver cancer patients who are unsuitable for standard therapies. Specific indications include patients with unresectable liver cancer who are not suitable for or refuse to receive standard

treatment and have not received systemic treatment in the past [414]. Additionally, these patients must meet at least two of the following peripheral blood composite marker criteria: AFP ≥400 ng/mL; TNF-α <2.5 pg/mL; IFN-γ ≥7.0 pg/mL.

In addition to Icaritin soft capsule, National Medical Products Administration has authorized several contemporary Chinese medicinal formulations for the treatment of advanced liver cancer, including Huaier Granules, Eleutherococcus injection, Huazhongin, Kanglait injection, Kang Ai injection, Aidi injection, Hepatic Fule capsule, Jinlong capsule, Crow's gallnut oil soft capsule, Compound Zangcang capsule, and Cidan capsule [415–419], etc. In addition, Huaier granules [202], cinobufacini combined with Jiedu granule [420], and Cidan capsule combined with TACE [421] may serve as adjunctive therapies following surgical resection of HCC.

Modern Chinese Medicine Diagnostics and Therapy

In addition to acupuncture treatment, appropriate Chinese herbal medicine, patent Chinese medicine, external application treatment, Chinese herbal bath, and Chinese herbal steaming treatment can be used according to the condition.

Antiviral Therapy and Other Liver-Protecting Treatment

Hepatitis B surface antigen (HBsAg), hepatitis B surface antibody (anti-HBs), and hepatitis B core antibody (anti-HBc) should be routinely screened for all patients diagnosed with HCC. In the case of patients positive for HBsAg or anti-HBc, quantitative serum HBV-DNA testing should be performed. First-line antiviral therapy is recommended for patients with HBV-associated HCC who are positive for HBsAg, regardless of whether HBV-DNA can be detectable [422]. For HCC patients with previous HBV infection, meaning positive for hepatitis B core antibody (HBcAb) and negative for HBsAg, changes in HBsAg, HBV-DNA, and alanine aminotransferase levels should be closely monitored [423]. The first-line antiviral agents including entecavir, tenofovir disoproxil, tenofovir alafenamide, or tenofovir alafenamide is recommended for patients with HBV-associated HCC [422, 423] (evidence level 1, Recommendation A). For HCC patients with HBV infection, antiviral therapy with oral nucleoside/nucleotide analogs should be performed throughout the entire treatment in addition to antitumor therapy. For patients receiving antiviral therapy with nucleoside/nucleotide analogs, virological, and blood biochemical parameters should be tested every 3–6 months. For patients not receiving antiviral therapy, HBV-DNA, HBsAg and alanine aminotransferase levels should be tested every 3–6 months. Highly sensitive serum HBV-DNA testing is recommended to timely identification of patients with low-level viremia, adjust antiviral therapy drugs, and maximize the suppression of viral replication. Antiviral treatment with DAAs is recommended in patients with HCV-related HCC who are positive for HCV-RNA [423, 424] (evidence level 1, recommendation A). The endpoint of antiviral therapy is the absence of HCV-RNA in serum or plasma by a sensitive assay at 12 weeks after completion of therapy.

Abnormal liver function may occur during the natural course of disease and/or treatment in patients with HCC. Therefore, it is essential to administer hepatoprotective agents in a timely and appropriate manner, ensuring they possess both anti-inflammatory and cholagogic properties, as well as exhibit antioxidative and detoxifying properties. Additionally, they should aid in the repair and protection of hepatocyte membranes. Liver-protecting drugs include magnesium isoglycyrrhizinate injection, diammonium glycyrrhizinate, compound glycyrrhizin, ademetionine, ursodeoxycholic acid, bicyclol, silymarin, reduced glutathione, polyene phosphatidylcholine, and ulinastatin [425], etc.

Symptomatic Supportive Treatment

Individuals with HCC frequently encounter complications such as cirrhosis, splenomegaly, and cytopenia affecting one or multiple blood cell lines as a result of anti-neoplastic therapy. Pharmacologic therapy or blood product transfusions may be considered in these cases. Patients with neutropenia may be administered granulocyte colony stimulating factor (G-CSF), including PEG-rhG-CSF and rhG-CSF, as appropriate [426]. Patients with hemoglobin <80 g/L may be treated with iron, folic acid, vitamin B12, and erythropoietin, depending on the etiology of the anemia. Patients with thrombocytopenia caused by chronic liver disease, recombinant human thrombopoietin or thrombopoietin receptor agonists (e.g., avatrombopag, lusutrombopag) can be used to increase platelet count [427] as needed.

For patients with end-stage HCC, best supportive care should be provided including analgesic treatment, correction of hypoalbuminemia, enhanced nutritional support, blood sugar control in patients with diabetes, and management of complications including ascites, jaundice, hepatic encephalopathy, gastrointestinal

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bleeding, and hepatorenal syndrome. Bisphosphonates or denosumab may be used in patients with bone metastasis. In addition, adequate rehabilitation exercises can increase patient immunity. Meanwhile, psychological interventions for patients should be emphasized to enhance patient confidence in overcoming the disease, transform negative thinking into positive thinking, and let patients enjoy a sense of security and comfort through palliative care while reducing depression and anxiety.

Summary:

1. The indications for systemic antitumor therapy in unresectable advanced HCC: HCC patients with CNLC stage IIIa and IIIb, HCC patients with CNLC stage IIb who are NOT suitable for surgical resection or TACE, and HCC patients who are resistant to TACE or failure TACE.
2. First-line systemic antitumor treatments may be preferred to atezolizumab in combination with bevacizumab, sintilimab in combination with a bevacizumab biosimilar, or apatinib mesylate in combination with camrelizumab. Donafenib, Lenvatinib, tislelizumab, sorafenib FOLFOX4 systemic chemotherapy are still used for the first-line treatment of HCC.
3. In China, second-line systemic antitumor treatments include: regorafenib, apatinib, pembrolizumab, ramucirumab (serum AFP level ≥400 μg/L), camrelizumab, and tislelizumab. For patients receiving first-line treatment with immune-based combination regiment, immune monotherapy, or tyrosine kinase inhibitor monotherapy, second-line treatment may choose approved second-line treatment agents or previously unused first-line treatment agents based on the type of disease progression and the specific first-line treatment.
4. Traditional Chinese medicine prescription can be selected based on the underlying etiology. In addition to Icaritin soft capsule, several modern Chinese medicinal formulations have received approval for the treatment of advanced HCC in China. In addition, Huaier granules may serve as an adjunctive therapy following surgical resection.
5. Antiviral therapy should be performed throughout the entire treatment in addition to antitumor therapy.

Treatment of Spontaneous Rupture of Liver Tumors

Spontaneous rupture of liver tumors is a potentially life-threatening complication associated with HCC. The in-hospital mortality rate for conservative management of a ruptured tumor is extremely high. Nevertheless, this factor does not serve as a definitive predictor of long-term patient survival; surgical resection can confer substantial survival advantages for select patients [428]. Therefore, after successful initial rescue measures have been implemented, a comprehensive assessment of the patient's hemodynamics, liver function, overall health, and the feasibility of tumor resection should be conducted. This evaluation will guide the development of an individualized treatment plan [429–433].

Surgical resection is the first choice in patients with a resectable liver tumor, adequate hepatic reserve function, and stable hemodynamic status [434, 435] (evidence level 3, recommendation B). TAE is a viable option for patients exhibiting compromised liver reserve function and unstable hemodynamics who are not candidates for surgical intervention [436] (evidence level 4, recommendation B).

In situations where a comprehensive assessment of liver function and tumors is hindered by the constraints of emergency conditions, TAE may be initially conducted. Subsequent treatment protocols can then be determined based on follow-up evaluations. Notably, significant survival advantages may be achieved if a secondary surgical resection is undertaken [434] (evidence level 3, recommendation B).

The spontaneous rupture of liver tumors constitutes a significant risk factor for postoperative recurrence and necessitates thorough intraoperative irrigation of the abdominal cavity. Adjuvant therapy following surgery may enhance survival outcomes [428, 437]. Postoperative surveillance should be intensified, and aggressive radical resection might be warranted in patients presenting with postoperative peritoneal metastases [438] (evidence level 3, recommendation C).

Summary:

1. Spontaneous rupture of liver tumors represents a potentially life-threatening complication of HCC. A comprehensive assessment of the patient's hemodynamics, liver function, overall health status, and the feasibility of tumor resection is essential for formulating an individualized treatment strategy.

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Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

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References

1 Han BF, Zheng RS, Zeng HM, Wang S, Sun K, Chen R, et al. Cancer incidence and mortality in China, 2022. J Natl Cancer Cent. 2024;4(1):47–53. https://doi.org/10.1016/j.jncc.2024.01.006

2 Zheng RS, Zhang SW, Zeng HM, Wang S, Sun K, Chen R, et al. Cancer incidence and mortality in China, 2016. J Natl Cancer Cent. 2022;2(1):1–9. https://doi.org/10.1016/j.jncc.2022.02.002

3 Zhou M, Wang H, Zeng X, Yin P, Zhu J, Chen W, et al. Mortality, morbidity, and risk factors in China and its provinces, 1990-2017: a systematic analysis for the Global Burden of Disease Study 2017. Lancet. 2019;394(10204):1145–58. https://doi.org/10.1016/s0140-6736(19)30427-1

4 Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA Cancer J Clin. 2018;68(6):394–424. https://doi.org/10.3322/caac.21492

5 Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. J Clin Epidemiol. 2011;64(4):383–94. https://doi.org/10.1016/j.jclinepi.2010.04.026

6 Balshem H, Helfand M, Schünemann HJ, Oxman AD, Kunz R, Brozek J, et al. GRADE guidelines: 3. Rating the quality of evidence. J Clin Epidemiol. 2011;64(4):401–6. https://doi.org/10.1016/j.jclinepi.2010.07.015

7 Andrews JC, Schünemann HJ, Oxman AD, Pottie K, Meerpohl JJ, Coello PA, et al. GRADE guidelines: 15. Going from evidence to recommendation-determinants of a recommendation's direction and strength. J Clin Epidemiol. 2013;66(7):726–35. https://doi.org/10.1016/j.jclinepi.2013.02.003

8 ASCO guidelines methodology manual [EB/OL]. (2021-09-09) [2023-12-15]. Available from: https://www.asco.org/sites/new-www.asco.org/files/content-files/advocacy-and-policy/documents/Guidelines-Methodology-Manual_0.pdf

9 Zhang BH, Yang BH, Tang ZY. Randomized controlled trial of screening for hepatocellular carcinoma. J Cancer Res Clin Oncol. 2004;130(7):417–22. https://doi.org/10.1007/s00432-004-0552-0

10 Zeng H, Cao M, Xia C, Wang D, Chen K, Zhu Z, et al. Performance and effectiveness of hepatocellular carcinoma screening in individuals with HBsAg seropositivity in China: a multicenter prospective study. Nat Cancer. 2023;4(9):1382–94. https://doi.org/10.1038/s43018-023-00618-8

11 Hou JL, Zhao W, Lee C, Hann HW, Peng CY, Tanwandee T, et al. Outcomes of long-term treatment of chronic HBV infection with entecavir or other agents from a randomized trial in 24 countries. Clin Gastroenterol Hepatol. 2020;18(2):457–67.e21. https://doi.org/10.1016/j.cgh.2019.07.010

12 Fan R, Papatheodoridis G, Sun J, Innes H, Toyoda H, Xie Q, et al. aMAP risk score predicts hepatocellular carcinoma development in patients with chronic hepatitis. J Hepatol. 2020;73(6):1368–78. https://doi.org/10.1016/j.jhep.2020.07.025

13 Fan R, Chen L, Zhao S, Yang H, Li Z, Qian Y, et al. Novel, high accuracy models for hepatocellular carcinoma prediction based on longitudinal data and cell-free DNA signatures. J Hepatol. 2023;79(4):933–44. https://doi.org/10.1016/j.jhep.2023.05.039

14 Xin H, Fan R, Guo Y, et al. Creation of a hospital-community integrated "pyramid" liver cancer screening model to realize early screening, early diagnosis and early treatment of liver cancer. Chin J Liver Dis. 2021;29(4):289–92. https://doi.org/10.3760/cma.j.cn501113-20210408-00174-1

15 Dong Y, Wang WP, Lee WJ, Meloni MF, Clevert DA, Chammas MC, et al. Contrast-enhanced ultrasound features of histopathologically proven hepatocellular carcinoma in the non-cirrhotic liver: a multicenter study. Ultrasound Med Biol. 2022;48(9):1797–805. https://doi.org/10.1016/j.ultrasmedbio.2022.05.005

16 Wang WP, Dong Y, Cao J, Mao F, Xu Y, Si Q, et al. Detection and characterization of small superficially located focal liver lesions by contrast-enhanced ultrasound with high frequency transducers. Med Ultrason. 2017;19(4):349–56. https://doi.org/10.11152/mu-1276

17 Dong Y, Wang WP, Mao F, Zhang Q, Yang D, Tannapfel A, et al. Imaging features of fibrolamellar hepatocellular carcinoma with contrast-enhanced ultrasound. Ultraschall Med. 2021;42(3):306–13. https://doi.org/10.1055/a-1110-7124
18 Fan PL, Xia HS, Ding H, Dong Y, Chen LL, Wang WP. Characterization of early hepatocellular carcinoma and high-grade dysplastic nodules on contrast-enhanced ultrasound. Correlation with histopathologic findings. J Ultrasound Med. 2020;39(9):1799–808. https://doi.org/10.1002/jum.15288
19 Shen YT, Yue WW, Xu HX. Non-invasive imaging in the diagnosis of combined hepatocellular carcinoma and cholangiocarcinoma. Abdom Radiol. 2023;48(6):2019–37. https://doi.org/10.1007/s00261-023-03879-0
20 Han H, Ji Z, Huang B, Mao L, Cao Q, Fan P, et al. The preliminary application of simultaneous display of contrast-enhanced ultrasound and micro-flow imaging technology in the diagnosis of hepatic tumors. J Ultrasound Med. 2023;42(3):729–37. https://doi.org/10.1002/jum.16111
21 Barr RG, Huang P, Luo Y, Xie X, Zheng R, Yan K, et al. Contrast-enhanced ultrasound imaging of the liver: a review of the clinical evidence for SonoVue and Sonazoid. Abdom Radiol. 2020;45(11):3779–88. https://doi.org/10.1007/s00261-020-02573-9
22 Dietrich CF, Nolsøe CP, Barr RG, Berzigotti A, Burns PN, Cantisani V, et al. Guidelines and good clinical practice recommendations for Contrast-Enhanced Ultrasound (CEUS) in the liver- update 2020 WFUMB in cooperation with EFSUMB, AFSUMB, AIUM, and FLAUS. Ultrasound Med Biol. 2020;46(10):2579–604. https://doi.org/10.1016/j.ultrasmedbio.2020.04.030
23 Lee JY, Minami Y, Choi BI, Lee WJ, Chou YH, Jeong WK, et al. The AFSUMB consensus statements and recommendations for the clinical practice of contrast-enhanced ultrasound using sonazoid. J Med Ultrasound. 2020;28(2):59–82. https://doi.org/10.4103/JMU.JMU_124_19
24 Zhao CK, Guan X, Pu YY, Zhou BY, Wang LF, Sun YK, et al. Response evaluation using contrast-enhanced ultrasound for unresectable advanced hepatocellular carcinoma treated with tyrosine kinase inhibitors plus anti-PD-1 antibody therapy. Ultrasound Med Biol. 2024;50(1):142–9. https://doi.org/10.1016/j.ultrasmedbio.2023.09.016
25 Zhou BY, Liu H, Pu YY, Wang LF, Sun YK, Yin HH, et al. Quantitative analysis of pretreatment dynamic contrast-enhanced ultrasound for assessing the response of colorectal liver metastases to chemotherapy plus targeted therapy: a dual-institutional study. Abdom Radiol. 2024;49(2):414–24. https://doi.org/10.1007/s00261-023-04055-0
26 Wen-Ping WANG, Zheng-Biao JI, Yi DONG, et al. Application study of real-time navigation ultrasonography in the diagnosis of small hepatocellular carcinoma. Chin J Med Ultrasound. 2016;13(1):56–60. https://doi.org/10.3877/cma.j.issn.1672-6448.2016.01.014
27 Dong Y, Wang WP, Mao F, Ji ZB, Huang BJ. Application of imaging fusion combining contrast-enhanced ultrasound and magnetic resonance imaging in detection of hepatic cellular carcinomas undetectable by conventional ultrasound. J Gastroenterol Hepatol. 2016;31(4):822–8. https://doi.org/10.1111/jgh.13202
28 Bo XW, Xu HX, Wang D, Guo LH, Sun LP, Li XL, et al. Fusion imaging of contrast-enhanced ultrasound and contrast-enhanced CT or MRI before radiofrequency ablation for liver cancers. Br J Radiol. 2016;89(1067):20160379. https://doi.org/10.1259/bjr.20160379
29 Bo XW, Xu HX, Guo LH, Sun LP, Li XL, Zhao CK, et al. Ablative safety margin depicted by fusion imaging with post-treatment contrast-enhanced ultrasound and pre-treatment CECT/CEMRI after radiofrequency ablation for liver cancers. Br J Radiol. 2017;90(1078):20170063. https://doi.org/10.1259/bjr.20170063
30 Dong Y, Wang WP, Xu Y, Cao J, Mao F, Dietrich CF. Point shear wave speed measurement in differentiating benign and malignant focal liver lesions. Med Ultrason. 2017;19(3):259–64. https://doi.org/10.11152/mu-1142
31 Zhuang Y, Ding H, Zhang Y, Sun H, Xu C, Wang W. Two-dimensional shear-wave elastography performance in the noninvasive evaluation of liver fibrosis in patients with chronic hepatitis B: comparison with serum fibrosis indexes. Radiology. 2017;283(3):873–82. https://doi.org/10.1148/radiol.2016160131
32 Guan X, Chen YC, Xu HX. New horizon of ultrasound for screening and surveillance of non-alcoholic fatty liver disease spectrum. Eur J Radiol. 2022;154:110450. https://doi.org/10.1016/j.ejrad.2022.110450
33 Huang YL, Bian H, Zhu YL, Yan HM, Wang WP, Xia MF, et al. Quantitative diagnosis of nonalcoholic fatty liver disease with ultrasound attenuation imaging in a biopsy-proven cohort. Acad Radiol. 2023;30(Suppl 1):S155–63. https://doi.org/10.1016/j.acra.2023.05.033
34 Hsu CL, Loomba R. From NAFLD to MASLD: implications of the new nomenclature for preclinical and clinical research. Nat Metab. 2024;6(4):600–2. https://doi.org/10.1038/s42255-024-00985-1
35 Kanwal F, Neuschwander-Tetri BA, Loomba R, Rinella ME. Metabolic dysfunction-associated steatotic liver disease: update and impact of new nomenclature on the American Association for the Study of Liver Diseases practice guidance on nonalcoholic fatty liver disease. Hepatology. 2024;79(5):1212–9. https://doi.org/10.1097/HEP.0000000000000670
36 European Association for the Study of the Liver EASLEuropean Association for the Study of Diabetes EASDEuropean Association for the Study of Obesity EASO. EASL-EASD-EASO Clinical Practice Guidelines on the management of metabolic dysfunction-associated steatotic liver disease (MASLD). J Hepatol. 2024;81(3):492–542. https://doi.org/10.1016/j.jhep.2024.04.031
37 Chen Y, Lu Q, Zhu Y, Huang B, Dong Y, Wang W. Prediction of microvascular invasion in combined hepatocellular-cholangiocarcinoma based on preoperative clinical data and contrast-enhanced ultrasound characteristics. Ultrasound Med Biol. 2022;48(7):1190–201. https://doi.org/10.1016/j.ultrasmedbio.2022.02.014
38 Zhang H, Guo L, Wang D, Wang J, Bao L, Ying S, et al. Multi-source transfer learning via multi-kernel support vector machine plus for B-mode ultrasound-based computer-aided diagnosis of liver cancers. IEEE J Biomed Health Inform. 2021;25(10):3874–85. https://doi.org/10.1109/JBHI.2021.3073812
39 Ding W, Wang Z, Liu FY, Cheng ZG, Yu X, Han Z, et al. A hybrid machine learning model based on semantic information can optimize treatment decision for naïve single 3-5-cm HCC patients. Liver Cancer. 2022;11(3):256–67. https://doi.org/10.1159/000522123
40 Liu F, Liu D, Wang K, Xie X, Su L, Kuang M, et al. Deep learning radiomics based on contrast-enhanced ultrasound might optimize curative treatments for very-early or early- stage hepatocellular carcinoma patients. Liver Cancer. 2020;9(4):397–413. https://doi.org/10.1159/000505694
41 Lee YJ, Lee JM, Lee JS, Lee HY, Park BH, Kim YH, et al. Hepatocellular carcinoma: diagnostic performance of multidetector CT and MR imaging-a systematic review and meta-analysis. Radiology. 2015;275(1):97–109. https://doi.org/10.1148/radiol.14140690
42 Liu X, Jiang H, Chen J, Zhou Y, Huang Z, Song B. Gadoxetic acid disodium-enhanced magnetic resonance imaging outperformed multidetector computed tomography in diagnosing small hepatocellular carcinoma: a meta-analysis. Liver Transpl. 2017;23(12):1505–18. https://doi.org/10.1002/lt.24867
43 Marrero JA, Kulik LM, Sirlin CB, Zhu AX, Finn RS, Abecassis MM, et al. Diagnosis, staging, and management of hepatocellular carcinoma: 2018 practice guidance by the American association for the study of liver diseases. Hepatology. 2018;68(2):723–50. https://doi.org/10.1002/hep.29913

44 Vogel A, Cervantes A, Chau I, Daniele B, Llovet JM, Meyer T, et al. Hepatocellular carcinoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Ann Oncol. 2018;29(Suppl 4): iv238–55. https://doi.org/10.1093/annonc/mdy308
45 Omata M, Cheng AL, Kokudo N, Kudo M, Lee JM, Jia J, et al. Asia-Pacific clinical practice guidelines on the management of hepatocellular carcinoma: a 2017 update. Hepatol Int. 2017;11(4):317–70. https://doi.org/10.1007/s12072-017-9799-9
46 Cho ES, Choi JY. MRI features of hepatocellular carcinoma related to biologic behavior. Korean J Radiol. 2015;16(3):449–64. https://doi.org/10.3348/kjr.2015.16.3.449
47 Hwang J, Kim YK, Jeong WK, Choi D, Rhim H, Lee WJ. Nonhypervascular hypointense nodules at gadoxetic acid-enhanced MR imaging in chronic liver disease: diffusion-weighted imaging for characterization. Radiology. 2015;276(1): 137–46. https://doi.org/10.1148/radiol.15141350
48 Huang P, Ni X, Zhou C, Shi Z, Wu F, Xiao Y, et al. Subcentimeter nodules with diagnostic hallmarks of hepatocellular carcinoma: comparison of pathological features and survival outcomes with nodules measuring 1-2 cm. J Hepatocell Carcinoma. 2023;10:169–80. https://doi.org/10.2147/JHC.S401027
49 Zeng MS, Ye HY, Guo L, Peng WJ, Lu JP, Teng GJ, et al. Gd-EOB-DTPA-enhanced magnetic resonance imaging for focal liver lesions in Chinese patients: a multicenter, open-label, phase III study. phase III study. Hepatobiliary Pancreat Dis Int. 2013;12(6): 607–16. https://doi.org/10.1016/s1499-3872(13)60096-x
50 Ichikawa T, Saito K, Yoshioka N, Tanimoto A, Gokan T, Takehara Y, et al. Detection and characterization of focal liver lesions: a Japanese phase III, multicenter comparison between gadoxetic acid disodium-enhanced magnetic resonance imaging and contrast-enhanced computed tomography predominantly in patients with hepatocellular carcinoma and chronic liver disease. Investig Radiol. 2010;45(3):133–41. https://doi.org/10.1097/RLI.0b013e3181caea5b
51 Wang W, Yang C, Zhu K, Yang L, Ding Y, Luo R, et al. Recurrence after curative resection of hepatitis B virus-related hepatocellular carcinoma: diagnostic algorithms on gadoxetic acid-enhanced magnetic resonance imaging. Liver Transpl. 2020;26(6): 751–63. https://doi.org/10.1002/lt.25713
52 Yoo SH, Choi JY, Jang JW, Bae SH, Yoon SK, Kim DG, et al. Gd-EOB-DTPA-enhanced MRI is better than MDCT in decision making of curative treatment for hepatocellular carcinoma. Ann Surg Oncol. 2013;20(9):2893–900. https://doi.org/10.1245/s10434-013-3001-y
53 Rao SX, Wang J, Wang J, Jiang XQ, Long LL, Li ZP, et al. Chinese consensus on the clinical application of hepatobiliary magnetic resonance imaging contrast agent: gadoxetic acid disodium. J Dig Dis. 2019;20(2):54–61. https://doi.org/10.1111/1751-2980.12707
54 Huang P, Zhou C, Wu F, Xiao Y, Qian X, Wang Y, et al. An improved diagnostic algorithm for subcentimeter hepatocellular carcinoma on gadoxetic acid-enhanced MRI. Eur Radiol. 2023;33(4):2735–45. https://doi.org/10.1007/s00330-022-09282-5
55 Renzulli M, Biselli M, Brocchi S, Granito A, Vasuri F, Tovoli F, et al. New hallmark of hepatocellular carcinoma, early hepatocellular carcinoma and high-grade dysplastic nodules on Gd-EOB-DTPA MRI in patients with cirrhosis: a new diagnostic algorithm. Gut. 2018;67(9):1674–82. https://doi.org/10.1136/gutjnl-2017-315384
56 Chang Y, Jeong SW, Young Jang J, Jae Kim Y. Recent updates of transarterial chemoembololization in hepatocellular carcinoma. Int J Mol Sci. 2020;21(21):8165. https://doi.org/10.3390/ijms21218165
57 Kudo M, Kawamura Y, Hasegawa K, Tateishi R, Kariyama K, Shiina S, et al. Management of hepatocellular carcinoma in Japan: JSH consensus statements and recommendations 2021 update. Liver Cancer. 2021;10(3):181–223. https://doi.org/10.1159/000514174
58 Shi J, Lai EC, Li N, Guo WX, Xue J, Lau WY, et al. Surgical treatment of hepatocellular carcinoma with portal vein tumor thrombus. Ann Surg Oncol. 2010;17(8):2073–80. https://doi.org/10.1245/s10434-010-0940-4
59 Chen M, Cao J, Hu J, Topatana W, Li S, Juengpanich S, et al. Clinical-radiomic analysis for pretreatment prediction of objective response to first transarterial chemoembolization in hepatocellular carcinoma. Liver Cancer. 2021;10(1):38–51. https://doi.org/10.1159/000512028
60 Xu X, Zhang HL, Liu QP, Sun SW, Zhang J, Zhu FP, et al. Radiomic analysis of contrast-enhanced CT predicts microvascular invasion and outcome in hepatocellular carcinoma. J Hepatol. 2019;70(6):1133–44. https://doi.org/10.1016/j.jhep.2019.02.023
61 Chong HH, Yang L, Sheng RF, Yu YL, Wu DJ, Rao SX, et al. Multi-scale and multi-parametric radiomics of gadoxetate disodium-enhanced MRI predicts microvascular invasion and outcome in patients with solitary hepatocellular carcinoma ≤5 cm. Eur Radiol. 2021;31(7):4824–38. https://doi.org/10.1007/s00330-020-07601-2
62 Yang L, Gu D, Wei J, Yang C, Rao S, Wang W, et al. A radiomics nomogram for preoperative prediction of microvascular invasion in hepatocellular carcinoma. Liver Cancer. 2019;8(5):373–86. https://doi.org/10.1159/000494099
63 Lei Z, Li J, Wu D, Xia Y, Wang Q, Si A, et al. Nomogram for preoperative estimation of microvascular invasion risk in hepatitis B virus-related hepatocellular carcinoma within the milan criteria. JAMA Surg. 2016;151(4): 356–63. https://doi.org/10.1001/jamasurg.2015.4257
64 Moustafa AS, Abdel Aal AK, Ertel N, Saad N, DuBay D, Saddekni S. Chemoembolization of hepatocellular carcinoma with extrahepatic collateral blood supply: anatomic and technical considerations. Radiographics. 2017;37(3):963–77. https://doi.org/10.1148/rg.2017160122
65 Pung L, Ahmad M, Mueller K, Rosenberg J, Stave C, Hwang GL, et al. The role of cone-beam CT in transcatheter arterial chemoembolization for hepatocellular carcinoma: a systematic review and meta-analysis. J Vasc Interv Radiol. 2017;28(3):334–41. https://doi.org/10.1016/j.jvir.2016.11.037
66 Lin CY, Chen JH, Liang JA, Lin CC, Jeng LB, Kao CH. 18F-FDG PET or PET/CT for detecting extrahepatic metastases or recurrent hepatocellular carcinoma: a systematic review and meta-analysis. Eur J Radiol. 2012; 81(9):2417–22. https://doi.org/10.1016/j.ejrad.2011.08.004
67 Park JW, Kim JH, Kim SK, Kang KW, Park KW, Choi JI, et al. A prospective evaluation of 18F-FDG and 11C-acetate PET/CT for detection of primary and metastatic hepatocellular carcinoma. J Nucl Med. 2008; 49(12):1912–21. https://doi.org/10.2967/jnumed.108.055087
68 Boellaard R, O'Doherty MJ, Weber WA, Mottaghy FM, Lonsdale MN, Stroobants SG, et al. FDG PET and PET/CT: EANM procedure guidelines for tumour PET imaging: version 1.0. Eur J Nucl Med Mol Imaging. 2010;37(1):181–200. https://doi.org/10.1007/s00259-009-1297-4
69 Chalian H, Töre HG, Horowitz JM, Salem R, Miller FH, Yaghmai V. Radiologic assessment of response to therapy: comparison of RECIST Versions 1.1 and 1.0. Radiographics. 2011;31(7):2093–105. https://doi.org/10.1148/rg.317115050
70 Wahl RL, Jacene H, Kasamon Y, Lodge, MA. From RECIST to PERCIST: evolving Considerations for PET response criteria in solid tumors. J Nucl Med. 2009;50(Suppl 1):122S–50S. https://doi.org/10.2967/jnumed.108.057307
71 Ferda J, Ferdová E, Baxa J, Kreuzberg B, Daum O, Třeška V, et al. The role of 18F-FDG accumulation and arterial enhancement as biomarkers in the assessment of typing, grading and staging of hepatocellular carcinoma using 18F-FDG-PET/CT with integrated dual-phase CT angiography. Anticancer Res. 2015;35(4):2241–6.
72 Hyun SH, Eo JS, Lee JW, Choi JY, Lee KH, Na SJ, et al. Prognostic value of 18F-fluorodeoxyglucose positron emission tomography/computed tomography in patients with Barcelona Clinic Liver Cancer stages 0 and A hepatocellular carcinomas: a multicenter retrospective cohort study. Eur J Nucl Med Mol Imaging. 2016;43(9):1638–45. https://doi.org/10.1007/s00259-016-3348-y

73 Na SJ, Oh JK, Hyun SH, Lee JW, Hong IK, Song BI, et al. (18)F-FDG PET/CT can predict survival of advanced hepatocellular carcinoma patients: a multicenter retrospective cohort study. J Nucl Med. 2017; 58(5):730–6. https://doi.org/10.2967/jnumed.116.182022

74 Bertagna F, Bertoli M, Bosio G, Biasiotto G, Sadeghi R, Giubbini R, et al. Diagnostic role of radiolabelled choline PET or PET/CT in hepatocellular carcinoma: a systematic review and meta-analysis. Hepatol Int. 2014; 8(4):493–500. https://doi.org/10.1007/s12072-014-9566-0

75 Cheung TT, Ho CL, Lo CM, Chen S, Chan SC, Chok KSH, et al. 11C-acetate and 18F-FDG PET/CT for clinical staging and selection of patients with hepatocellular carcinoma for liver transplantation on the basis of Milan criteria: surgeon's perspective. J Nucl Med. 2013;54(2):192–200. https://doi.org/10.2967/jnumed.112.107516

76 Siripongsatian D, Promteangtrong C, Kunawudhi A, Kiatkittikul P, Boonkawin N, Chinnanthachai C, et al. Comparisons of quantitative parameters of Ga-68-Labelled fibroblast activating protein inhibitor (FAPI) PET/CT and [^{18} F] F-FDG PET/CT in patients with liver malignancies. Mol Imaging Biol. 2022;24(5):818–29. https://doi.org/10.1007/s11307-022-01732-2

77 Lan L, Zhang S, Xu T, Liu H, Wang W, Feng Y, et al. Prospective comparison of^{68} Ga-FAPI versus^{18} F-FDG PET/CT for tumor staging in biliary tract cancers. Radiology. 2022;304(3):648–57. https://doi.org/10.1148/radiol.213118

78 Zhang Y, Shi H, Cheng D, Jiang L, Xiu Y, Li B, et al. Added value of SPECT/spiral CT versus SPECT in diagnosing solitary spinal lesions in patients with extraskeletal malignancies. Nucl Med Commun. 2013;34(5): 451–8. https://doi.org/10.1097/MNM.0b013e32835fa552

79 Hectors SJ, Wagner M, Besa C, Huang W, Taouli B. Multiparametric FDG-PET/MRI of hepatocellular carcinoma: initial experience. Contrast Media Mol Imaging. 2018; 2018:5638283. https://doi.org/10.1155/2018/5638283

80 Zhou J, Yu L, Gao X, Hu J, Wang J, Dai Z, et al. Plasma microRNA panel to diagnose hepatitis B virus-related hepatocellular carcinoma. J Clin Oncol. 2011;29(36): 4781–8. https://doi.org/10.1200/JCO.2011.38.2697

81 Best J, Bechmann LP, Sowa JP, Sydor S, Dechêne A, Pflanz K, et al. GALAD score detects early hepatocellular carcinoma in an international cohort of patients with nonalcoholic steatohepatitis. Clin Gastroenterol Hepatol. 2020;18(3):728–35. e4. https://doi.org/10.1016/j.cgh.2019.11.012

82 Piratvisuth T, Hou J, Tanwandee T, Berg T, Vogel A, Trojan J, et al. Development and clinical validation of a novel algorithmic score (GAAD) for detecting HCC in prospective cohort studies. Hepatol Commun. 2023;7(11):e0317. https://doi.org/10.1097/HC9.0000000000000317

83 Yang T, Xing H, Wang G, Wang N, Liu M, Yan C, et al. A novel online calculator based on serum biomarkers to detect hepatocellular carcinoma among patients with hepatitis B. Clin Chem. 2019;65(12):1543–53. https://doi.org/10.1373/clinchem.2019.308965

84 Guo W, Sun YF, Shen MN, Ma XL, Wu J, Zhang CY, et al. Circulating tumor cells with stem-like phenotypes for diagnosis, prognosis, and therapeutic response evaluation in hepatocellular carcinoma. Clin Cancer Res. 2018;24(9):2203–13. https://doi.org/10.1158/1078-0432.CCR-17-1753

85 Zhang X, Wang Z, Tang W, Wang X, Liu R, Bao H, et al. Ultrasensitive and affordable assay for early detection of primary liver cancer using plasma cell-free DNA fragmentomics. Hepatology. 2022;76(2): 317–29. https://doi.org/10.1002/hep.32308

86 Qu C, Wang Y, Wang P, Chen K, Wang M, Zeng H, et al. Detection of early-stage hepatocellular carcinoma in asymptomatic HBsAg-seropositive individuals by liquid biopsy. Proc Natl Acad Sci U S A. 2019; 116(13):6308–12. https://doi.org/10.1073/pnas.1819799116

87 Cai J, Chen L, Zhang Z, Zhang X, Lu X, Liu W, et al. Genome-wide mapping of 5-hydroxymethylcytosines in circulating cell-free DNA as a non-invasive approach for early detection of hepatocellular carcinoma. Gut. 2019;68(12):2195–205. https://doi.org/10.1136/gutjnl-2019-318882

88 Forner A, Vilana R, Ayuso C, Bianchi L, Solé M, Ayuso JR, et al. Diagnosis of hepatic nodules 20 mm or smaller in cirrhosis: prospective validation of the noninvasive diagnostic criteria for hepatocellular carcinoma. Hepatology. 2008;47(1):97–104. https://doi.org/10.1002/hep.21966

89 Roberts LR, Sirlin CB, Zaiem F, Almasri J, Prokop LJ, Heimbach JK, et al. Imaging for the diagnosis of hepatocellular carcinoma: a systematic review and meta-analysis. Hepatology. 2018;67(1):401–21. https://doi.org/10.1002/hep.29487

90 European Association for the Study of the Liver. EASL Clinical Practice Guidelines: management of hepatocellular carcinoma. J Hepatol. 2018;69(1):182–236. https://doi.org/10.1016/j.jhep.2018.03.019

91 Cong WM, Bu H, Chen J, Dong H, Zhu YY, Feng LH, et al. Practice guidelines for the pathological diagnosis of primary liver cancer: 2015 update. World J Gastroenterol. 2016;22(42):9279–87. https://doi.org/10.3748/wjg.v22.i42.9279

92 The WHO Classification of Tumours Editorial Board. WHO classification of tumours, 5th edition, digestive system tumours [M]. Lyon: IARC Press; 2019; p. 254–9.

93 Wang H, Chen J, Zhang X, Sheng X, Chang XY, Chen J, et al. Expert consensus on the pathologic diagnosis of intrahepatic cholangiocarcinoma (2022 edition). Chin J Pathol. 2022;51(9):819–27. https://doi.org/10.3760/cma.j.ca112151-20220517-00423

94 Huang YH, Zhang CZ, Huang QS, Yeong J, Wang F, Yang X, et al. Clinicopathologic features, tumor immune microenvironment and genomic landscape of Epstein-Barr virus-associated intrahepatic cholangiocarcinoma. J Hepatol. 2021;74(4):838–49. https://doi.org/10.1016/j.jhep.2020.10.037

95 Yu WL, Yu G, Dong H, Chen K, Xie J, Yu H, et al. Proteomics analysis identified TPI1 as a novel biomarker for predicting recurrence of intrahepatic cholangiocarcinoma. J Gastroenterol. 2020; 55(12):1171–82. https://doi.org/10.1007/s00535-020-01729-0

96 Cong WM. Surgical pathology of hepatobiliary tumors[M]. Singapore: Springer; 2017; p. 145–216.

97 Chen L, Chen S, Zhou Q, Cao Q, Dong Y, Feng S, et al. Microvascular invasion status and its survival impact in hepatocellular carcinoma depend on tissue sampling protocol. Ann Surg Oncol. 2021;28(11): 6747–57. https://doi.org/10.1245/s10434-021-09673-w

98 Nara S, Shimada K, Sakamoto Y, Esaki M, Kishi Y, Kosuge T, et al. Prognostic impact of marginal resection for patients with solitary hepatocellular carcinoma: evidence from 570 hepatectomies. Surgery. 2012; 151(4):526–36. https://doi.org/10.1016/j.surg.2011.12.002

99 Cong W-M. Surgical pathology of hepatobiliary tumors. Beijing: People's Health Press; 2015; p. 276–339.

100 Lu XY, Xi T, Lau WY, Dong H, Zhu Z, Shen F, et al. Hepatocellular carcinoma expressing cholangiocyte phenotype is a novel subtype with highly aggressive behavior. Ann Surg Oncol. 2011;18(8):2210–7. https://doi.org/10.1245/s10434-011-1585-7

101 Zhuo J, Lu D, Lin Z, Yang X, Yang M, Wang J, et al. The distinct responsiveness of cytokeratin 19-positive hepatocellular carcinoma to regorafenib. Cell Death Dis. 2021; 12(12):1084. https://doi.org/10.1038/s41419-021-04320-4

102 Scheuer PJ. Classification of chronic viral hepatitis: a need for reassessment. J Hepatol. 1991;13(3):372–4. https://doi.org/10.1016/0168-8278(91)90084-o

103 Prevention and control program of viral hepatitis. Chin J Infect Dis. 2001;19(1): 56–62. https://doi.org/10.3760/j.issn:1000-6680.2001.01.027

104 World Health Organization. World Health Organization. Guidelines for the prevention, care and treatment of persons with chronic hepatitis B infection [EB/OL]. 2015. http://www.who.int/hiv/pub/hepatitis/hepatitis-b-guidelines/en/

105 Rodríguez-Perálvarez M, Luong TV, Andreana L, Meyer T, Dhillon AP, Burroughs AK. A systematic review of microvascular invasion in hepatocellular carcinoma: diagnostic and prognostic variability. Ann Surg Oncol. 2013;20(1):325–39. https://doi.org/10.1245/s10434-012-2513-1
106 Wang H, Chen JJ, Yin SY, Sheng X, Wang HX, Lau WY, et al. A grading system of microvascular invasion for patients with hepatocellular carcinoma undergoing liver resection with curative intent: a multicenter study. J Hepatocell Carcinoma. 2024;11:191–206. https://doi.org/10.2147/JHC.S447731
107 Zheng Z, Guan R, Jianxi W, Zhao Z, Peng T, Liu C, et al. Microvascular invasion in hepatocellular carcinoma: a review of its definition, clinical significance, and comprehensive management. J Oncol. 2022;2022:9567041. https://doi.org/10.1155/2022/9567041
108 Sheng X, Ji Y, Ren GP, Lu CL, Yun JP, Chen LH, et al. A standardized pathological proposal for evaluating microvascular invasion of hepatocellular carcinoma: a multicenter study by LCPGC. Hepatol Int. 2020;14(6):1034–47. https://doi.org/10.1007/s12072-020-10111-4
109 Isik B, Gonultas F, Sahin T, Yilmaz S. Microvascular venous invasion in hepatocellular carcinoma: why do recurrences occur? J Gastrointest Cancer. 2020;51(4):1133–6. https://doi.org/10.1007/s12029-020-00487-9
110 Kendall T, Verheij J, Gaudio E, Evert M, Guido M, Goeppert B, et al. Anatomical, histomorphological and molecular classification of cholangiocarcinoma. Liver Int. 2019;39(Suppl 1):7–18. https://doi.org/10.1111/liv.14093
111 Zou Y, Zhu K, Pang Y, Han J, Zhang X, Jiang Z, et al. Molecular detection of FGFR2 rearrangements in resected intrahepatic cholangiocarcinomas: FISH could Be an ideal method in patients with histological small duct subtype. J Clin Transl Hepatol. 2023;11(6):1355–67. https://doi.org/10.14218/JCTH.2022.00060S
112 Dong L, Lu D, Chen R, Lin Y, Zhu H, Zhang Z, et al. Proteogenomic characterization identifies clinically relevant subgroups of intrahepatic cholangiocarcinoma. Cancer Cell. 2022;40(1):70–87. e15. https://doi.org/10.1016/j.ccell.2021.12.006
113 Travis WD, Dacic S, Wistuba I, Sholl L, Adusumilli P, Bubendorf L, et al. IASLC multidisciplinary recommendations for pathologic assessment of lung cancer resection specimens after neoadjuvant therapy. J Thorac Oncol. 2020;15(5):709–40. https://doi.org/10.1016/j.jtho.2020.01.005
114 Allard MA, Sebagh M, Ruiz A, Guettier C, Paule B, Vibert E, et al. Does pathological response after transarterial chemoembolization for hepatocellular carcinoma in cirrhotic patients with cirrhosis predict outcome after liver resection or transplantation? J Hepatol. 2015;63(1):83–92. https://doi.org/10.1016/j.jhep.2015.01.023
115 Stein JE, Lipson EJ, Cottrell TR, Forde PM, Anders RA, Cimino-Mathews A, et al. Pan-tumor pathologic scoring of response to PD-(L)1 blockade. Clin Cancer Res. 2020;26(3):545–51. https://doi.org/10.1158/1078-0432.CCR-19-2379
116 Imamura H, Seyama Y, Kokudo N, Maema A, Sugawara Y, Sano K, et al. One thousand fifty-six hepatectomies without mortality in 8 years. Arch Surg. 2003;138(11):1198–206; discussion 1206. https://doi.org/10.1001/archsurg.138.11.1198
117 Kubota K, Makuuchi M, Kusaka K, Kobayashi T, Miki K, Hasegawa K, et al. Measurement of liver volume and hepatic functional reserve as a guide to decision-making in resectional surgery for hepatic tumors. Hepatology. 1997;26(5):1176–81. https://doi.org/10.1053/jhep.1997.v26.pm0009362359
118 Bruix J, Castells A, Bosch J, Feu F, Fuster J, Garcia-Pagan JC, et al. Surgical resection of hepatocellular carcinoma in cirrhotic patients: prognostic value of preoperative portal pressure. Gastroenterology. 1996;111(4):1018–22. https://doi.org/10.1016/s0016-5085(96)70070-7
119 Cescon M, Colecchia A, Cucchetti A, Peri E, Montrone L, Ercolani G, et al. Value of transient elastography measured with FibroScan in predicting the outcome of hepatic resection for hepatocellular carcinoma. Ann Surg. 2012;256(5):706–13; discussion 712-713. https://doi.org/10.1097/SLA.0b013e3182724ce8
120 Shen Y, Zhou C, Zhu G, Shi G, Zhu X, Huang C, et al. Liver stiffness assessed by shear wave elastography predicts postoperative liver failure in patients with hepatocellular carcinoma. J Gastrointest Surg. 2017;21(9):1471–9. https://doi.org/10.1007/s11605-017-3443-9
121 Rajakannu M, Cherqui D, Ciacio O, Golse N, Pittau G, Allard MA, et al. Liver stiffness measurement by transient elastography predicts late posthepatectomy outcomes in patients undergoing resection for hepatocellular carcinoma. Surgery. 2017;162(4):766–74. https://doi.org/10.1016/j.surg.2017.06.006
122 Zhong JH, Ke Y, Gong WF, Xiang B, Ma L, Ye X, et al. Hepatic resection associated with good survival for selected patients with intermediate and advanced-stage hepatocellular carcinoma. Ann Surg. 2014;260(2):329–40. https://doi.org/10.1097/sla.0000000000000236
123 Xiao H, Zhang B, Mei B, Zuo C, Wei G, Wang R, et al. Hepatic resection for hepatocellular carcinoma in patients with portal hypertension: a long-term benefit compared with transarterial chemoembolization and thermal ablation. Medicine. 2015;94(7):e495. https://doi.org/10.1097/MD.0000000000000495
124 Bosch J, Abraldes JG, Berzigotti A, García-Pagan JC. The clinical use of HVPG measurements in chronic liver disease. Nat Rev Gastroenterol Hepatol. 2009;6(10):573–82. https://doi.org/10.1038/nrgastro.2009.149
125 Chen X, Zhai J, Cai X, Zhang Y, Wei L, Shi L, et al. Severity of portal hypertension and prediction of postoperative liver failure after liver resection in patients with Child-Pugh grade A cirrhosis. Br J Surg. 2012;99(12):1701–10. https://doi.org/10.1002/bjs.8951
126 Chin KM, Prieto M, Cheong CK, Di Martino M, Ielpo B, Goh BKP, et al. Outcomes after curative therapy for hepatocellular carcinoma in patients with non-alcoholic fatty liver disease: a meta-analysis and review of current literature. HPB. 2021;23(8):1164–74. https://doi.org/10.1016/j.hpb.2021.01.009
127 Mirdad RS, Madison Hyer J, Diaz A, Tsilimigras DI, Azap RA, Paro A, et al. Postoperative imaging surveillance for hepatocellular carcinoma: how much is enough? J Surg Oncol. 2021;123(7):1568–77. https://doi.org/10.1002/jso.26433
128 Chen MS, Li JQ, Zheng Y, Guo RP, Liang HH, Zhang YQ, et al. A prospective randomized trial comparing percutaneous local ablative therapy and partial hepatectomy for small hepatocellular carcinoma. Ann Surg. 2006;243(3):321–8. https://doi.org/10.1097/01.sla.0000201480.65519.b8
129 Takayama T, Hasegawa K, Izumi N, Kudo M, Shimada M, Yamanaka N, et al. Surgery versus radiofrequency ablation for small hepatocellular carcinoma: a randomized controlled trial (SURF trial). Liver Cancer. 2022;11(3):209–18. https://doi.org/10.1159/000521665
130 Yamashita T, Kawaguchi Y, Kaneko S, Takayama T, Izumi N, Yamanaka N, et al. A multicenter, non-randomized, controlled trial to evaluate the efficacy of surgery versus radiofrequency ablation for small hepatocellular carcinoma (SURF-Cohort Trial): analysis of overall survival. J Clin Oncol. 2022;40(16_Suppl l):4095. https://doi.org/10.1200/jco.2022.40.16_suppl.4095
131 Mohkam K, Dumont PN, Manichon AF, Jouvet JC, Boussel L, Merle P, et al. No-touch multibipolar radiofrequency ablation vs. surgical resection for solitary hepatocellular carcinoma ranging from 2 to 5 cm. J Hepatol. 2018;68(6):1172–80. https://doi.org/10.1016/j.jhep.2018.01.014
132 Xu XL, Liu XD, Liang M, Luo BM. Radiofrequency ablation versus hepatic resection for small hepatocellular carcinoma: systematic review of randomized controlled trials with meta-analysis and trial sequential analysis. Radiology. 2018;287(2):461–72. https://doi.org/10.1148/radiol.2017162756

133 Liu PH, Hsu CY, Hsia CY, Lee YH, Huang YH, Chiou YY, et al. Surgical resection versus radiofrequency ablation for single hepatocellular carcinoma ≤ 2 cm in a propensity score model. Ann Surg. 2016;263(3):538–45. https://doi.org/10.1097/sla.0000000000001178
134 Feng K, Yan J, Li X, Xia F, Ma K, Wang S, et al. A randomized controlled trial of radiofrequency ablation and surgical resection in the treatment of small hepatocellular carcinoma. J Hepatol. 2012;57(4):794–802. https://doi.org/10.1016/j.jhep.2012.05.007
135 Xu Q, Kobayashi S, Ye X, Meng X. Comparison of hepatic resection and radiofrequency ablation for small hepatocellular carcinoma: a meta-analysis of 16, 103 patients. Sci Rep. 2014;4:7252. https://doi.org/10.1038/srep07252
136 Xia Y, Li J, Liu G, Wang K, Qian G, Lu Z, et al. Long-term effects of repeat hepatectomy vs percutaneous radiofrequency ablation among patients with recurrent hepatocellular carcinoma: a randomized clinical trial. JAMA Oncol. 2020;6(2):255–63. https://doi.org/10.1001/jamaoncol.2019.4477
137 Yin L, Li H, Li AJ, Lau WY, Pan ZY, Lai ECH, et al. Partial hepatectomy vs. transcatheter arterial chemoembolization for resectable multiple hepatocellular carcinoma beyond Milan Criteria: a RCT. J Hepatol. 2014;61(1):82–8. https://doi.org/10.1016/j.jhep.2014.03.012
138 Torzilli G, Belghiti J, Kokudo N, Takayama T, Capussotti L, Nuzzo G, et al. A snapshot of the effective indications and results of surgery for hepatocellular carcinoma in tertiary referral centers: is it adherent to the EASL/AASLD recommendations? centers: is it adherent to the EASL/AASLD recommendations? an observational study of the HCC East-West study group. Ann Surg. 2013;257(5):929–37. https://doi.org/10.1097/SLA.0b013e31828329b8
139 Hyun MH, Lee YS, Kim JH, Lee CU, Jung YK, Seo YS, et al. Hepatic resection compared to chemoembolization in intermediate-to advanced-stage hepatocellular carcinoma: a meta-analysis of high-quality studies. Hepatology. 2018;68(3):977–93. https://doi.org/10.1002/hep.29883
140 Tsilimigras DI, Mehta R, Paredes AZ, Moris D, Sahara K, Bagante F, et al. Overall tumor burden dictates outcomes for patients undergoing resection of multinodular hepatocellular carcinoma beyond the milan criteria. Ann Surg. 2020;272(4):574–81. https://doi.org/10.1097/SLA.0000000000004346
141 Famularo S, Donadon M, Cipriani F, Giuliante F, Ferri S, Celsa C, et al. Hepatectomy versus sorafenib in advanced nonmetastatic hepatocellular carcinoma: a real-life multicentric weighted comparison. Ann Surg. 2022;275(4):743–52. https://doi.org/10.1097/SLA.0000000000005373
142 Kokudo T, Hasegawa K, Matsuyama Y, Takayama T, Izumi N, Kadoya M, et al. Survival benefit of liver resection for hepatocellular carcinoma associated with portal vein invasion. J Hepatol. 2016;65(5):938–43. https://doi.org/10.1016/j.jhep.2016.05.044
143 Zhang XP, Gao YZ, Chen ZH, Chen MS, Li LQ, Wen TF, et al. An eastern hepatobiliary surgery hospital/portal vein tumor thrombus scoring system as an aid to decision making on hepatectomy for hepatocellular carcinoma patients with portal vein tumor thrombus: a multicenter study. Hepatology. 2019;69(5):2076–90. https://doi.org/10.1002/hep.30490
144 Govalan R, Lauzon M, Luu M, Ahn JC, Kosari K, Todo T, et al. Comparison of surgical resection and systemic treatment for hepatocellular carcinoma with vascular invasion. National cancer database analysis. Liver Cancer. 2021;10(5):407–18. https://doi.org/10.1159/000515554
145 Pawlik TM, Poon RT, Abdalla EK, Ikai I, Nagorney DM, Belghiti J, et al. Hepatectomy for hepatocellular carcinoma with major portal or hepatic vein invasion: results of a multicenter study. Surgery. 2005;137(4):403–10. https://doi.org/10.1016/j.surg.2004.12.012
146 Lu J, Zhang XP, Zhong BY, Lau WY, Madoff DC, Davidson JC, et al. Management of patients with hepatocellular carcinoma and portal vein tumour thrombosis: comparing east and west. Lancet Gastroenterol Hepatol. 2019;4(9):721–30. https://doi.org/10.1016/S2468-1253(19)30178-5
147 Fan J, Zhou J, Wu ZQ, Qiu SJ, Wang XY, Shi YH, et al. Efficacy of different treatment strategies for hepatocellular carcinoma with portal vein tumor thrombosis. World J Gastroenterol. 2005;11(8):1215–9. https://doi.org/10.3748/wjg.v11.i8.1215
148 Wei X, Jiang Y, Zhang X, Feng S, Zhou B, Ye X, et al. Neoadjuvant three-dimensional conformal radiotherapy for resectable hepatocellular carcinoma with portal vein tumor thrombus: a randomized, open-label, multicenter controlled study. J Clin Oncol. 2019;37(24):2141–51. https://doi.org/10.1200/JCO.18.02184
149 Wang K, Guo WX, Chen MS, Mao YL, Sun BC, Shi J, et al. Multimodality treatment for hepatocellular carcinoma with portal vein tumor thrombus: a large-scale, multicenter, propensity mathching score analysis. Medicine. 2016;95(11):e3015. https://doi.org/10.1097/MD.0000000000003015
150 Li XL, Zhu XD, Cai H, Li Y, Zhou J, Fan J, et al. Postoperative α-fetoprotein response predicts tumor recurrence and survival after hepatectomy for hepatocellular carcinoma: a propensity score matching analysis. Surgery. 2019;165(6):1161–7. https://doi.org/10.1016/j.surg.2019.01.009
151 Yang J, Tao HS, Cai W, Zhu W, Zhao D, Hu HY, et al. Accuracy of actual resected liver volume in anatomical liver resections guided by 3-dimensional parenchymal staining using fusion indocyanine green fluorescence imaging. J Surg Oncol. 2018;118(7):1081–7. https://doi.org/10.1002/jso.25258
152 Mise Y, Hasegawa K, Satou S, Shindoh J, Miki K, Akamatsu N, et al. How has virtual hepatectomy changed the practice of liver surgery? experience of 1194 virtual hepatectomy before liver resection and living donor liver transplantation. Ann Surg. 2018;268(1):127–33. https://doi.org/10.1097/SLA.0000000000002213
153 Digital Medicine Branch of Chinese Medical Association; Digital Intelligent Surgery Professional Committee of Chinese Research Hospital Society; Liver Cancer Professional Committee of Chinese Physicians Association. Guidelines for the application of computer-assisted combined indocyanine green molecular fluorescence imaging technology in the diagnosis and surgical navigation of liver tumors (2019 edition). Chin J Pract Surg. 2019;39(7):641–50. https://doi.org/10.19538/j.cjps.issn1005-2208.2019.07.01
154 Jiang HT, Cao JY. Impact of laparoscopic versus open hepatectomy on perioperative clinical outcomes of patients with primary hepatic carcinoma. Chin Med Sci J. 2015;30(2):80–3. https://doi.org/10.1016/s1001-9294(15)30016-x
155 Hepatobiliary and Pancreatic Surgery Specialized Committee of the Chinese Society of Research Hospitals. Chinese expert consensus on laparoscopic hepatectomy for hepatocellular carcinoma (2020 edition). Chin J Gastrointest Surg. 2020;19(11):1119–34. https://doi.org/10.3760/cma.j.cn115610-20201029-00682
156 Zhu P, Liao W, Zhang WG, Chen L, Shu C, Zhang ZW, et al. A prospective study using propensity score matching to compare long-term survival outcomes after robotic-assisted, laparoscopic, or open liver resection for patients with BCLC stage 0-A hepatocellular carcinoma. Ann Surg. 2023;277(1):e103–11. https://doi.org/10.1097/SLA.0000000000005380
157 Wang Q, Li HJ, Dai XM, Xiang ZQ, Zhu Z. Laparoscopic versus open liver resection for hepatocellular carcinoma in elderly patients: systematic review and meta-analysis of propensity-score matched studies. Int J Surg. 2022;105:106821. https://doi.org/10.1016/j.ijsu.2022.106821
158 Wang X, Teh CSC, Ishizawa T, Aoki T, Cavallucci D, Lee SY, et al. Consensus guidelines for the use of fluorescence imaging in hepatobiliary surgery. Ann Surg. 2021;274(1):97–106. https://doi.org/10.1097/SLA.0000000000004718

159 Di Benedetto F, Magistri P, Di Sandro S, Sposito C, Oberkofler C, Brandon E, et al. Safety and efficacy of robotic vs open liver resection for hepatocellular carcinoma. JAMA Surg. 2023;158(1):46–54. https://doi.org/10.1001/jamasurg.2022.5697
160 Yongxiang XIA, Feng ZHANG, Xiangcheng LI, et al. Analysis of 10966 cases of primary hepatocellular carcinoma with exceptional treatment. Chin J Surg. 2021;59(1):6–17. https://doi.org/10.3760/cma.j.cn112139-20201110-00791
161 Hidaka M, Eguchi S, Okuda K, Beppu T, Shirabe K, Kondo K, et al. Impact of anatomical resection for hepatocellular carcinoma with microportal invasion (vp1): a multi-institutional study by the kyushu study group of liver surgery. Ann Surg. 2020;271(2):339–46. https://doi.org/10.1097/SLA.0000000000002981
162 Zhong FP, Zhang YJ, Liu Y, Zou SB. Prognostic impact of surgical margin in patients with hepatocellular carcinoma: a meta-analysis. Medicine. 2017;96(37):e8043. https://doi.org/10.1097/MD.0000000000008043
163 Shi M, Guo RP, Lin XJ, Zhang YQ, Chen MS, Zhang CQ, et al. Partial hepatectomy with wide versus narrow resection margin for solitary hepatocellular carcinoma: a prospective randomized trial. Ann Surg. 2007;245(1):36–43. https://doi.org/10.1097/01.sla.0000231758.07868.71
164 Yang P, Si A, Yang J, Cheng Z, Wang K, Li J, et al. A wide-margin liver resection improves long-term outcomes for patients with HBV-related hepatocellular carcinoma with microvascular invasion. Surgery. 2019;165(4):721–30. https://doi.org/10.1016/j.surg.2018.09.016
165 Liao K, Yang K, Cao L, Lu Y, Zheng B, Li X, et al. Laparoscopic anatomical versus non-anatomical hepatectomy in the treatment of hepatocellular carcinoma: a randomised controlled trial. Int J Surg. 2022;102:106652. https://doi.org/10.1016/j.ijsu.2022.106652
166 Liu CL, Fan ST, Lo CM, Tung-Ping Poon R, Wong J. Anterior approach for major right hepatic resection for large hepatocellular carcinoma. Ann Surg. 2000;232(1):25–31. https://doi.org/10.1097/00000658-200007000-00004
167 Zhou C, Peng Y, Zhou K, Zhang L, Zhang X, Yu L, et al. Surgical resection plus radiofrequency ablation for the treatment of multifocal hepatocellular carcinoma. Hepatobiliary Surg Nutr. 2019;8(1):19–28. https://doi.org/10.21037/hbsn.2018.11.19
168 Zhang ZM, Lai EC, Zhang C, Yu H, Liu Z, Wan B, et al. The strategies for treating primary hepatocellular carcinoma with portal vein tumor thrombus. Int J Surg. 2015;20:8–16. https://doi.org/10.1016/j.ijsu.2015.05.009
169 Fu SY, Lau WY, Li AJ, Yang Y, Pan ZY, Sun YM, et al. Liver resection under total vascular exclusion with or without preceding Pringle manoeuvre. Br J Surg. 2010;97(1):50–5. https://doi.org/10.1002/bjs.6841
170 Satoh S, Ikai I, Honda G, Okabe H, Takeyama O, Yamamoto Y, et al. Clinicopathologic evaluation of hepatocellular carcinoma with bile duct thrombi. Surgery. 2000;128(5):779–83. https://doi.org/10.1067/msy.2000.108659
171 Kim DS, Kim BW, Hatano E, Hwang S, Hasegawa K, Kudo A, et al. Surgical outcomes of hepatocellular carcinoma with bile duct tumor thrombus: a korea-Japan multicenter study. Ann Surg. 2020;271(5):913–21. https://doi.org/10.1097/SLA.0000000000003014
172 Zhu XD, Huang C, Shen YH, Ji Y, Ge NL, Qu XD, et al. Downstaging and resection of initially unresectable hepatocellular carcinoma with tyrosine kinase inhibitor and anti- PD-1 antibody combinations. Liver Cancer. 2021;10(4):320–9. https://doi.org/10.1159/000514313
173 Hepatobiliary and Pancreatic Surgery Specialized Committee of the Chinese Society of Research Hospitals. Expert consensus on precision hepatectomy. Chin J Dig Surg. 2017;16(9):883–93. https://doi.org/10.3760/cma.j.issn.1673-9752.2017.09.001
174 Aloia TA. Associating liver partition and portal vein ligation for staged hepatectomy: portal vein embolization should remain the gold standard. JAMA Surg. 2015;150(10):927–8. https://doi.org/10.1001/jamasurg.2015.1646
175 Piron L, Deshayes E, Escal L, Souche R, Herrero A, Pierredon-Foulongne MA, et al. [Portal vein embolization: present and future]. Bull Cancer. 2017;104(5):407–16. https://doi.org/10.1016/j.bulcan.2017.03.009
176 Ogata S, Belghiti J, Farges O, Varma D, Sibert A, Vilgrain V. Sequential arterial and portal vein embolizations before right hepatectomy in patients with cirrhosis and hepatocellular carcinoma. Br J Surg. 2006;93(9):1091–8. https://doi.org/10.1002/bjs.5341
177 Hwang S, Ha TY, Ko GY, Kwon DI, Song GW, Jung DH, et al. Preoperative sequential portal and hepatic vein embolization in patients with hepatobiliary malignancy. World J Surg. 2015;39(12):2990–8. https://doi.org/10.1007/s00268-015-3194-2
178 Dupré A, Hitier M, Peyrat P, Chen Y, Meeus P, Rivoire M. Associating portal embolization and artery ligation to induce rapid liver regeneration in staged hepatectomy. Br J Surg. 2015;102(12):1541–50. https://doi.org/10.1002/bjs.9900
179 Glantzounis GK, Tokidis E, Basourakos SP, Ntzani EE, Lianos GD, Pentheroudakis G. The role of portal vein embolization in the surgical management of primary hepatobiliary cancers. a systematic review. Eur J Surg Oncol. 2017;43(1):32–41. https://doi.org/10.1016/j.ejso.2016.05.026
180 Schnitzbauer AA, Lang SA, Goessmann H, Nadalin S, Baumgart J, Farkas SA, et al. Right portal vein ligation combined with in situ splitting induces rapid left lateral liver lobe hypertrophy enabling 2-staged extended right hepatic resection in small-for-size settings. Ann Surg. 2012;255(3):405–14. https://doi.org/10.1097/SLA.0b013e31824856f5
181 Wang Z, Peng Y, Hu J, Wang X, Sun H, Sun J, et al. Associating liver partition and portal vein ligation for staged hepatectomy for unresectable hepatitis B virus-related hepatocellular carcinoma: a single center study of 45 patients. Ann Surg. 2020;271(3):534–41. https://doi.org/10.1097/SLA.0000000000002942
182 Peng Y, Wang Z, Qu X, Chen F, Sun H, Wang X, et al. Transcatheter arterial embolization-salvaged ALPPS, a novel ALPPS procedure especially for patients with hepatocellular carcinoma and severe fibrosis/cirrhosis. Hepatobiliary Surg Nutr. 2022;11(4):504–14. https://doi.org/10.21037/hbsn-21-466
183 Li PP, Huang G, Jia NY, Pan ZY, Liu H, Yang Y, et al. Associating liver partition and portal vein ligation for staged hepatectomy versus sequential transarterial chemoembolization and portal vein embolization in staged hepatectomy for HBV-related hepatocellular carcinoma: a randomized comparative study. Hepatobiliary Surg Nutr. 2022;11(1):38–51. https://doi.org/10.21037/hbsn-20-264
184 Zhang Y, Huang G, Wang Y, Liang L, Peng B, Fan W, et al. Is salvage liver resection necessary for initially unresectable hepatocellular carcinoma patients downstaged by transarterial chemoembolization? Ten years of experience. Oncologist. 2016;21(12):1442–9. https://doi.org/10.1634/theoncologist.2016-0094
185 Lyu N, Kong Y, Mu L, Lin Y, Li J, Liu Y, et al. Hepatic arterial infusion of oxaliplatin plus fluorouracil/leucovorin vs. sorafenib for advanced hepatocellular carcinoma. J Hepatol. 2018;69(1):60–9. https://doi.org/10.1016/j.jhep.2018.02.008
186 He M, Li Q, Zou R, Shen J, Fang W, Tan G, et al. Sorafenib plus hepatic arterial infusion of oxaliplatin, fluorouracil, and leucovorin vs sorafenib alone for hepatocellular carcinoma with portal vein invasion: a randomized clinical trial. JAMA Oncol. 2019;5(7):953–60. https://doi.org/10.1001/jamaoncol.2019.0250
187 Peng Z, Fan W, Zhu B, Wang G, Sun J, Xiao C, et al. Lenvatinib combined with transarterial chemoembolization as first-line treatment for advanced hepatocellular carcinoma: a phase III, randomized clinical trial (LAUNCH). J Clin Oncol. 2023;41(1):117–27. https://doi.org/10.1200/JCO.22.00392
188 Kudo M, Aoki T, Ueshima K, Tsuchiya K, Morita M, Chishina H, et al. Achievement of complete response and drug-free status by atezolizumab plus bevacizumab combined with or without curative conversion in patients with transarterial chemoembolization-unsuitable, intermediate-stage hepatocellular carcinoma: a multicenter proof-of-concept study. Liver Cancer. 2023;12(4):321–38. https://doi.org/10.1159/000529574

189 Sangro B, Kudo M, Erinjeri JP, Qin S, Ren Z, Chan SL, et al. Durvalumab with or without bevacizumab with transarterial chemoembolisation in hepatocellular carcinoma (EMERALD-1): a multiregional, randomised, double-blind, placebo-controlled, phase 3 study. Lancet. 2025;405(10474):216–32. https://doi.org/10.1016/S0140-6736(24)02551-0
190 Li B, Qiu J, Zheng Y, Shi Y, Zou R, He W, et al. Conversion to resectability using transarterial chemoembolization combined with hepatic arterial infusion chemotherapy for initially unresectable hepatocellular carcinoma. Ann Surg Open. 2021;2(2):e057. https://doi.org/10.1097/AS9.0000000000000057
191 Byun HK, Kim HJ, Im YR, Kim DY, Han KH, Seong J. Dose escalation by intensity modulated radiotherapy in liver-directed concurrent chemoradiotherapy for locally advanced BCLC stage C hepatocellular carcinoma. Radiother Oncol. 2019;133:1–8. https://doi.org/10.1016/j.radonc.2018.12.025
192 Pinna AD, Yang T, Mazzaferro V, De Carlis L, Zhou J, Roayaie S, et al. Liver transplantation and hepatic resection can achieve cure for hepatocellular carcinoma. Ann Surg. 2018;268(5):868–75. https://doi.org/10.1097/sla.0000000000002889
193 Tabrizian P, Jibara G, Shrager B, Schwartz M, Roayaie S. Recurrence of hepatocellular cancer after resection: patterns, treatments, and prognosis. Ann Surg. 2015;261(5):947–55. https://doi.org/10.1097/sla.0000000000000710
194 Chan A, Zhong J, Berhane S, Toyoda H, Cucchetti A, Shi K, et al. Development of pre- and post-operative models to predict early recurrence of hepatocellular carcinoma after surgical resection. J Hepatol. 2018;69(6):1284–93. https://doi.org/10.1016/j.jhep.2018.08.027
195 Wu JC, Huang YH, Chau GY, Su CW, Lai CR, Lee PC, et al. Risk factors for early and late recurrence in hepatitis B-related hepatocellular carcinoma. J Hepatol. 2009;51(5):890–7. https://doi.org/10.1016/j.jhep.2009.07.009
196 Imamura H, Matsuyama Y, Tanaka E, Ohkubo T, Hasegawa K, Miyagawa S, et al. Risk factors contributing to early and late phase intrahepatic recurrence of hepatocellular carcinoma after hepatectomy. J Hepatol. 2003;38(2):200–7. https://doi.org/10.1016/s0168-8278(02)00360-4
197 Wang Z, Ren Z, Chen Y, Hu J, Yang G, Yu L, et al. Adjuvant transarterial chemoembolization for HBV-related hepatocellular carcinoma after resection: a randomized controlled study. Clin Cancer Res. 2018;24(9):2074–81. https://doi.org/10.1158/1078-0432.CCR-17-2899
198 Wei W, Jian PE, Li SH, Guo ZX, Zhang YF, Ling YH, et al. Adjuvant transcatheter arterial chemoembolization after curative resection for hepatocellular carcinoma patients with solitary tumor and microvascular invasion: a randomized clinical trial of efficacy and safety. Cancer Commun. 2018;38(1):61. https://doi.org/10.1186/s40880-018-0331-y
199 Li SH, Mei J, Cheng Y, Li Q, Wang QX, Fang CK, et al. Postoperative adjuvant hepatic arterial infusion chemotherapy with FOLFOX in hepatocellular carcinoma with microvascular invasion: a multicenter, phase III, randomized study. J Clin Oncol. 2023;41(10):1898–908. https://doi.org/10.1200/JCO.22.01142
200 Lee JH, Lee JH, Lim YS, Yeon JE, Song TJ, Yu SJ, et al. Adjuvant immunotherapy with autologous cytokine-induced killer cells for hepatocellular carcinoma. Gastroenterology. 2015;148(7):1383–91.e6. https://doi.org/10.1053/j.gastro.2015.02.055
201 He C, Peng W, Li C, Wen TF. Thymalfasin, a promising adjuvant therapy in small hepatocellular carcinoma after liver resection. Medicine. 2017;96(16):e6606. https://doi.org/10.1097/MD.0000000000006606
202 Chen Q, Shu C, Laurence AD, Chen Y, Peng BG, Zhen ZJ, et al. Effect of Huaier granule on recurrence after curative resection of HCC: a multicentre, randomised clinical trial. Gut. 2018;67(11):2006–16. https://doi.org/10.1136/gutjnl-2018-315983
203 Huang G, Li PP, Lau WY, Pan ZY, Zhao LH, Wang ZG, et al. Antiviral therapy reduces hepatocellular carcinoma recurrence in patients with low HBV-DNA levels: a randomized controlled trial. Ann Surg. 2018;268(6):943–54. https://doi.org/10.1097/SLA.0000000000002727
204 Wu J, Yin Z, Cao L, Xu X, Yan T, Liu C, et al. Adjuvant pegylated interferon therapy improves the survival outcomes in patients with hepatitis-related hepatocellular carcinoma after curative treatment: a meta-analysis. Medicine. 2018;97(28):e11295. https://doi.org/10.1097/MD.0000000000011295
205 Singal AG, Lim JK, Kanwal F. AGA clinical practice update on interaction between oral direct-acting antivirals for chronic hepatitis C infection and hepatocellular carcinoma: expert review. Gastroenterology. 2019;156(8):2149–57. https://doi.org/10.1053/j.gastro.2019.02.046
206 Cabibbo G, Celsa C, Calvaruso V, Petta S, Cacciola I, Cannavò MR, et al. Direct-acting antivirals after successful treatment of early hepatocellular carcinoma improve survival in HCV- cirrhotic patients. J Hepatol. 2019;71(2):265–73. https://doi.org/10.1016/j.jhep.2019.03.027
207 Bruix J, Takayama T, Mazzaferro V, Chau GY, Yang J, Kudo M, et al. Adjuvant sorafenib for hepatocellular carcinoma after resection or ablation (STORM): a phase 3, randomised, double-blind, placebo-controlled trial. Lancet Oncol. 2015;16(13):1344–54. https://doi.org/10.1016/S1470-2045(15)00198-9
208 Qin S, Chen M, Cheng AL, Kaseb AO, Kudo M, Lee HC, et al. Atezolizumab plus bevacizumab versus active surveillance in patients with resected or ablated high-risk hepatocellular carcinoma (IMbrave050): a randomised, open-label, multicentre, phase 3 trial. Lancet. 2023;402(10415):1835–47. https://doi.org/10.1016/S0140-6736(23)01796-8
209 Yopp A, Kudo M, Chen M, Cheng AL, Kaseb A, Lee H, et al. LBA39: updated efficacy and safety data from IMbrave050: phase III study of adjuvant atezolizumab (atezo) + bevacizumab (bev) vs active surveillance in patients (pts) with resected or ablated high-risk hepatocellular carcinoma (HCC). Ann Oncol. 2024;35(Suppl_2):S1230–72. https://doi.org/10.1016/j.annonc.2024.08.2279
210 Sapisochin G, Bruix J. Liver transplantation for hepatocellular carcinoma: outcomes and novel surgical approaches. Nat Rev Gastroenterol Hepatol. 2017;14(4):203–17. https://doi.org/10.1038/nrgastro.2016.193
211 Fan J, Yang GS, Fu ZR, Peng ZH, Xia Q, Peng CH, et al. Liver transplantation outcomes in 1,078 hepatocellular carcinoma patients: a multi-center experience in Shanghai, China. J Cancer Res Clin Oncol. 2009;135(10):1403–12. https://doi.org/10.1007/s00432-009-0584-6
212 Zheng SS, Xu X, Wu J, Chen J, Wang WL, Zhang M, et al. Liver transplantation for hepatocellular carcinoma: Hangzhou experiences. Transplantation. 2008;85(12):1726–32. https://doi.org/10.1097/TP.0b013e31816b67e4
213 Li J, Yan LN, Yang J, Chen ZY, Li B, Zeng Y, et al. Indicators of prognosis after liver transplantation in Chinese hepatocellular carcinoma patients. World J Gastroenterol. 2009;15(33):4170–6. https://doi.org/10.3748/wjg.15.4170
214 Zhuo SHAO, Yang G, Yang N, et al. Application of Sanya consensus in liver transplantation for primary liver cancer. Chin J Pract Surg. 2008;28(6):466–9. https://doi.org/10.3321/j.issn:1005-2208.2008.06.018
215 Kulik L, Heimbach JK, Zaiem F, Almasri J, Prokop LJ, Wang Z, et al. Therapies for patients with hepatocellular carcinoma awaiting liver transplantation: a systematic review and meta -analysis. Hepatology. 2018;67(1):381–400. https://doi.org/10.1002/hep.29485
216 Lee S, Kim KW, Song GW, Kwon JH, Hwang S, Kim KH, et al. The real impact of bridging or downstaging on survival outcomes after liver transplantation for hepatocellular carcinoma. Liver Cancer. 2020;9(6):721–33. https://doi.org/10.1159/000507887

217 Nordness MF, Hamel S, Godfrey CM, Shi C, Johnson DB, Goff LW, et al. Fatal hepatic necrosis after nivolumab as a bridge to liver transplant for HCC: are checkpoint inhibitors safe for the pretransplant patient? Am J Transpl. 2020;20(3):879–83. https://doi.org/10.1111/ajt.15617

218 Mazzaferro V, Citterio D, Bhoori S, Bongini M, Miceli R, De Carlis L, et al. Liver transplantation in hepatocellular carcinoma after tumour downstaging (XXL): a randomised, controlled, phase 2b/3 trial. Lancet Oncol. 2020;21(7):947–56. https://doi.org/10.1016/S1470-2045(20)30224-2

219 Mehta N, Guy J, Frenette CT, Dodge JL, Osorio RW, Minteer WB, et al. Excellent outcomes of liver transplantation following down-staging of hepatocellular carcinoma to within milan criteria: a multicenter study. Clin Gastroenterol Hepatol. 2018;16(6):955–64. https://doi.org/10.1016/j.cgh.2017.11.037

220 Tabrizian P, Holzner ML, Mehta N, Halazun K, Agopian VG, Yao F, et al. Ten-year outcomes of liver transplant and downstaging for hepatocellular carcinoma. JAMA Surg. 2022;157(9):779–88. https://doi.org/10.1001/jamasurg.2022.2800

221 Llovet JM, Pavel M, Rimola J, Diaz MA, Colmenero J, Saavedra-Perez D, et al. Pilot study of living donor liver transplantation for patients with hepatocellular carcinoma exceeding Milan Criteria (Barcelona Clinic Liver Cancer extended criteria). Liver Transpl. 2018;24(3):369–79. https://doi.org/10.1002/lt.24977

222 Pinheiro RS, Waisberg DR, Nacif LS, Rocha-Santos V, Arantes RM, Ducatti L, et al. Living donor liver transplantation for hepatocellular cancer: an (almost) exclusive Eastern procedure? Transl Gastroenterol Hepatol. 2017;2:68. https://doi.org/10.21037/tgh.2017.08.02

223 Kulik LM, Fisher RA, Rodrigo DR, Brown RS Jr, Freise CE, Shaked A, et al. Outcomes of living and deceased donor liver transplant recipients with hepatocellular carcinoma: results of the A2ALL cohort. Am J Transpl. 2012;12(11):2997–3007. https://doi.org/10.1111/j.1600-6143.2012.04272.x

224 Wong T, Ng K, Fung J, Chan AAC, Cheung TT, Chok KSH, et al. Long-term survival outcome between living donor and deceased donor liver transplant for hepatocellular carcinoma: intention -to-Treat and propensity score matching analyses. Ann Surg Oncol. 2019;26(5):1454–62. https://doi.org/10.1245/s10434-019-07206-0

225 Goldaracena N, Gorgen A, Doyle A, Hansen BE, Tomiyama K, Zhang W, et al. Live donor liver transplantation for patients with hepatocellular carcinoma offers increased survival vs. deceased donation. J Hepatol. 2019;70(4):666–73. https://doi.org/10.1016/j.jhep.2018.12.029

226 Sposito C, Cucchetti A, Mazzaferro V. Assessing competing risks for death following liver transplantation for hepatocellular carcinoma. Dig Dis Sci. 2019;64(4):1001–7. https://doi.org/10.1007/s10620-019-05538-1

227 Segev DL, Sozio SM, Shin EJ, Nazarian SM, Nathan H, Thuluvath PJ, et al. Steroid avoidance in liver transplantation: meta-analysis and meta-regression of randomized trials. Liver Transpl. 2008;14(4):512–25. https://doi.org/10.1002/lt.21396

228 Rodríguez-Perálvarez M, Tsochatzis E, Naveas MC, Pieri G, García-Caparrós C, O'Beirne J, et al. Reduced exposure to calcineurin inhibitors early after liver transplantation prevents recurrence of hepatocellular carcinoma. J Hepatol. 2013;59(6):1193–9. https://doi.org/10.1016/j.jhep.2013.07.012

229 Liang W, Wang D, Ling X, Kao AA, Kong Y, Shang Y, et al. Sirolimus-based immunosuppression in liver transplantation for hepatocellular carcinoma: a meta-analysis. Liver Transpl. 2012;18(1):62–9. https://doi.org/10.1002/lt.22441

230 Zhou J, Wang Z, Wu ZQ, Qiu SJ, Yu Y, Huang XW, et al. Sirolimus-based immunosuppression therapy in liver transplantation for patients with hepatocellular carcinoma exceeding the Milan criteria. Transpl Proc. 2008;40(10):3548–53. https://doi.org/10.1016/j.transproceed.2008.03.165

231 Geissler EK, Schnitzbauer AA, Zülke C, Lamby PE, Proneth A, Duvoux C, et al. Sirolimus use in liver transplant recipients with hepatocellular carcinoma: a randomized, multicenter. Open-label phase 3 trial. Transplantation. 2016;100(1):116–25. https://doi.org/10.1097/TP.0000000000000965

232 Thorat A, Jeng LB, Yang HR, Yeh CC, Hsu SC, Chen TH, et al. Assessing the role of everolimus in reducing hepatocellular carcinoma recurrence after living donor liver transplantation for patients within the UCSF criteria: re-inventing the role of mammalian target of rapamycin inhibitors. Ann Hepatobiliary Pancreat Surg. 2017;21(4):205–11. https://doi.org/10.14701/ahbps.2017.21.4.205

233 Schnitzbauer AA, Filmann N, Adam R, Bachellier P, Bechstein WO, Becker T, et al. mTOR inhibition is most beneficial after liver transplantation for hepatocellular carcinoma in patients with active tumors. Ann Surg. 2020;272(5):855–62. https://doi.org/10.1097/SLA.0000000000004280

234 Filgueira NA. Hepatocellular carcinoma recurrence after liver transplantation: risk factors, screening and clinical presentation. World J Hepatol. 2019;11(3):261–72. https://doi.org/10.4254/wjh.v11.i3.261

235 Bodzin AS, Lunsford KE, Markovic D, Harlander-Locke MP, Busuttil RW, Agopian VG. Predicting mortality in patients developing recurrent hepatocellular carcinoma after liver transplantation: impact of treatment modality and recurrence characteristics. Ann Surg. 2017;266(1):118–25. https://doi.org/10.1097/sla.0000000000001894

236 Au KP, Chok K. Multidisciplinary approach for post-liver transplant recurrence of hepatocellular carcinoma: a proposed management algorithm. World J Gastroenterol. 2018;24(45):5081–94. https://doi.org/10.3748/wjg.v24.i45.5081

237 Iavarone M, Invernizzi F, Czauderna C, Sanduzzi-Zamparelli M, Bhoori S, Amaddeo G, et al. Preliminary experience on safety of regorafenib after sorafenib failure in recurrent hepatocellular carcinoma after liver transplantation. Am J Transpl. 2019;19(11):3176–84. https://doi.org/10.1111/ajt.15551

238 Shi GM, Wang J, Huang XW, Huang XY, He YF, Ji Y, et al. Graft programmed death ligand 1 expression as a marker for transplant rejection following anti-programmed death 1 immunotherapy for recurrent liver tumors. Liver Transpl. 2021;27(3):444–9. https://doi.org/10.1002/lt.25887

239 Sapisochin G, Lee WC, Joo DJ, Joh JW, Hata K, Soin AS, et al. Long-term effects of everolimus-facilitated tacrolimus reduction in living-donor liver transplant recipients with hepatocellular carcinoma. Ann Transpl. 2022;27:e937988. https://doi.org/10.12659/AOT.937988

240 Lee DD, Sapisochin G, Mehta N, Gorgen A, Musto KR, Hajda H, et al. Surveillance for HCC after liver transplantation: increased monitoring may yield aggressive treatment options and improved postrecurrence survival. Transplantation. 2020;104(10):2105–12. https://doi.org/10.1097/TP.0000000000003117

241 Zhong JH, Xing BC, Zhang WG, Chan AWH, Chong CCN, Serenari M, et al. Repeat hepatic resection versus radiofrequency ablation for recurrent hepatocellular carcinoma: retrospective multicentre study. Br J Surg. 2021;109(1):71–8. https://doi.org/10.1093/bjs/znab340

242 Wang Z, Liu M, Zhang DZ, Wu SS, Hong ZX, He GB, et al. Microwave ablation versus laparoscopic resection as first-line therapy for solitary 3-5-cm HCC. Hepatology. 2022;76(1):66–77. https://doi.org/10.1002/hep.32323

243 Li L, Zhang J, Liu X, Li X, Jiao B, Kang T. Clinical outcomes of radiofrequency ablation and surgical resection for small hepatocellular carcinoma: a meta-analysis. J Gastroenterol Hepatol. 2012;27(1):51–8. https://doi.org/10.1111/j.1440-1746.2011.06947.x

244 Huang J, Yan L, Cheng Z, Wu H, Du L, Wang J, et al. A randomized trial comparing radiofrequency ablation and surgical resection for HCC conforming to the Milan criteria. Ann Surg. 2010;252(6):903–12. https://doi.org/10.1097/SLA.0b013e3181efc656

245 Feng Q, Chi Y, Liu Y, Zhang L, Liu Q. Efficacy and safety of percutaneous radiofrequency ablation versus surgical resection for small hepatocellular carcinoma: a meta-analysis of 23 studies. J Cancer Res Clin Oncol. 2015;141(1):1–9. https://doi.org/10.1007/s00432-014-1708-1
246 Chen QW, Ying HF, Gao S, Shen YH, Meng ZQ, Chen H, et al. Radiofrequency ablation plus chemoembolization versus radiofrequency ablation alone for hepatocellular carcinoma: a systematic review and meta-analysis. Clin Res Hepatol Gastroenterol. 2016;40(3):309–14. https://doi.org/10.1016/j.clinre.2015.07.008
247 Zhang YJ, Chen MS, Chen Y, Lau WY, Peng Z. Long-term outcomes of transcatheter arterial chemoembolization combined with radiofrequency ablation as an initial treatment for early-stage hepatocellular carcinoma. JAMA Netw Open. 2021;4(9):e2126992. https://doi.org/10.1001/jamanetworkopen.2021.26992
248 Peng ZW, Zhang YJ, Chen MS, Xu L, Liang HH, Lin XJ, et al. Radiofrequency ablation with or without transcatheter arterial chemoembolization in the treatment of hepatocellular carcinoma: a prospective randomized trial. J Clin Oncol. 2013;31(4):426–32. https://doi.org/10.1200/JCO.2012.42.9936
249 Wang L, Ke Q, Lin N, Huang Q, Zeng Y, Liu J. The efficacy of transarterial chemoembolization combined with microwave ablation for unresectable hepatocellular carcinoma: a systematic review and meta-analysis. Int J Hyperthermia. 2019;36(1):1288–96. https://doi.org/10.1080/02656736.2019.1692148
250 Zhou C, Zhang X, Peng Y, Zhou K, Hu J, Yu L, et al. Surgical resection plus radiofrequency ablation versus radical surgery for hepatocellular carcinoma: a propensity score matching analysis. J Cancer. 2019;10(17):3933–40. https://doi.org/10.7150/jca.29501
251 Zhuang BW, Li W, Wang W, Li B, Lu M, Kuang M, et al. Treatment effect of radiofrequency ablation versus liver transplantation and surgical resection for hepatocellular carcinoma within Milan criteria: a population-based study. Eur Radiol. 2021;31(7):5379–89. https://doi.org/10.1007/s00330-020-07551-9
252 Livraghi T, Meloni F, Di Stasi M, Rolle E, Solbiati L, Tinelli C, et al. Sustained complete response and complications rates after radiofrequency ablation of very early hepatocellular carcinoma in cirrhosis: is resection still the treatment of choice? Hepatology. 2008;47(1):82–9. https://doi.org/10.1002/hep.21933
253 Peng ZW, Lin XJ, Zhang YJ, Liang HH, Guo RP, Shi M, et al. Radiofrequency ablation versus hepatic resection for the treatment of hepatocellular carcinomas 2 cm or smaller: a retrospective comparative study. Radiology. 2012;262(3):1022–33. https://doi.org/10.1148/radiol.11110817
254 Vietti Violi N, Duran R, Guiu B, Cercueil JP, Aubé C, Digklia A, et al. Efficacy of microwave ablation versus radiofrequency ablation for the treatment of hepatocellular carcinoma in patients with chronic liver disease: a randomised controlled phase 2 trial. Lancet Gastroenterol Hepatol. 2018;3(5):317–25. https://doi.org/10.1016/S2468-1253(18)30029-3
255 An C, Li WZ, Huang ZM, Yu XL, Han YZ, Liu FY, et al. Small single perivascular hepatocellular carcinoma: comparisons of radiofrequency ablation and microwave ablation by using propensity score analysis. Eur Radiol. 2021;31(7):4764–73. https://doi.org/10.1007/s00330-020-07571-5
256 Yu J, Yu XL, Han ZY, Cheng ZG, Liu FY, Zhai HY, et al. Percutaneous cooled-probe microwave versus radiofrequency ablation in early-stage hepatocellular carcinoma: a phase III randomised controlled trial. randomized controlled trial. Gut. 2017;66(6):1172–3. https://doi.org/10.1136/gutjnl-2016-312629
257 Tan W, Deng Q, Lin S, Wang Y, Xu G. Comparison of microwave ablation and radiofrequency ablation for hepatocellular carcinoma: a systematic review and meta-analysis. Int J Hyperthermia. 2019;36(1):264–72. https://doi.org/10.1080/02656736.2018.1562571
258 Yu J, Cheng ZG, Han ZY, Liu FY, Zheng RQ, Cheng W, et al. Period-dependent survival benefit of percutaneous microwave ablation for hepatocellular carcinoma: a 12-year real-world, multicentric experience. Liver Cancer. 2022;11(4):341–53. https://doi.org/10.1159/000522134
259 Lin SM, Lin CJ, Lin CC, Hsu CW, Chen YC. Randomised controlled trial comparing percutaneous radiofrequency thermal ablation, percutaneous ethanol injection, and percutaneous acetic acid injection to treat hepatocellular carcinoma of 3 cm or less. and percutaneous acetic acid injection to treat hepatocellular carcinoma of 3 cm or less. Gut. 2005;54(8):1151–6. https://doi.org/10.1136/gut.2004.045203
260 Ming LIU, Chao LIU, Chengli LI, et al. Expert consensus on image-guided cryoablation for hepatocellular carcinoma (2020 edition). Chin Med J. 2020;55(5):489–92. https://doi.org/10.3969/j.issn.1008-1070.2020.05.008
261 Hasegawa K, Aoki T, Ishizawa T, Kaneko J, Sakamoto Y, Sugawara Y, et al. Comparison of the therapeutic outcomes between surgical resection and percutaneous ablation for small hepatocellular carcinoma. Ann Surg Oncol. 2014;21(Suppl 3):S348–355. https://doi.org/10.1245/s10434-014-3585-x
262 Ahmed M, Solbiati L, Brace CL, Breen DJ, Callstrom MR, Charboneau JW, et al. Image-guided tumor ablation: standardization of terminology and reporting criteria--a 10-year update. J Vasc Interv Radiol. 2014;25(11):1691–705.e4. https://doi.org/10.1016/j.jvir.2014.08.027
263 Li L, Wang W, Pan H, Ma G, Shi X, Xie H, et al. Microwave ablation combined with OK-432 induces Th1-type response and specific antitumor immunity in a murine model of breast cancer. J Transl Med. 2017;15(1):23. https://doi.org/10.1186/s12967-017-1124-9
264 Mizukoshi E, Yamashita T, Arai K, Sunagozaka H, Ueda T, Arihara F, et al. Enhancement of tumor-associated antigen-specific T cell responses by radiofrequency ablation of hepatocellular carcinoma. Hepatology. 2013;57(4):1448–57. https://doi.org/10.1002/hep.26153
265 Slovak R, Ludwig JM, Gettinger SN, Herbst RS, Kim HS. Immuno-thermal ablations - boosting the anticancer immune response. J Immunother Cancer. 2017;5(1):78. https://doi.org/10.1186/s40425-017-0284-8
266 Duan X, Wang M, Han X, Ren J, Huang G, Ju S, et al. Combined use of microwave ablation and cell immunotherapy induces nonspecific immunity of hepatocellular carcinoma model mice. Cell Cycle. 2020;19(24):3595–607. https://doi.org/10.1080/15384101.2020.1853942
267 Rozeman EA, Prevoo W, Meier M, Sikorska K, Van TM, van de Wiel BA, et al. Phase Ib/II trial testing combined radiofrequency ablation and ipilimumab in uveal melanoma (SECIRA-UM). Melanoma Res. 2020;30(3):252–60. https://doi.org/10.1097/CMR.0000000000000653
268 Chinese Anti-Cancer Association Tumor Intervention Expert Committee. Principles of transcatheter arterial infusion of chemotherapeutic drugs application: Chinese oncology intervention expert consensus. J Interv Radiol. 2017;26(11):963–70. https://doi.org/10.3969/j.issn.1008-794X.2017.11.001
269 Lencioni R, de Baere T, Soulen MC, Rilling WS, Geschwind JFH. Lipiodol transarterial chemoembolization for hepatocellular carcinoma: a systematic review of efficacy and safety data. Hepatology. 2016;64(1):106–16. https://doi.org/10.1002/hep.28453
270 Pelletier G, Ducreux M, Gay F, Luboinski M, Hagège H, Dao T, et al. Treatment of unresectable hepatocellular carcinoma with lipiodol chemoembolization: a multicenter randomized trial. Groupe CHC. J Hepatol. 1998;29(1):129–34. https://doi.org/10.1016/s0168-8278(98)80187-6
271 Lo CM, Ngan H, Tso WK, Liu CL, Lam CM, Poon RTP, et al. Randomized controlled trial of transarterial lipiodol chemoembolization for unresectable hepatocellular carcinoma. Hepatology. 2002;35(5):1164–71. https://doi.org/10.1053/jhep.2002.33156

272 Llovet JM, Real MI, Montaña X, Planas R, Coll S, Aponte J, et al. Arterial embolisation or chemoembolisation versus symptomatic treatment in patients with unresectable hepatocellular carcinoma: a randomised controlled trial. Lancet. 2002;359(9319):1734–9. https://doi.org/10.1016/S0140-6736(02)08649-X
273 Cammà C, Schepis F, Orlando A, Albanese M, Shahied L, Trevisani F, et al. Transarterial chemoembolization for unresectable hepatocellular carcinoma: meta-analysis of randomized controlled trials. Radiology. 2002;224(1):47–54. https://doi.org/10.1148/radiol.2241011262
274 Llovet JM, Bruix J. Systematic review of randomized trials for unresectable hepatocellular carcinoma: chemoembolization improves survival. Hepatology. 2003;37(2):429–42. https://doi.org/10.1053/jhep.2003.50047
275 Chinese Medical Association Radiology Branch Interventional Group Collaborative Group. Expert consensus on the technical specifications of transcatheter hepatic artery chemoembolization for primary hepatocellular carcinoma. Chin J Radiol. 2011;45(10):908–12. https://doi.org/10.3760/cma.j.issn.1005-1201.2011.10.003
276 Clinical Guidelines Committee of Chinese Interventionalists College. Clinical practice guidelines for transarterial chemoembolization (TACE) treatment of hepatocellular carcinoma in China (2021 edition). Zhonghua Nei Ke Za Zhi. 2021;60(7):599–614. https://doi.org/10.3760/cma.j.cn112137-20210425-00991
277 Zhi GUO, Gaoyun TENG, Zou Y, et al. Recommended technical operation of drug-loaded microspheres for primary and metastatic hepatocellular carcinoma. Chin J Radiol. 2019;53(5):336–40. https://doi.org/10.3760/cma.j.issn.1005-1201.2019.05.002
278 Shao G, Zou Y, Lucatelli P, Tsilimigras DI, Shimise S, Kawaguchi T. Chinese expert consensus on technical recommendations for the standard operation of drug-eluting beads for transvascular embolization. Ann Transl Med. 2021;9(8):714. https://doi.org/10.21037/atm-21-1678
279 Liang B, Makamure J, Shu S, Zhang L, Sun T, Zheng C. Treatment response, survival, and safety of transarterial chemoembolization with CalliSpheres(®) microspheres versus conventional transarterial chemoembolization in hepatocellular carcinoma: a meta-analysis. Front Oncol. 2021;11:576232. https://doi.org/10.3389/fonc.2021.576232
280 Miyayama S, Matsui O. Superselective conventional transarterial chemoembolization for hepatocellular carcinoma: rationale, technique, and outcome. J Vasc Interv Radiol. 2016;27(9):1269–78. https://doi.org/10.1016/j.jvir.2016.04.014
281 Iwazawa J, Ohue S, Hashimoto N, Muramoto O, Mitani T. Survival after C-arm CT-assisted chemoembolization of unresectable hepatocellular carcinoma. Eur J Radiol. 2012;81(12):3985–92. https://doi.org/10.1016/j.ejrad.2012.08.012
282 Takayasu K, Arii S, Ikai I, Kudo M, Matsuyama Y, Kojiro M, et al. Overall survival after transarterial lipiodol infusion chemotherapy with or without embolization for unresectable hepatocellular carcinoma: propensity score analysis. AJR Am J Roentgenol. 2010;194(3):830–7. https://doi.org/10.2214/AJR.09.3308
283 Miyayama S, Matsui O, Yamashiro M, Ryu Y, Kaito K, Ozaki K, et al. Ultraselective transcatheter arterial chemoem-bolization with a 2-f tip microcatheter for small hepatocellular carcinomas: relationship between local tumor recurrence and visualization of the portal vein with iodized oil. J Vasc Interv Radiol. 2007;18(3):365–76. https://doi.org/10.1016/j.jvir.2006.12.004
284 de Baere T, Ronot M, Chung JW, Golfieri R, Kloeckner R, Park JW, et al. Initiative on superselective conventional transarterial chemoembolization results (INSPIRE). Cardiovasc Interv Radiol. 2022;45(10):1430–40. https://doi.org/10.1007/s00270-022-03233-9
285 Prajapati HJ, Spivey JR, Hanish SI, El-Rayes BF, Kauh JS, Chen Z, et al. mRECIST and EASL responses at early time point by contrast-enhanced dynamic MRI predict survival in patients with unresectable hepatocellular carcinoma (HCC) treated by doxorubicin drug-eluting beads transarterial chemoembolization (DEB TACE). Ann Oncol. 2013;24(4):965–73. https://doi.org/10.1093/annonc/mds605
286 Xia D, Wang Q, Bai W, Wang E, Wang Z, Mu W, et al. Optimal time point of response assessment for predicting survival is associated with tumor burden in hepatocellular carcinoma receiving repeated transarterial chemoembolization. Eur Radiol. 2022;32(9):5799–810. https://doi.org/10.1007/s00330-022-08716-4
287 Kim BK, Kim KA, Park JY, Ahn SH, Chon CY, Han KH, et al. Prospective comparison of prognostic values of modified Response Evaluation Criteria in Solid Tumours with European Association for the Study of the Liver criteria in hepatocellular carcinoma following chemoembolisation. Eur J Cancer. 2013;49(4):826–34. https://doi.org/10.1016/j.ejca.2012.08.022
288 Memon K, Kulik L, Lewandowski RJ, Wang, E, Riaz, A, Ryu, RK, et al, Radiographic response to locoregional therapy in hepatocellular carcinoma predicts patient survival times. Gastroenterology. 2011;141(2):526–35.e5352. https://doi.org/10.1053/j.gastro.2011.04.054
289 Lu J, Zhao M, Arai Y, Zhong BY, Zhu HD, Qi XL, et al. Clinical practice of transarterial chemoembolization for hepatocellular carcinoma: consensus statement from an international expert panel of International Society of Multidisciplinary Interventional Oncology (ISMIO). Hepatobiliary Surg Nutr. 2021;10(5):661–71. https://doi.org/10.21037/hbsn-21-260
290 Terzi E, Golfieri R, Piscaglia F, Galassi M, Dazzi A, Leoni S, et al. Response rate and clinical outcome of HCC after first and repeated cTACE performed on demand. J Hepatol. 2012;57(6):1258–67. https://doi.org/10.1016/j.jhep.2012.07.025
291 Lu J, Guo JH, Ji JS, Li YL, Lv WF, Zhu HD, et al. Irradiation stent with 125 I plus TACE versus sorafenib plus TACE for hepatocellular carcinoma with major portal vein tumor thrombosis: a multicenter randomized trial. Int J Surg. 2023;109(5):1188–98. https://doi.org/10.1097/JS9.0000000000000295
292 Luo JJ, Zhang ZH, Liu QX, Zhang W, Wang JH, Yan ZP. Endovascular brachytherapy combined with stent placement and TACE for treatment of HCC with main portal vein tumor thrombus. Hepatol Int. 2016;10(1):185–95. https://doi.org/10.1007/s12072-015-9663-8
293 Lu J, Guo JH, Zhu HD, Zhu GY, Chen L, Teng GJ. Safety and efficacy of irradiation stent placement for malignant portal vein thrombus combined with transarterial chemoembolization for hepatocellular carcinoma: a single-center experience. J Vasc Interv Radiol. 2017;28(6):786–94.e3. https://doi.org/10.1016/j.jvir.2017.02.014
294 Hongtao HU, Hailiang LAI, Guo C, et al. 125I particle implantation combined with arterial chemoembolization for the treatment of primary liver cancer combined with portal vein thrombosis. Chin J Radiol. 2012;46(6):552–6. https://doi.org/10.3760/cma.j.issn.1005-1201.2012.06.016
295 Zhang ZH, Zhang W, Gu JY, Liu QX, Ma JQ, Liu LX, et al. Treatment of hepatocellular carcinoma with tumor thrombus with the use of iodine-125 seed strand implantation and transarterial chemoembolization: a propensity-score analysis. J Vasc Interv Radiol. 2018;29(8):1085–93. https://doi.org/10.1016/j.jvir.2018.02.013
296 Yang SB, Zhang JH, Fu YF, Wang R. TACE with portal vein radioactive seeds for HCC with portal vein tumor thrombus: a meta-analysis. Minim Invasive Ther Allied Technol. 2022;31(6):856–64. https://doi.org/10.1080/13645706.2022.2045326
297 Si ZM, Wang GZ, Qian S, Qu XD, Yan ZP, Liu R, et al. Combination therapies in the management of large (≥5 cm) hepatocellular carcinoma: microwave ablation immediately followed by transarterial chemoembolization. J Vasc Interv Radiol. 2016;27(10):1577–83. https://doi.org/10.1016/j.jvir.2016.02.014

298 Lewis AR, Padula CA, Mckinney JM, Toskich BB. Ablation plus transarterial embolic therapy for hepatocellular carcinoma larger than 3 cm: science, evidence, and future directions. And future directions. Semin Interv Radiol. 2019;36(4):303–9. https://doi.org/10.1055/s-0039-1697641
299 Morimoto M, Numata K, Kondou M, Nozaki A, Morita S, Tanaka K. Midterm outcomes in patients with intermediate-sized hepatocellular carcinoma: a randomized controlled trial for determining the efficacy of radiofrequency ablation combined with transcatheter arterial chemoembolization. Cancer. 2010;116(23):5452–60. https://doi.org/10.1002/cncr.25314
300 Yuan P, Wang F, Zhu G, Chen B. The clinical efficiency of TACE combined with simultaneous computed tomography-guided radiofrequency ablation for advanced hepatocellular carci-noma. Invest New Drugs. 2021;39(5):1383–8. https://doi.org/10.1007/s10637-021-01101-w
301 Huo YR, Eslick GD. Transcatheter arterial chemoembolization plus radiotherapy compared with chemoembolization alone for hepatocellular carcinoma. A systematic review and meta-analysis. JAMA Oncol. 2015;1(6):756–65. https://doi.org/10.1001/jamaoncol.2015.2189
302 Chinese Medical Association Radiation Oncology Branch. Chinese biomedical engineering society precision radiotherapy branch liver cancer group and digestive system tumor expert committee, Chinese society of research hospitals radiation oncology branch liver cancer group. Consensus on radiotherapy for primary liver cancer in 2016. Chin J Radiat Oncol. 2016;25(11):1141–50. https://doi.org/10.3760/cma.j.issn.1004-4221.2016.11.001
303 Kim Y, Stahl CC, Makramalla A, Olowokure OO, Ristagno RL, Dhar VK, et al. Downstaging therapy followed by liver transplantation for hepatocellular carcinoma beyond Milan criteria. Surgery. 2017;162(6):1250–8. https://doi.org/10.1016/j.surg.2017.08.007
304 Mehta N, Frenette C, Tabrizian P, Hoteit M, Guy J, Parikh N, et al. Downstaging outcomes for hepatocellular carcinoma: results from the multicenter evaluation of reduction in tumor size before liver transplantation (MERITS-LT) consortium. Gastroenterology. 2021;161(5):1502–12. https://doi.org/10.1053/j.gastro.2021.07.033
305 Shi F, Wu M, Lian SS, Mo ZQ, Gou Q, Xu RD, et al. Radiofrequency ablation following downstaging of hepatocellular carcinoma by using transarterial chemoembolization: long -term outcomes. Radiology. 2019;293(3):707–15. https://doi.org/10.1148/radiol.2019181991
306 Wu JY, Yin ZY, Bai YN, Chen YF, Zhou SQ, Wang SJ, et al. Lenvatinib combined with anti-PD-1 antibodies plus transcatheter arterial chemoembolization for unresectable hepatocellular carcinoma: a multicenter retrospective study. J Hepatocell Carcinoma. 2021;8:1233–40. https://doi.org/10.2147/JHC.S332420
307 Chiang CL, Chiu K, Chan K, Lee FAS, Li JCB, Wan CWS, et al. Sequential transarterial chemoembolisation and stereotactic body radiotherapy followed by immunotherapy as conversion therapy for patients with locally advanced, unresectable hepatocellular carcinoma (START-FIT): a single-arm, phase 2 trial. Lancet Gastroenterol Hepatol. 2023;8(2):169–78. https://doi.org/10.1016/S2468-1253(22)00339-9
308 Li L, Li B, Zhang M. Postoperative adjuvant transarterial chemoembolization improves the prognosis of hepatocellular carcinoma patients with microvascular invasion: a systematic review and meta-analysis. Acta Radiol. 2020;61(6):723–31. https://doi.org/10.1177/0284185119878357
309 Esagian SM, Kakos CD, Giorgakis E, Burdine L, Barreto JC, Mavros MN. Adjuvant transarterial chemoembolization fol-lowing curative-intent hepatectomy versus hepatectomy alone for hepatocellular carcinoma: a systematic review and meta-analysis of randomized controlled trials. Cancers. 2021;13(12):2984. https://doi.org/10.3390/cancers13122984
310 Chen W, Ma T, Zhang J, Zhang X, Chen W, Shen Y, et al. A systematic review and meta-analysis of adjuvant transarterial chemoembolization after curative resection for patients with hepatocellular carcinoma. HPB. 2020;22(6):795–808. https://doi.org/10.1016/j.hpb.2019.12.013
311 Liu BJ, Gao S, Zhu X, Guo JH, Kou FX, Liu SX, et al. Combination therapy of chemoembolization and hepatic arterial infusion chemotherapy in hepatocellular carcinoma with portal vein tumor thrombosis compared with chemoembolization alone: a propensity score-matched analysis. BioMed Res Int. 2021;2021:6670367. https://doi.org/10.1155/2021/6670367
312 Huang J, Huang W, Zhan M, Guo Y, Liang L, Cai M, et al. Drug-eluting bead transarterial chemoembolization combined with FOLFOX-based hepatic arterial infusion chemotherapy for large or huge hepatocellular carcinoma. J Hepatocell Carcinoma. 2021;8:1445–58. https://doi.org/10.2147/JHC.S339379
313 Wang Q, Xia D, Bai W, Wang E, Sun J, Huang M, et al. Development of a prognostic score for recommended TACE candidates with hepatocellular carcinoma: a multicentre observational study. J Hepatol. 2019;70(5):893–903. https://doi.org/10.1016/j.jhep.2019.01.013
314 Wang Z, Wang E, Bai W, Xia D, Ding R, Li J, et al. Exploratory analysis to identify candidates benefitting from combination therapy of transarterial chemoembolization and sorafenib for first-line treatment of unresectable hepatocellular carcinoma: a multicenter retrospective observational study. Liver Cancer. 2020;9(3):308–25. https://doi.org/10.1159/000505692
315 Xia D, Bai W, Wang E, Li J, Chen X, Wang Z, et al. Lenvatinib with or without concurrent drug-eluting beads transarterial chemoembolization in patients with unresectable, advanced hepatocellular carcinoma: a real-world, multicenter, retrospective study. Advanced hepatocellular carcinoma: a real-world, multicenter, retrospective study. Liver Cancer. 2022;11(4):368–82. https://doi.org/10.1159/000523849
316 Zhu HD, Li HL, Huang MS, Yang WZ, Yin GW, Zhong BY, et al. Transarterial chemoembolization with PD-(L)1 inhibitors plus molecular targeted therapies for hepatocellular carcinoma (CHANCE001). Signal Transduct Target Ther. 2023;8(1):58. https://doi.org/10.1038/s41392-022-01235-0
317 Jin ZC, Zhong BY, Chen JJ, Zhu HD, Sun JH, Yin GW, et al. Real-world efficacy and safety of TACE plus camrelizumab and apatinib in patients with HCC (CHANCE2211): a propensity score matching study. Eur Radiol. 2023;33(12):8669–81. https://doi.org/10.1007/s00330-023-09754-2
318 Li S, Wu J, Wu J, Fu Y, Zeng Z, Li Y, et al. Prediction of early treatment response to the combination therapy of TACE plus lenvatinib and anti-PD-1 antibody immunotherapy for unresectable hepatocellular carcinoma: multicenter retrospective study. Front Immunol. 2023;14:1109771. https://doi.org/10.3389/fimmu.2023.1109771
319 Kudo M, Ren Z, Guo Y, Han G, Lin H, Zheng J, et al. Transarterial chemoembolisation combined with lenvatinib plus pembrolizumab versus dual placebo for unresectable, non-metastatic hepatocellular carcinoma (LEAP-012): a multicentre, randomised, double-blind, phase 3 study. Lancet. 2025;405(10474):203–15. https://doi.org/10.1016/S0140-6736(24)02575-3
320 Jang JW, Choi JY, Bae SH, Kim CW, Yoon SK, Cho SH, et al. Transarterial chemolipiodolization can reactivate hepatitis B virus replication in patients with hepatocellular carcinoma. J Hepatol. 2004;41(3):427–35. https://doi.org/10.1016/j.jhep.2004.05.014
321 Chinese Medical Association Infectious Diseases Branch; Chinese Medical Association Hepatology Branch.Guidelines for the prevention and treatment of chronic hepatitis B. J Clin Hepatobiliary Dis. 2019;35(12):2648–69. https://doi.org/10.3969/j.issn.1001-5256.2019.12.007

322 Lyu N, Wang X, Li JB, Lai JF, Chen QF, Li SL, et al. Arterial chemotherapy of oxaliplatin plus fluorouracil versus sorafenib in advanced hepatocellular carcinoma: a biomolecular exploratory, randomized, phase III trial (FOHAIC-1). J Clin Oncol. 2022;40(5):468–80. https://doi.org/10.1200/JCO.21.01963
323 Lyu N, Lin Y, Kong Y, Zhang Z, Liu L, Zheng L, et al. FOXAI: a phase II trial evaluating the efficacy and safety of hepatic arterial infusion of oxaliplatin plus fluorouracil/leucovorin for advanced hepatocellular carcinoma. Gut. 2018;67(2):395–6. https://doi.org/10.1136/gutjnl-2017-314138
324 Long Y, Liang Y, Li S, Guo J, Wang Y, Luo Y, et al. Therapeutic outcome and related predictors of stereotactic body radiotherapy for small liver-confined HCC: a systematic review and meta-analysis of observational studies. Radiat Oncol. 2021;16(1):68. https://doi.org/10.1186/s13014-021-01761-1
325 Chen YX, Zhuang Y, Yang P, Fan J, Zhou J, Hu Y, et al. Helical IMRT-based stereotactic body radiation therapy using an abdominal compression technique and modified fractionation regimen for small hepatocellular carcinoma. Technol Cancer Res Treat. 2020;19:1533033820937002. https://doi.org/10.1177/1533033820937002
326 Chino F, Stephens SJ, Choi SS, Marin D, Kim CY, Morse MA, et al. The role of external beam radiotherapy in the treatment of hepatocellular cancer. Cancer. 2018;124(17):3476–89. https://doi.org/10.1002/cncr.31334
327 Hara K, Takeda A, Tsurugai Y, Saigusa Y, Sanuki N, Eriguchi T, et al. Radiotherapy for hepatocellular carcinoma results in comparable survival to radiofrequency ablation: a propensity score analysis. Hepatology. 2019;69(6):2533–45. https://doi.org/10.1002/hep.30591
328 Jang WI, Bae SH, Kim MS, Han CJ, Park SC, Kim SB, et al. A phase 2 multicenter study of stereotactic body radiotherapy for hepatocellular carcinoma: safety and efficacy. Cancer. 2020;126(2):363–72. https://doi.org/10.1002/cncr.32502
329 Kim N, Cheng J, Jung I, Liang JD, Shih YL, Huang WY, et al. Stereotactic body radiation therapy vs. radiofrequency ablation in Asian patients with hepatocellular carcinoma. J Hepatol. 2020;73(1):121–9. https://doi.org/10.1016/j.jhep.2020.03.005
330 Su TS, Liang P, Liang J, Lu HZ, Jiang HY, Cheng T, et al. Long-term survival analysis of stereotactic ablative radiotherapy versus liver resection for small hepatocellular carcinoma. Int J Radiat Oncol Biol Phys. 2017;98(3):639–46. https://doi.org/10.1016/j.ijrobp.2017.02.095
331 Wahl DR, Stenmark MH, Tao Y, Pollom EL, Caoili EM, Lawrence TS, et al. Outcomes after stereotactic body radiotherapy or radiofrequency ablation for hepatocellular carcinoma. J Clin Oncol. 2016;34(5):452–9. https://doi.org/10.1200/JCO.2015.61.4925
332 Comito T, Loi M, Franzese C, Clerici E, Franceschini D, Badalamenti M, et al. Stereotactic radiotherapy after incomplete transarterial (chemo-) embolization (TAE\-TACE) versus exclusive TAE or TACE for treatment of inoperable HCC: a phase III trial (NCT02323360). Curr Oncol. 2022;29(11):8802–13. https://doi.org/10.3390/curroncol29110692
333 Yoon SM, Kim SY, Lim YS, Kim KM, Shim JH, Lee D, et al. Stereotactic body radiation therapy for small (≤5 cm) hepatocellular carcinoma not amenable to curative treatment. Results of a single-arm, phase II clinical trial. Clin Mol Hepatol. 2020;26(4):506–15. https://doi.org/10.3350/cmh.2020.0038
334 Meng MB, Cui YL, Lu Y, She B, Chen Y, Guan YS, et al. Transcatheter arterial chemoembolization in combination with radiotherapy for unresectable hepatocellular carcinoma: a systematic review and meta-analysis. Radiother Oncol. 2009;92(2):184–94. https://doi.org/10.1016/j.radonc.2008.11.002
335 Ohri N, Dawson LA, Krishnan S, Seong J, Cheng JC, Sarin SK, et al. Radiotherapy for hepatocellular carcinoma: new indications and directions for future study. J Natl Cancer Inst. 2016;108(9):djw133. https://doi.org/10.1093/jnci/djw133
336 Yoon SM, Ryoo BY, Lee SJ, Kim JH, Shin JH, An JH, et al. Efficacy and safety of transarterial chemoembolization plus external beam radiotherapy vs sorafenib in hepatocellular carcinoma with macroscopic vascular invasion: a randomized clinical trial. JAMA Oncol. 2018;4(5):661–9. https://doi.org/10.1001/jamaoncol.2017.5847
337 Zeng ZC, Fan J, Tang ZY, Zhou J, Qin LX, Wang JH, et al. A comparison of treatment combinations with and without radiotherapy for hepatocellular carcinoma with portal vein and/or inferior vena cava tumor thrombus. Int J Radiat Oncol Biol Phys. 2005;61(2):432–43. https://doi.org/10.1016/j.ijrobp.2004.05.025
338 Shen L, Xi M, Zhao L, Zhang X, Wang X, Huang Z, et al. Combination therapy after TACE for hepatocellular carcinoma with macroscopic vascular invasion: stereotactic body radiotherapy versus sorafenib. Cancers. 2018;10(12):516. https://doi.org/10.3390/cancers10120516
339 Wei Z, Zhao J, Bi X, Zhang Y, Zhou J, Li Z, et al. Neoadjuvant radiotherapy for resectable hepatocellular carcinoma with portal vein tumor thrombus: a systematic review. Hepatobiliary Surg Nutr. 2022;11(5):709–17. https://doi.org/10.21037/hbsn-20-854
340 Sun J, Yang L, Shi J, Liu C, Zhang X, Chai Z, et al. Postoperative adjuvant IMRT for patients with HCC and portal vein tumor thrombus: an open-label randomized controlled trial. Radiother Oncol. 2019;140:20–5. https://doi.org/10.1016/j.radonc.2019.05.006
341 Su K, Gu T, Xu K, Wang J, Liao H, Li X, et al. Gamma knife radiosurgery versus transcatheter arterial chemoembolization for hepatocellular carcinoma with portal vein tumor thrombus: a propensity score matching study. Hepatol Int. 2022;16(4):858–67. https://doi.org/10.1007/s12072-022-10339-2
342 Guo L, Wei X, Feng S, Zhai J, Guo W, Shi J, et al. Radiotherapy prior to or after transcatheter arterial chemoembolization for the treatment of hepatocellular carcinoma with portal vein tumor thrombus: a randomized controlled trial. Hepatol Int. 2022;16(6):1368–78. https://doi.org/10.1007/s12072-022-10423-7
343 Rim CH, Park S, Yoon WS, Shin IS, Park HC. Radiotherapy for bone metastases of hepatocellular carcinoma: a hybrid systematic review with meta-analyses. Int J Radiat Biol. 2023;99(3):419–30. https://doi.org/10.1080/09553002.2022.2094020
344 Jihye C, Jinsil S. Application of radiotherapeutic strategies in the BCLC-defined stages of hepatocellular carcinoma. Liver Cancer. 2012;1(3–4):216–25. https://doi.org/10.1159/000343836
345 Soliman H, Ringash J, Jiang H, Singh K, Kim J, Dinniwell R, et al. Phase II trial of palliative radiotherapy for hepatocellular carcinoma and liver metastases. J Clin Oncol. 2013;31(31):3980–6. https://doi.org/10.1200/JCO.2013.49.9202
346 Wong TC, Lee VH, Law AL, Pang HH, Lam KO, Lau V, et al. Prospective study of stereotactic body radiation therapy for hepatocellular carcinoma on waitlist for liver transplant. Hepatology. 2021;74(5):2580–94. https://doi.org/10.1002/hep.31992
347 Sapisochin G, Barry A, Doherty M, Fischer S, Goldaracena N, Rosales R, et al. Stereotactic body radiotherapy vs. TACE or RFA as a bridge to transplant in patients with hepatocellular carcinoma. an intention-to-treat analysis. J Hepatol. 2017;67(1):92–9. https://doi.org/10.1016/j.jhep.2017.02.022
348 Wu F, Chen B, Dong D, Rong W, Wang H, Wang L, et al. Phase 2 evaluation of neoadjuvant intensity-modulated radiotherapy in centrally located hepatocellular carcinoma: a nonrandomized controlled trial. JAMA Surg. 2022;157(12):1089–96. https://doi.org/10.1001/jamasurg.2022.4702
349 Chen B, Wu JX, Cheng SH, Wang LM, Rong WQ, Wu F, et al. Phase 2 study of adjuvant radiotherapy following narrow-margin hepatectomy in patients with HCC. Hepatology. 2021;74(5):2595–604. https://doi.org/10.1002/hep.31993

350 Shi C, Li Y, Geng L, Shen W, Sui C, Dai B, et al. Adjuvant stereotactic body radiotherapy after marginal resection for hepatocellular carcinoma with microvascular invasion: a randomised controlled trial. Eur J Cancer. 2022;166:176–84. https://doi.org/10.1016/j.ejca.2022.02.012

351 Wang WH, Wang Z, Wu JX, Zhang T, Rong WQ, Wang LM, et al. Survival benefit with IMRT following narrow-margin hepatectomy in patients with hepatocellular carcinoma close to major vessels. Liver Int. 2015;35(12):2603–10. https://doi.org/10.1111/liv.12857

352 Wang L, Wang W, Rong W, Li Z, Wu F, Liu Y, et al. Postoperative adjuvant treatment strategy for hepatocellular carcinoma with microvascular invasion: a non-randomized interventional clinical study. BMC Cancer. 2020;20(1):614. https://doi.org/10.1186/s12885-020-07087-7

353 Chen J, He K, Han Y, Guo L, Su K, Wu Z. Clinical efficacy and safety of external radiotherapy combined with sorafenib in the treatment of hepatocellular carcinoma: a systematic review and meta-analysis. Ann Hepatol. 2022;27(4):100710. https://doi.org/10.1016/j.aohep.2022.100710

354 Munoz-Schuffenegger P, Barry A, Atenafu EG, Kim J, Brierley J, Ringash J, et al. Stereotactic body radiation therapy for hepatocellular carcinoma with Macrovascular invasion. Radiother Oncol. 2021;156:120–6. https://doi.org/10.1016/j.radonc.2020.11.033

355 Chang WI, Kim BH, Kim YJ, Yoon JH, Jung YJ, Chie EK. Role of radiotherapy in Barcelona Clinic Liver Cancer stage C hepatocellular carcinoma treated with sorafenib. J Gastroenterol Hepatol. 2022;37(2):387–94. https://doi.org/10.1111/jgh.15722

356 Li H, Wu Z, Chen J, Su K, Guo L, Xu K, et al. External radiotherapy combined with sorafenib has better efficacy in unresectable hepatocellular carcinoma: a systematic review and meta-analysis. Clin Exp Med. 2023;23(5):1537–49. https://doi.org/10.1007/s10238-022-00972-4

357 Brade AM, Ng S, Brierley J, Kim J, Dinniwell R, Ringash J, et al. Phase 1 trial of sorafenib and stereotactic body radiation therapy for hepatocellular carcinoma. Int J Radiat Oncol Biol Phys. 2016;94(3):580–7. https://doi.org/10.1016/j.ijrobp.2015.11.048

358 Wang H, Zhu X, Zhao Y, Dong D, Li L, Cai Y, et al. Phase 1 trial of apatinib combined with intensity-modulated radiotherapy in unresectable hepatocellular carcinoma. BMC Cancer. 2022;22(1):771. https://doi.org/10.1186/s12885-022-09819-3

359 Huang Y, Zhang Z, Liao W, Hu K, Wang Z. Combination of sorafenib, camrelizumab, transcatheter arterial chemoembolization, and stereotactic body radiation therapy as a novel downstaging strategy in advanced hepatocellular carcinoma with portal vein tumor thrombus: a case series study. Front Oncol. 2021;11:650394. https://doi.org/10.3389/fonc.2021.650394

360 Li J, Xuan S, Dong P, Xiang Z, Gao C, Li M, et al. Immunotherapy of hepatocellular carcinoma: recent progress and new strategy. Front Immunol. 2023;14:1192506. https://doi.org/10.3389/fimmu.2023.1192506

361 Kimura T, Fujiwara T, Kameoka T, Adachi Y, Kariya S. The current role of stereotactic body radiation therapy (SBRT) in hepatocellular carcinoma (HCC). Cancers. 2022;14(18):4383. https://doi.org/10.3390/cancers14184383

362 Zhong L, Wu D, Peng W, Sheng H, Xiao Y, Zhang X, et al. Safety of PD-1/PD-L1 inhibitors combined with palliative radiotherapy and anti-angiogenic therapy in advanced hepatocellular carcinoma. Front Oncol. 2021;11:686621. https://doi.org/10.3389/fonc.2021.686621

363 Chen YX, Yang P, Du SS, Zhuang Y, Huang C, Hu Y, et al. Stereotactic body radiotherapy combined with sintilimab in patients with recurrent or oligometastatic hepatocellular carcinoma: a phase II clinical trial. World J Gastroenterol. 2023;29(24):3871–82. https://doi.org/10.3748/wjg.v29.i24.3871

364 Bujold A, Massey CA, Kim JJ, Brierley J, Cho C, Wong RKS, et al. Sequential phase I and II trials of stereotactic body radiotherapy for locally advanced hepatocellular carcinoma. J Clin Oncol. 2013;31(13):1631–9. https://doi.org/10.1200/JCO.2012.44.1659

365 Zeng Z. Stereotactic radiation therapy for hepatocellular carcinoma. Chin J Oncol. 2015;37(9):650–2. https://doi.org/10.3760/cma.j.issn.0253-3766.2015.09.004

366 He J, Shi S, Ye L, Ma G, Pan X, Huang Y, et al. A randomized trial of conventional fraction versus hypofraction radiotherapy for bone metastases from hepatocellular carcinoma. J Cancer. 2019;10(17):4031–7. https://doi.org/10.7150/jca.28674

367 Hou JZ, Zeng ZC, Wang BL, Yang P, Zhang JY, Mo HF. High dose radiotherapy with image-guided hypo-IMRT for hepatocellular carcinoma with portal vein and/or inferior vena cava tumor thrombi is more feasible and efficacious than conventional 3D-CRT. Jpn J Clin Oncol. 2016;46(4):357–62. https://doi.org/10.1093/jjco/hyv205

368 Zhang H, Chen Y, Hu Y, Yang P, Wang B, Zhang J, et al. Image-guided intensity-modulated radiotherapy improves short-term survival for abdominal lymph node metastases from hepatocellular carcinoma. Ann Palliat Med. 2019;8(5):717–27. https://doi.org/10.21037/apm.2019.11.17

369 Byun HK, Kim HJ, Im YR, Kim DY, Han KH, Seong J. Dose escalation in radiotherapy for incomplete transarterial chemoembolization of hepatocellular carcinoma. Strahlenther Onkol. 2020;196(2):132–41. https://doi.org/10.1007/s00066-019-01488-9

370 D'Avola D, Granito A, Torre-Aláez M, Piscaglia F. The importance of liver functional reserve in the non-surgical treatment of hepatocellular carcinoma. J Hepatol. 2022;76(5):1185–98. https://doi.org/10.1016/j.jhep.2021.11.013

371 Hu Y, Zhou YK, Chen YX, Shi SM, Zeng ZC. 4D-CT scans reveal reduced magnitude of respiratory liver motion achieved by different abdominal compression plate positions in patients with intrahepatic tumors undergoing helical tomotherapy. Med Phys. 2016;43(7):4335. https://doi.org/10.1118/1.4953190

372 Song SH, Jeong WK, Choi D, Kim YK, Park HC, Yu JI. Evaluation of early treatment response to radiotherapy for HCC using pre- and post-treatment MRI. Acta Radiol. 2019;60(7):826–35. https://doi.org/10.1177/0284185118805253

373 Gatti M, Maino C, Darvizeh F, Serafini A, Tricarico E, Guarneri A, et al. Role of gadoxetic acid-enhanced liver magnetic resonance imaging in the evaluation of hepatocellular carcinoma after locoregional treatment. World J Gastroenterol. 2022;28(26):3116–31. https://doi.org/10.3748/wjg.v28.i26.3116

374 Sanuki-Fujimoto N, Takeda A, Ohashi T, Kunieda E, Iwabuchi S, Takatsuka K, et al. CT evaluations of focal liver reactions following stereotactic body radiotherapy for small hepatocellular carcinoma with cirrhosis: relationship between imaging appearance and baseline liver function. Br J Radiol. 2010;83(996):1063–71. https://doi.org/10.1259/bjr/74105551

375 Guangyuan H, Pang J, Yan C, et al. Follow-up observation of CT imaging after stereotactic ablative radiotherapy for primary hepatocellular carcinoma. J Clin Mil Med. 2014;42(4):393–5. https://doi.org/10.3969/j.issn.1671-3826.2014.04.21

376 Guha C, Kavanagh BD. Hepatic radiation toxicity: avoidance and amelioration. Semin Radiat Oncol. 2011;21(4):256–63. https://doi.org/10.1016/j.semradonc.2011.05.003

377 Zeng ZC, Seong J, Yoon SM, Cheng JCH, Lam KO, Lee AS, et al. Consensus on stereotactic body radiation therapy for small-sized hepatocellular carcinoma at the 7th asia-pacific primary liver cancer expert meeting. Liver Cancer. 2017;6(4):264–74. https://doi.org/10.1159/000475768

378 Kim TH, Koh YH, Kim BH, Kim MJ, Lee JH, Park B, et al. Proton beam radiotherapy vs. radiofrequency ablation for recurrent hepatocellular carcinoma: a randomized phase III trial. J Hepatol. 2021;74(3):603–12. https://doi.org/10.1016/j.jhep.2020.09.026

379 Bian H, Zheng JS, Nan G, Li R, Chen C, Hu CX, et al. Randomized trial of [131I] metuximab in treatment of hepatocellular carcinoma after percutaneous radiofrequency ablation. J Natl Cancer Inst. 2014;106(9):dju239. https://doi.org/10.1093/jnci/dju239
380 Working committee on treatment of metastatic bone tumors of the Chinese medical association nuclear medicine branch. Expert consensus on strontium chloride [89 Sr] treatment of metastatic bone tumors (2017 edition). Chin J Nucl Med Mol Imaging, 2018;38(6):412–5. https://doi.org/10.3760/cma.j.issn.2095-2848.2018.06.008
381 Finn RS, Qin S, Ikeda M, Galle PR, Ducreux M, Kim TY, et al. Atezolizumab plus bevacizumab in unresectable hepatocellular carcinoma. N Engl J Med. 2020;382(20):1894–905. https://doi.org/10.1056/NEJMoa1915745
382 Cheng AL, Qin S, Ikeda M, Galle PR, Ducreux M, Kim TY, et al. Updated efficacy and safety data from IMbrave150: atezolizumab plus bevacizumab vs. sorafenib for unresectable hepatocellular carcinoma. J Hepatol. 2022;76(4):862–73. https://doi.org/10.1016/j.jhep.2021.11.030
383 Ren Z, Xu J, Bai Y, Xu A, Cang S, Du C, et al. Sintilimab plus a bevacizumab biosimilar (IBI305) versus sorafenib in unresectable hepatocellular carcinoma (ORIENT-32): a randomised, open-label, phase 2-3 study. Lancet Oncol. 2021;22(7):977–90. https://doi.org/10.1016/S1470-2045(21)00252-7
384 Qin S, Chan SL, Gu S, Bai Y, Ren Z, Lin X, et al. Camrelizumab plus rivoceranib versus sorafenib as first-line therapy for unresectable hepatocellular carcinoma (CARES-310): a randomised, open-label, international phase 3 study. Lancet. 2023;402(10408):1133–46. https://doi.org/10.1016/S0140-6736(23)00961-3
385 Qin S, Bi F, Gu S, Bai Y, Chen Z, Wang Z, et al. Donafenib versus sorafenib in first-line treatment of unresectable or metastatic hepatocellular carcinoma: a randomized, open- label, parallel-controlled phase II-III trial. J Clin Oncol. 2021;39(27):3002–11. https://doi.org/10.1200/JCO.21.00163
386 Kudo M, Finn RS, Qin S, Han KH, Ikeda K, Piscaglia F, et al. Lenvatinib versus sorafenib in first-line treatment of patients with unresectable hepatocellular carcinoma: a randomised phase 3 non-inferiority trial. Lancet. 2018;391(10126):1163–73. https://doi.org/10.1016/s0140-6736(18)30207-1
387 Qin S, Kudo M, Meyer T, Bai Y, Guo Y, Meng Z, et al. Tislelizumab vs sorafenib as first-line treatment for unresectable hepatocellular carcinoma: a phase 3 randomized clinical trial. JAMA Oncol. 2023;9(12):1651–9. https://doi.org/10.1001/jamaoncol.2023.4003
388 Llovet JM, Ricci S, Mazzaferro V, Hilgard P, Gane E, Blanc JF, et al. Sorafenib in advanced hepatocellular carcinoma. N Engl J Med. 2008;359(4):378–90. https://doi.org/10.1056/NEJMoa0708857
389 Cheng AL, Kang YK, Chen Z, Tsao CJ, Qin S, Kim JS, et al. Efficacy and safety of sorafenib in patients in the Asia-Pacific region with advanced hepatocellular carcinoma: a phase III randomised, double-blind, placebo-controlled trial. Lancet Oncol. 2009;10(1):25–34. https://doi.org/10.1016/S1470-2045(08)70285-7
390 Pressiani T, Boni C, Rimassa L, Labianca R, Fagiuoli S, Salvagni S, et al. Sorafenib in patients with Child-Pugh class A and B advanced hepatocellular carcinoma: a prospective feasibility analysis. Ann Oncol. 2013;24(2):406–11. https://doi.org/10.1093/annonc/mds343
391 Qin S, Bai Y, Lim HY, Thongprasert S, Chao Y, Fan J, et al. Randomized, multicenter, open-label study of oxaliplatin plus fluorouracil/leucovorin versus doxorubicin as palliative chemotherapy in patients with advanced hepatocellular carcinoma from Asia. J Clin Oncol. 2013;31(28):3501–8. https://doi.org/10.1200/JCO.2012.44.5643
392 Qin S, Cheng Y, Liang J, Shen L, Bai Y, Li J, et al. Efficacy and safety of the FOLFOX4 regimen versus doxorubicin in Chinese patients with advanced hepatocellular carcinoma. A subgroup analysis of the EACH study. Oncologist. 2014;19(11):1169–78. https://doi.org/10.1634/theoncologist.2014-0190
393 Fenglian QU, Xuezhi HAO, Qin S, et al. Phase II multicenter clinical study of arsenite injection in the treatment of primary hepatocellular carcinoma. Chin J Cancer. 2011;33(9):697–701. https://doi.org/10.3760/cma.j.issn.0253-3766.2011.09.013
394 Abou-Alfa GK, Lau G, Kudo M, Chan SL, Kelley RK, Furuse J, et al. Tremelimumab plus durvalumab in unresectable hepatocellular carcinoma. NEJM Evid. 2022;1(8):EVIDoa2100070. https://doi.org/10.1056/EVIDoa2100070
395 Galle PR, Decaens T, Kudo M, Qin S, Fonseca L, Sangro B, et al. Nivolumab (NIVO) plus ipilimumab (IPI) vs lenvatinib (LEN) or sorafenib (SOR) as first-line treatment for unresectable hepatocellular carcinoma (uHCC): first results from CheckMate 9DW. J Clin Oncol. 2024;42(17_Suppl l):LBA4008. abstr LBA4008).https://doi.org/10.1200/JCO.2024.42.17_suppl.LBA4008
396 Qiao Q, Han C, Ye S, Li J, Shao G, Bai Y, et al. The efficacy and safety of cadonilimab combined with lenvatinib for first-line treatment of advanced hepatocellular carcinoma (COMPASSION-08): a phase Ib/II single-arm clinical trial. Front Immunol. 2023;14:1238667. https://doi.org/10.3389/fimmu.2023.1238667
397 Bruix J, Qin S, Merle P, Granito A, Huang YH, Bodoky G, et al. Regorafenib for patients with hepatocellular carcinoma who progressed on sorafenib treatment (RESORCE): a randomised, double-blind, placebo-controlled, phase 3 trial. Lancet. 2017;389(10064):56–66. https://doi.org/10.1016/s0140-6736(16)32453-9
398 Qin S, Li Q, Gu S, Chen X, Lin L, Wang Z, et al. Apatinib as second-line or later therapy in patients with advanced hepatocellular carcinoma (AHELP): a multicentre, double-blind, randomised, placebo-controlled, phase 3 trial. Lancet Gastroenterol Hepatol. 2021;6(7):559–68. https://doi.org/10.1016/S2468-1253(21)00109-6
399 Zhu AX, Kang YK, Yen CJ, Finn RS, Galle PR, Llovet JM, et al. Ramucirumab after sorafenib in patients with advanced hepatocellular carcinoma and increased α-fetoprotein concentrations (REACH-2): a randomised, double-blind, placebo-controlled, phase 3 trial. Lancet Oncol. 2019;20(2):282–96. https://doi.org/10.1016/S1470-2045(18)30937-9
400 Shao G, Bai Y, Yuan X, Chen X, Gu S, Gu K, et al. Ramucirumab as second-line treatment in Chinese patients with advanced hepatocellular carcinoma and elevated alpha-fetoprotein after sorafenib (REACH-2 China): a randomised, multicentre, double-blind study. eClinicalMedicine. 2022;54:101679. https://doi.org/10.1016/j.eclinm.2022.101679
401 Qin S, Chen Z, Fang W, Ren Z, Xu R, Ryoo BY, et al. Pembrolizumab versus placebo as second-line therapy in patients from Asia with advanced hepatocellular carcinoma: a randomized, double-blind, phase III trial. J Clin Oncol. 2023;41(7):1434–43. https://doi.org/10.1200/JCO.22.00620
402 Qin S, Ren Z, Meng Z, Chen Z, Chai X, Xiong J, et al. Camrelizumab in patients with previously treated advanced hepatocellular carcinoma: a multicentre, open-label, parallel-group, randomised, phase 2 trial. Lancet Oncol. 2020;21(4):571–80. https://doi.org/10.1016/S1470-2045(20)30011-5
403 Xu J, Shen J, Gu S, Zhang Y, Wu L, Wu J, et al. Camrelizumab in combination with apatinib in patients with advanced hepatocellular carcinoma (RESCUE): a nonrandomized, open-label, phase II trial. Clin Cancer Res. 2021;27(4):1003–11. https://doi.org/10.1158/1078-0432.CCR-20-2571
404 Xu J, Zhang Y, Jia R, Yue C, Chang L, Liu R, et al. Anti-PD-1 antibody SHR-1210 combined with apatinib for advanced hepatocellular carcinoma, gastric, or esophagogastric junction cancer: an open-label, dose escalation and expansion study. Clin Cancer Res. 2019;25(2):515–23. https://doi.org/10.1158/1078-0432.CCR-18-2484
405 Ren Z, Ducreux M, Abou-Alfa GK, Merle P, Fang W, Edeline J, et al. Tislelizumab in patients with previously treated advanced hepatocellular carcinoma (RATIONALE-208): a multicenter, non-randomized, open-label, phase 2 trial. Liver Cancer. 2023;12(1):72–84. https://doi.org/10.1159/000527175

406 Yau T, Kang YK, Kim TY, El-Khoueiry AB, Santoro A, Sangro B, et al. Efficacy and safety of nivolumab plus ipilimumab in patients with advanced hepatocellular carcinoma previously treated with sorafenib: the CheckMate 040 randomized clinical trial. JAMA Oncol. 2020;6(11):e204564. https://doi.org/10.1001/jamaoncol.2020.4564
407 Abou-Alfa GK, Meyer T, Cheng AL, El-Khoueiry AB, Rimassa L, Ryoo BY, et al. Cabozantinib in patients with advanced and progressing hepatocellular carcinoma. N Engl J Med. 2018;379(1):54–63. https://doi.org/10.1056/NEJMoa1717002
408 NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines). Hepatocellular carcinoma. version 1. 2023.
409 Hasegawa K, Takemura N, Yamashita T, Watadani T, Kaibori M, Kubo S, et al. Clinical practice guidelines for hepatocellular carcinoma: the Japan society of Hepatology 2021 version (5th JSH-HCC guidelines). Hepatol Res. 2023;53(5):383–90. https://doi.org/10.1111/hepr.13892
410 Singal AG, Llovet JM, Yarchoan M, Mehta N, Heimbach JK, Dawson LA, et al. AASLD Practice Guidance on prevention, diagnosis, and treatment of hepatocellular carcinoma. Hepatology. 2023;78(6):1922–65. https://doi.org/10.1097/HEP.0000000000000466
411 Seymour L, Bogaerts J, Perrone A, Ford R, Schwartz LH, Mandrekar S, et al. iRECIST: guidelines for response criteria for use in trials testing immunotherapeutics. Lancet Oncol. 2017;18(3):e143–52. https://doi.org/10.1016/S1470-2045(17)30074-8
412 Ding-Fang C. Disease identification and treatment to create a Chinese system of integrated Chinese and Western clinical medicine. Chin J Integr Med. 2019;39(9):1034–5. https://doi.org/10.7661/j.cjim.20190815.241
413 Cai D. On the clinical diagnosis and treatment model of disease evidence combination. Chin J Integr Med. 2019;39(2):133–5. https://doi.org/10.7661/j.cjim.20190114.040
414 Sun Y, Qin SK, Li W, Guo Y, Zhang Y, Meng L, et al. A randomized, double-blinded, phase III study of icaritin versus huachashu as the first-line therapy in biomarker-enriched HBV -related advanced hepatocellular carcinoma with poor conditions: interim analysis result. J Clin Oncol. 2021;39(15_Suppl l):abstract 4077. https://doi.org/10.1200/JCO.2021.39.15_suppl.4077
415 Cai W, Yin C, Fan Q. Clinical observation of AIDI combined with hepatic artery chemoembolization for the treatment of intermediate and advanced primary hepatocellular carcinoma. Chin J Physicians. 2018;20(11):1723–5. https://doi.org/10.3760/cma.j.issn.1008-1372.2018.11.032
416 Cheng Y, Hua H. Research progress of elemene in the treatment of primary hepatocellular carcinoma. J Clin Oncol. 2017;22(10):950–3. https://doi.org/10.3969/j.issn.1009-0460.2017.10.018
417 Ying F, Qingyuan L, Zhou Z, et al. Study on the effect of TACE combined with Jinlong capsule in the treatment of primary hepatocellular carcinoma. China Pract Med. 2019;14(21):42–4. https://doi.org/10.14163/j.cnki.11-5547/r.2019.21.022
418 Ji-liang G. Prospective, randomized controlled clinical study of Liver Flex formula for advanced primary liver cancer. Chin J Traditional Chin Med. 2014;39(12):2367–9. https://doi.org/10.4268/cjcmm20141243
419 Lu D, Wang Y, Weilin Z, et al. Clinical study of Conrad combined with hepatic artery chemoembolization for hepatocellular carcinoma. World Clin Med. 2017;11(5):70–2.
420 Zhai XF, Liu XL, Shen F, Fan J, Ling CQ. Traditional herbal medicine prevents postoperative recurrence of small hepatocellular carcinoma: a randomized controlled study. Cancer. 2018;124(10):2161–8. https://doi.org/10.1002/cncr.30915
421 Zheng DH, Yang JM, Wu JX, Cheng SQ, Zhang SG, Wu D, et al. Cidan capsule in combination with adjuvant transarterial chemoembolization reduces recurrence rate after curative resection of hepatocellular carcinoma: a multicenter, randomized controlled trial. Chin J Integr Med. 2023;29(1):3–9. https://doi.org/10.1007/s11655-022-3537-4
422 International Hepatobiliary and Pancreatic Association China. Chinese Anti-Cancer Association Liver Cancer Committee, Chinese Society of Research Hospitals Hepatobiliary and Pancreatic Surgery Committee, et al. Chinese expert consensus on antiviral therapy for hepatitis B virus-associated hepatocellular carcinoma (2023 edition). Chin J Gastrointest Surg. 2023;22(1):29–41. https://doi.org/10.3760/cma.j.cn115610-20221024-00612
423 Liver Cancer Study Group Chinese Society of Hepatology Chinese Medical Association. Expert consensus on antiviral therapy for HBV/HCV-associated hepatocellular carcinoma (2021 update). Chin J Liver Dis. 2021;29(10):948–66. https://doi.org/10.3760/cma.j.cn501113-20210907-00456
424 Chinese Medical Association Hepatology Branch, Chinese Medical Association Infectious Diseases Branch. Guidelines for the prevention and treatment of hepatitis C Chin J Infect Dis. 2023(1):29–46. https://doi.org/10.3760/cma.j.cn311365-20230217-00045
425 Yang H, Mao Y, Lu X, Sang X, Du S, Zhao H, et al. The effects of urinary trypsin inhibitor on liver function and inflammatory factors in patients undergoing hepatectomy: a prospective, randomized, controlled clinical study. Am J Surg. 2011;202(2):151–7. https://doi.org/10.1016/j.amjsurg.2010.07.044
426 Qin S. Ma Jun. Chinese Society of Clinical Oncology (CSCO) guidelines for standardized management of radiotherapy-associated neutropenia in oncology (2021). J Clin Oncol. 2021;26(7):638–48. https://doi.org/10.3969/j.issn.1009-0460.2021.07.011
427 Thrombosis and Hemostasis Group Chinese Society of Hematology Chinese Medical Association. Chinese expert consensus on the management of clinical application of thrombopoietic drugs (2023 edition). Chin J Hematol. 2023;44(7):535–42. https://doi.org/10.3760/cma.j.issn.0253-2727.2023.07.002
428 Huang A, Guo DZ, Wang YP, Fan J, Yang XR, Zhou J. The treatment strategy and outcome for spontaneously ruptured hepatocellular carcinoma: a single-center experience in 239 patients. J Cancer Res Clin Oncol. 2022;148(11):3203–14. https://doi.org/10.1007/s00432-022-03916-3
429 Moris D, Chakedis J, Sun SH, Spolverato G, Tsilimigras DI, Ntanasis-Stathopoulos I, et al. Management, outcomes, and prognostic factors of ruptured hepatocellular carcinoma: a systematic review. J Surg Oncol. 2018;117(3):341–53. https://doi.org/10.1002/jso.24869
430 Sahu SK, Chawla YK, Dhiman RK, Singh V, Duseja A, Taneja S, et al. Rupture of hepatocellular carcinoma: a review of literature. J Clin Exp Hepatol. 2019;9(2):245–56. https://doi.org/10.1016/j.jceh.2018.04.002
431 Tan NP, Majeed A, Roberts SK, Gow PJ, Hey P, Mah X, et al. Survival of patients with ruptured and non-ruptured hepatocellular carcinoma. Med J Aust. 2020;212(6):277–8. https://doi.org/10.5694/mja2.50483
432 Yoshida H, Mamada Y, Taniai N, Uchida E. Spontaneous ruptured hepatocellular carcinoma. Hepatol Res. 2016;46(1):13–21. https://doi.org/10.1111/hepr.12498
433 Zhong F, Cheng XS, He K, Sun SB, Zhou J, Chen HM. Treatment outcomes of spontaneous rupture of hepatocellular carcinoma with hemorrhagic shock: a multicenter study. Springerplus. 2016;5(1):1101. https://doi.org/10.1186/s40064-016-2762-8
434 Aoki T, Kokudo N, Matsuyama Y, Izumi N, Ichida T, Kudo M, et al. Prognostic impact of spontaneous tumor rupture in patients with hepatocellular carcinoma: an analysis of 1160 cases from a nationwide survey. Ann Surg. 2014;259(3):532–42. https://doi.org/10.1097/SLA.0b013e31828846de

435 Lai EC, Lau WY. Spontaneous rupture of hepatocellular carcinoma: a systematic review. Arch Surg. 2006;141(2):191–8. https://doi.org/10.1001/archsurg.141.2.191
436 Shin BS, Park MH, Jeon GS. Outcome and prognostic factors of spontaneous ruptured hepatocellular carcinoma treated with transarterial embolization. Acta Radiol. 2011;52(3):331–5. https://doi.org/10.1258/ar.2010.100369
437 Park J, Jeong YS, Suh YS, Kim HC, Chung JW, Choi JW. Clinical course and role of embolization in patients with spontaneous rupture of hepatocellular carcinoma. Front Oncol. 2022;12:999557. https://doi.org/10.3389/fonc.2022.999557
438 Roussel E, Bubenheim M, Le Treut YP, Laurent A, Herrero A, Muscari F, et al. Peritoneal carcinomatosis risk and long-term survival following hepatectomy for spontaneous hepatocellular carcinoma rupture: results of a multicenter French study (French-AFC). Ann Surg Oncol. 2020;27(9):3383–92. https://doi.org/10.1245/s10434-020-08442-5